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Exercise for depression (Review)

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Exercise for depression (Review)

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[Intervention Review]

Exercise for depression

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ABSTRACT

Rationale

Depression is a common cause of morbidity and mortality worldwide. Depression is often treated with antidepressants or psychological therapy, or both, but some people may prefer alternative approaches such as exercise. This review updates one first published in 2008 and last updated in 2013.

Objectives

To determine the effectiveness of exercise in the treatment of depression in adults compared with no intervention, waiting list control or placebo, or where exercise is used as an adjunct to an established treatment that is received by both exercising and non-exercising groups.

To determine the effectiveness of exercise compared with other active interventions for depression in adults (psychological therapies, pharmacological treatments or alternative interventions such as light therapy).

Search methods

We searched the Cochrane Depression, Anxiety and Neurosis Review Group's Controlled Trials Register (CCDANCTR) to November 2013. We searched MEDLINE, Embase, PsycINFO and the Cochrane Central Register of Controlled Trials (CENTRAL) from 2013 to November 2023. No date or language restrictions were applied.

Eligibility criteria

We included randomised controlled trials (RCTs) in which exercise was compared to no treatment, inactive treatment or active treatment in adults (aged 18 years and over) with depression. We included trials that randomised individual participants or clusters. We excluded trials of postnatal depression. Two authors independently undertook study selection.

Outcomes

The primary outcome we assessed was a measure of depression or mood at the end of treatment and at any longer-term follow-up. Other outcomes were treatment acceptability, quality of life, cost and adverse events.

Risk of bias

We assessed the risk of bias using the Cochrane risk of bias tool RoB 1. Two authors independently performed the risk of bias assessment.

Synthesis methods

Two authors independently extracted data on outcomes at the end of the trial and end of follow-up (if available). We calculated effect sizes for each trial using Hedges' *g* method and a mean difference (MD) or standardised mean difference (SMD) for the overall pooled effect for continuous data, and risk ratios for dichotomous data. Where trials used a number of different tools to assess depression, we included only the main outcome measure in our meta-analyses. Where trials provided several 'doses' of exercise, we used data from the largest dose and performed sensitivity analysis using the lower dose. We performed subgroup analysis to explore the influence of diagnostic method, exercise intensity, number of exercise sessions, type of exercise and type of control (i.e. placebo, no treatment, waiting list, usual care and self monitoring). Through our sensitivity analyses, we explored the influence of study risk of bias.

Included studies

We included 73 RCTs (at least 4985 participants) in the review, 69 of which contributed data to our meta-analyses.

Synthesis of results

For the 57 trials (2189 participants) comparing exercise with no treatment or a control intervention, the pooled SMD for depressive symptoms at the end of treatment was -0.67 (95% confidence interval (CI) -0.82 to -0.52 ; low-certainty evidence), showing that exercise may result in a reduction in depressive symptoms. When we included only the seven trials (447 participants) with adequate allocation concealment, intention-to-treat analysis and blinded outcome assessment, the pooled SMD was smaller (SMD -0.46 , 95% CI -0.88 to -0.04). Pooled data from the nine trials (405 participants) with long-term follow-up provided very uncertain evidence about the effect of exercise on depressive symptoms (SMD -0.53 , 95% CI -1.11 to 0.06 ; very low certainty evidence).

Ten trials (414 participants) compared exercise with psychological therapy, finding there is probably little to no difference in their effect on depressive symptoms at the end of treatment (SMD 0.03 , 95% CI -0.16 to 0.23 ; moderate-certainty evidence). There were similar results at long-term follow-up (SMD -0.11 , 95% CI -0.48 to 0.26 ; 4 studies, 114 participants; low-certainty evidence).

Five trials (330 participants) compared exercise with pharmacological treatment, finding there may be little to no difference in their effect on depressive symptoms at the end of treatment (SMD -0.11 , 95% CI -0.33 to 0.10 ; low-certainty evidence). The evidence was very uncertain at long-term follow-up (SMD -0.40 , 95% CI -0.80 to 0.00 ; 1 study, 58 participants; very low certainty evidence).

There did not appear to be a difference between exercise and other interventions in terms of treatment acceptability, as measured by participants completing the study (moderate to low certainty evidence).

Results for the outcome 'quality of life' were inconsistent (low to very low certainty evidence).

Adverse events were not common in any comparison, but included musculoskeletal injuries and depression affecting those undertaking exercise, and diarrhoea, sexual dysfunction and fatigue reported by those receiving sertraline.

Many trials were affected by multiple sources of bias: randomisation was adequately concealed in only 22 studies, only 31 used intention-to-treat analyses, and only 23 used blinded outcome assessors. Blinding of those receiving and those delivering the interventions is inherently difficult; we judged all studies to be at high risk of performance bias. Many trials used participant self-report rating scales, which have the potential to bias findings.

Authors' conclusions

Exercise may be moderately more effective than a control intervention for reducing symptoms of depression. Exercise appears to be no more or less effective than psychological or pharmacological treatments, though this conclusion is based on a few small trials. Long-term follow-up was rare.

The addition of 35 RCTs (at least 2526 participants) to this update has had very little effect on the estimate of the benefit of exercise on symptoms of depression. If further research is to take place, it should focus on improving trial quality, assessing which characteristics of exercise are effective for different people, and exploring health equity.

Funding

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Registration

Protocols and previous versions: DOI 10.1002/14651858.CD004366; DOI 10.1002/14651858.CD4366.pub2; DOI 10.1002/14651858.CD4366.pub3; DOI 10.1002/14651858.CD4366.pub4; DOI 10.1002/14651858.CD4366.pub5; DOI 10.1002/14651858.CD4366.pub6

PLAIN LANGUAGE SUMMARY

Is exercise effective for treating depression?

Key messages

- Exercise may be moderately effective compared to no therapy for reducing symptoms of depression.
- The evidence suggests there is little to no difference in the reduction of symptoms of depression provided by exercise compared to those provided by psychological therapies or antidepressants, but this conclusion is based on a few small studies.
- The studies measured the outcomes at the end of treatment, and most of them did not follow up participants in the longer term.
- Unwanted effects from exercise were not common, affecting only a small number of participants.

What is depression?

Depression is a common illness, affecting over 100 million people worldwide. Depression can have a significant impact on people's physical health, as well as reducing their quality of life.

How is depression treated?

Research has shown that both pharmacological treatment (antidepressant medication) and psychological therapies (i.e. talking treatments aimed at changing people's thoughts, emotions, or behaviours) can be effective for treating depression. However, many people prefer to try alternative approaches. Some health guidelines suggest that exercise could be used as an alternative treatment.

What did we want to find out?

We wanted to find out if exercise reduces the symptoms of depression and improves quality of life in people with depression, and we wanted to find out how exercise compares to medication, psychological therapy and alternative treatments. We also wanted to find out if exercise is associated with any unwanted effects and if it provides good value for money. This review updates one last published in 2013.

What did we do?

We searched for randomised controlled trials (RCTs) (i.e. studies where people are assigned to a treatment group randomly) that assessed the effectiveness of exercise for treating depression in adults (18 years of age and over). Studies had to compare exercise with either another active treatment (e.g. medication or psychological therapy) or an inactive intervention (e.g. no treatment, being put on a waiting list, or being given a placebo treatment (i.e. an inactive treatment that seems like a real treatment)). All studies had to include adults with a diagnosis of depression, and the physical activity carried out had to fit our definition of 'exercise'.

We described, evaluated and summarised the results of the studies. We made a judgement about our confidence in the evidence based on factors such as the size of the studies and the methods used to conduct them. We searched medical databases for studies up to November 2023.

What did we find?

We found 73 studies that involved at least 4985 adults with depression. The risk of bias in some of the studies was high, which lowered our confidence in the findings.

Exercise may result in a reduction in depressive symptoms compared to no therapy, although the evidence about long-term effects is uncertain.

There is probably little to no difference in depressive symptoms between people undertaking exercise and those receiving psychological therapy. There may be little to no difference in depressive symptoms between people doing exercise and those taking antidepressants.

There does not seem to be a difference between the different interventions in terms of their acceptability as treatments, as measured by the number of participants completing the studies.

The benefits of exercise compared to no therapy, psychological therapy or pharmacological treatments on quality of life are inconsistent and uncertain.

Adverse events from exercise were not common. The small number of participants who experienced them usually reported muscle and joint problems or worsening of depression.

What are the limitations of the evidence?

Many of the studies included a relatively small number of people and had a high risk of bias in terms of the research methods they used. Also, most studies only assessed the effects of exercise over a short period of time. These factors limit our confidence in the findings of the

review. Future research should focus on improving the quality of the studies, working out which characteristics of exercise are effective for different people, and ensuring different types of people are included in the studies so that health equity issues can be considered.

SUMMARY OF FINDINGS

Summary of findings 1. Summary of findings table - Exercise compared to control for adults with depression

Exercise compared to control for adults with depression

Patient or population: adults with depression

Setting: any setting

Intervention: exercise

Comparison: control (placebo, no treatment (including waiting list, usual care, self-monitoring), exercise plus treatment versus treatment alone, stretching, meditation, relaxation or casual conversation)

Outcomes	Anticipated absolute effects* (95% CI)		Relative effect (95% CI)	Nº of participants (studies)	Certainty of the evidence (GRADE)	Comments
	Risk with control (placebo, no treatment (including waiting list, usual care, self-monitoring), exercise plus treatment versus treatment alone, stretching, meditation, relaxation or casual conversation)	Risk with exercise				
Reduction in depressive symptoms post-treatment	-	SMD 0.67 SD lower (0.82 lower to 0.52 lower)	-	2189 (57 RCTs)	⊕⊕⊕⊕ Low ^{a,b,c}	Exercise may result in a reduction in depressive symptoms.
Reduction in depressive symptoms follow-up	-	SMD 0.53 SD lower (1.11 lower to 0.06 higher)	-	405 (9 RCTs)	⊕⊕⊕⊕ Very low ^{a,d,e}	The evidence is very uncertain about the effect of exercise on depressive symptoms.
Completed intervention or control	955 per 1000	955 per 1000 (935 to 974)	RR 1.00 (0.98 to 1.02)	2311 (51 RCTs)	⊕⊕⊕⊕ Moderate ^a	Exercise probably results in little to no difference in completion of intervention.
Quality of life - mental	-	SMD 0.05 SD lower (0.85 lower to 0.76 higher)	-	98 (3 RCTs)	⊕⊕⊕⊕ Very low ^{a,f,g}	The evidence is very uncertain about the effect of exercise on mental quality of life.

Quality of life - physical	-	SMD 0.57 SD higher (0.15 higher to 0.98 higher)	-	178 (5 RCTs)	⊕⊕⊕⊕ Very low ^{a,h,i}	Exercise may increase physical quality of life but the evidence is very uncertain.
Quality of life	-	SMD 0.56 SD higher (0.08 lower to 1.2 higher)	-	39 (1 RCT)	⊕⊕⊕⊕ Very low ^{a,j,k}	The evidence is very uncertain about the effect of exercise on quality of life.
Adverse events	Due to the diversity of different adverse events reported, a meta-analysis was not performed. Adverse events were reported in both the exercise and control groups, but the nature and frequency of events varied across studies. Adverse events within the intervention group affected a relatively small number of participants, mainly related to muscle and joint problems or worsening the severity of depression. Adverse events within the control group affected a relatively small number of participants and included the need for gastric lavage, gastrointestinal symptoms such as diarrhoea and loose stools, hospital admission, physical incapacity, worsening of pre-existing chronic neck pain and withdrawal for personal reasons.			551 (10 RCTs)	⊕⊕⊕⊕ Moderate ^a	Exercise likely results in little to no difference in adverse events compared to 'control'.

***The risk in the intervention group** (and its 95% confidence interval) is based on the assumed risk in the comparison group and the **relative effect** of the intervention (and its 95% CI).

CI: confidence interval; **RR:** risk ratio; **SMD:** standardised mean difference

GRADE Working Group grades of evidence

High certainty: we are very confident that the true effect lies close to that of the estimate of the effect.

Moderate certainty: we are moderately confident in the effect estimate: the true effect is likely to be close to the estimate of the effect, but there is a possibility that it is substantially different.

Low certainty: our confidence in the effect estimate is limited: the true effect may be substantially different from the estimate of the effect.

Very low certainty: we have very little confidence in the effect estimate: the true effect is likely to be substantially different from the estimate of effect.

See interactive version of this table: https://gdt.gradeapro.org/presentations/#/isof/isof_question_revman_web_453424651261012128.

^a Downgraded once because concerns about an unclear or high risk of bias in the areas of random sequence generation, allocation concealment, incomplete outcome data and selective reporting.

^b Downgraded once due to substantial statistically significant heterogeneity of $I^2 = 61\%$.

^c The lower bound of the 95% confidence interval for the effect estimate reaches the point of clinical significance. Based upon a minimal clinically important difference (MCID) of reduction of 5 points on the Beck Depression Inventory, which is equivalent to a SMD of 0.45 (Button 2015; Hengartner 2022).

^d Downgraded twice due to considerable statistically significant heterogeneity of $I^2 = 87\%$.

^e Downgraded once because the lower bound of the 95% confidence interval for the effect estimate does not reach the point of clinical significance. Based upon a MCID of reduction of 5 points on the Beck Depression Inventory which, is equivalent to a SMD of 0.45 (Button 2015; Hengartner 2022).

- ^f Downgraded once due to substantial statistically significant heterogeneity of $I^2 = 74\%$.
- ^g Downgraded once because the lower bound of the 95% confidence interval for the effect estimate does not reach the point of MCID. Based upon a MCID of reduction of 5 points on the Short-Form 36 mental component score, which is equivalent to a SMD of 0.25 (Hengartner 2022; Ogura 2020).
- ^h Downgraded once due to moderate non-statistically significant heterogeneity $I^2 = 44\%$. One study Brenes 2007 report a non-statistical significant negative effect.
- ⁱ Downgraded once because the lower bound of the 95% confidence interval for the effect estimate does not reach the point of MCID. Based upon a MCID of reduction of 5 points on the Short-Form 36 physical component score, which is equivalent to a SMD of 0.25 (Ogura 2020).
- ^j The study population was limited to community-dwelling older adults, which may limit the applicability of the findings to other populations.
- ^k Downgraded once because the lower bound of the 95% confidence interval for the effect estimate does not reach the point of MCID. Based upon a MCID of reduction of 5 points on the Short-Form 36 averaged combined scores, which is equivalent to a SMD of 0.25 (Ogura 2020).

Summary of findings 2. Summary of findings table - Exercise compared to psychological therapies for adults with depression

Exercise compared to psychological therapies for adults with depression

Patient or population: adults with depression

Setting: any setting

Intervention: exercise

Comparison: psychological therapies

Outcomes	Anticipated absolute effects* (95% CI)		Relative effect (95% CI)	Nº of participants (studies)	Certainty of the evidence (GRADE)	Comments
	Risk with psychological therapies	Risk with exercise				
Reduction in depressive symptoms post-treatment	-	SMD 0.03 SD higher (0.16 lower to 0.23 higher)	-	414 (10 RCTs)	⊕⊕⊕⊖ Moderate ^a	Exercise probably results in little to no difference in depressive symptoms.
Reduction in depressive symptoms follow-up	-	SMD 0.11 SD lower (0.48 lower to 0.26 higher)	-	114 (4 RCTs)	⊕⊕⊕⊖ Low ^{a,b}	Exercise may result in little to no difference in depressive symptoms.
Completed exercise or psychological therapies	738 per 1000	782 per 1000 (730 to 834)	RR 1.06 (0.99 to 1.13)	433 (7 RCTs)	⊕⊕⊕⊖ Moderate ^a	Exercise probably results in little to no difference in completion of intervention.
Quality of life - physical	-	SMD 0.25 SD higher (0.21 lower to 0.72 higher)	-	72 (2 RCTs)	⊕⊕⊕⊖ Low ^{a,c}	Exercise may result in a slight increase in physical quality of life.
Quality of life - mental	-	SMD 0.05 SD higher (0.41 lower to 0.52 higher)	-	72 (2 RCTs)	⊕⊕⊕⊖ Low ^{a,b}	The evidence suggests that exercise results in little to no difference in mental quality of life.

Quality of life	-	SMD 0.37 SD higher (0.28 lower to 1.02 higher)	-	37 (1 RCT)	⊕⊕⊕⊕ Very low ^{d,e}	The evidence is very uncertain about the effect of exercise on quality of life.
Adverse events	One trial by Karg (2020) reported that eight participants dropped out of the exercise intervention group, including one due to physical complaints, while 12 participants dropped out of the comparator group (cognitive behavioural therapy), with none identified as experiencing an adverse event related to the intervention.			(1 RCT)	⊕⊕⊕⊕ Low ^f	Exercise may result in little to no difference in adverse events compared to psychological therapies.

***The risk in the intervention group** (and its 95% confidence interval) is based on the assumed risk in the comparison group and the **relative effect** of the intervention (and its 95% CI).

CI: confidence interval; **RR:** risk ratio; **SMD:** standardised mean difference

GRADE Working Group grades of evidence

High certainty: we are very confident that the true effect lies close to that of the estimate of the effect.

Moderate certainty: we are moderately confident in the effect estimate: the true effect is likely to be close to the estimate of the effect, but there is a possibility that it is substantially different.

Low certainty: our confidence in the effect estimate is limited: the true effect may be substantially different from the estimate of the effect.

Very low certainty: we have very little confidence in the effect estimate: the true effect is likely to be substantially different from the estimate of effect.

See interactive version of this table: https://gdt.gradepro.org/presentations/#/isof/isof_question_revman_web_453430473936187190.

^a Downgraded once because concerns about an unclear or high risk of bias in the areas of random sequence generation, allocation concealment, incomplete outcome data, and selective reporting.

^b The certainty of the evidence was downgraded due to imprecision, as the confidence interval encompasses both increased and decreased effectiveness, with the possibility of a clinically significant benefit favoring both intervention. Based upon a minimal clinically important difference (MCID) of reduction off 5 points on the Short-Form 36 mental component score, which is equivalent to a SMD of 0.25 (Hengartner 2022; Ogura 2020).

^c The certainty of the evidence was downgraded due to imprecision, as the confidence interval encompasses both increased and decreased effectiveness, with the possibility of a clinically significant benefit favoring psychological therapy. Based upon a MCID of reduction of 5 points on the Short-Form 36 physical component score, which is equivalent to a SMD of 0.25 (Ogura 2020).

^d Downgraded once due to lack of blinding of participants and those delivering the intervention.

^e The certainty of evidence was downgraded by two levels for imprecision due to reliance on a single, small RCT with wide confidence intervals that included the possibility of both substantial benefit and minimal or no effect, resulting in considerable uncertainty about the true effect size. Based upon a MCID of reduction of 5 points on the Short-Form 36 averaged combined scores, which is equivalent to a SMD of 0.25 (Ogura 2020).

^f Downgraded by two level for indirectness due to the evidence being derived from a single RCT conducted in Germany with a sample that may not be representative of other contexts. The population was predominantly female (69.2%) with a mean age of 42 years and included both clinical and non-clinical individuals, limiting applicability to broader or more diverse populations.

Summary of findings 3. Summary of findings table - Exercise compared to pharmacological treatment (antidepressants) for adults with depression

Exercise compared to pharmacological treatment (antidepressants) for adults with depression

Patient or population: adults with depression

Setting: any setting

Intervention: exercise

Comparison: pharmacological treatment (antidepressants)

Outcomes	Anticipated absolute effects* (95% CI)		Relative effect (95% CI)	Nº of participants (studies)	Certainty of the evidence (GRADE)	Comments
	Risk with pharmacological treatment (antidepressants)	Risk with exercise				
Reduction in depressive symptoms post-treatment	-	SMD 0.11 SD lower (0.33 lower to 0.1 higher)	-	330 (5 RCTs)	⊕⊕⊕⊕ Low ^{a,b}	The evidence suggests that exercise results in little to no difference in depressive symptoms.
Reduction in depressive symptoms follow-up	-	SMD 0.4 SD lower (0.8 lower to 0)	-	58 (1 RCT)	⊕⊕⊕⊕ Very low ^{a,b,c}	Exercise may result in a slight reduction on depressive symptoms but the evidence is very uncertain.
Completed exercise or antidepressants	424 per 1000	390 per 1000 (327 to 467)	RR 0.92 (0.77 to 1.10)	655 (5 RCTs)	⊕⊕⊕⊕ Low ^{d,e}	Exercise may result in little to no difference in adherence.
Quality of Life - mental	The mean quality of Life - mental was 43.2 Short Form (SF) 36 mental health component (score)	MD 11.9 Short Form (SF) 36 mental health component (score) lower (24.04 lower to 0.24 higher)	-	25 (1 RCT)	⊕⊕⊕⊕ Very low ^{a,f,g}	The evidence is very uncertain about the effect of exercise on mental quality of life.
Quality of Life - physical	The mean quality of Life - physical was 8.1 Established Populations for Epidemiologic Studies of the Elderly Short Physical Performance Battery (EPESE) Short Physical Performance Battery	MD 1.3 Established Populations for Epidemiologic Studies of the Elderly Short Physical Performance Battery (EPESE) Short Physical Performance Battery higher (0.67 lower to 3.27 higher)	-	25 (1 RCT)	⊕⊕⊕⊕ Very low ^{a,f,h}	The evidence is very uncertain about the effect of exercise on physical quality of life.

Adverse events	Blumenthal (1999) reported musculoskeletal injuries in 3 of 53 participants in the exercise group; five participants from the medication group experienced medication adverse events. Blumenthal (2007) used a 36-item somatic symptom checklist and found that a few participants reported worsening symptoms. Only one side effect showed a significant group difference ($P = 0.03$): post-treatment diarrhoea and loose stools were more common in the sertraline group (31%) than in the home-based (21%) and supervised exercise (10%) groups. Blumenthal (2012a), using the same checklist, found two significantly different side effects: the sertraline group reported greater post-treatment fatigue (20%) and sexual problems (26%) compared to the exercise group (2.4% for both). Hidalgo (2021) found adverse events in 8.9% of the exercise group (osteomuscular pain, mild contusions, dizziness) and in 22.5% of the antidepressant group (dizziness 6.5%, epigastric pain 5.2%, drowsiness 3.2%).	625 (4 RCTs)	⊕⊕⊕⊕ Low ^{a,i}	Exercise may result in fewer adverse events than pharmacological treatments.
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***The risk in the intervention group** (and its 95% confidence interval) is based on the assumed risk in the comparison group and the **relative effect** of the intervention (and its 95% CI).

CI: confidence interval; **MD:** mean difference; **RR:** risk ratio; **SMD:** standardised mean difference

GRADE Working Group grades of evidence

High certainty: we are very confident that the true effect lies close to that of the estimate of the effect.

Moderate certainty: we are moderately confident in the effect estimate: the true effect is likely to be close to the estimate of the effect, but there is a possibility that it is substantially different.

Low certainty: our confidence in the effect estimate is limited: the true effect may be substantially different from the estimate of the effect.

Very low certainty: we have very little confidence in the effect estimate: the true effect is likely to be substantially different from the estimate of effect.

See interactive version of this table: https://gdt.gradeapro.org/presentations/#/isof/isof_question_revman_web_453701683027835030.

^a Downgraded once because concerns about an unclear of bias in the areas of random sequence generation, allocation concealment, incomplete outcome data, and selective reporting.

^b Downgraded once because the lower bound of the 95% confidence interval for the effect estimate does not reach the point of clinical significance. Based upon a minimal clinically important difference (MCID) of reduction of five points on the Beck Depression Inventory, which is equivalent to a SMD of 0.45 (Button, 2015; Hengartner 2022).

^c Study only included participants aged 50 years and over. This limits the applicability of the findings to younger populations or to those with a wider age distribution.

^d Downgraded once because lack of blinding of participants and those delivering the intervention

^e Downgraded once because substantial statistically significant heterogeneity of $I^2 = 86\%$.

^f Downgraded once because only older adults were included in this RCT.

^g Downgraded once because the lower bound of the 95% confidence interval for the effect estimate does not reach the point of MCID. Based upon a MCID of reduction of 5 points on the Short-Form 36 mental component score, which is equivalent to a SMD of 0.25 (Hengartner 2022; Ogura 2020).

^h Downgraded once because the lower bound of the 95% confidence interval for the effect estimate does not reach the point of MCID. Based upon a MCID of reduction of 5 points on the Short-Form 36 physical component score, which is equivalent to a SMD of 0.25 (Ogura 2020).

ⁱ Downgraded once because variability in the reporting of adverse events across studies

Summary of findings 4. Summary of findings table - Exercise compared to type of control (subgroup analyses) in adults with depression

Exercise compared to type of control (subgroup analyses) in adults with depression

Patient or population: adults with depression

Setting: any setting

Intervention: exercise

Comparison: type of control (subgroup analyses)

Outcomes	Anticipated absolute effects* (95% CI)		Relative effect (95% CI)	Nº of participants (studies)	Certainty of the evidence (GRADE)	Comments
	Risk with type of control (subgroup analyses)	Risk with exercise				
Depressive symptoms (exercise vs control subgroup analysis: type of control - placebo)	-	SMD 0.21 SD lower (0.81 lower to 0.4 higher)	-	172 (3 RCTs)	⊕⊕⊕⊕ Very low ^{a,b,c}	The evidence is very uncertain about the effect of exercise on depressive symptoms.
Depressive symptoms (exercise vs control subgroup analysis: type of control - no treatment, waiting list, usual care, self monitoring)	-	SMD 0.85 SD lower (1.06 lower to 0.64 lower)	-	897 (27 RCTs)	⊕⊕⊕⊕ Low ^{d,e,f}	Exercise may result in a large reduction in depressive symptoms.
Depressive symptoms (exercise vs control subgroup analysis: type of control - exercise plus treatment vs treatment)	-	SMD 0.57 SD lower (0.83 lower to 0.3 lower)	-	812 (19 RCTs)	⊕⊕⊕⊕ Very low ^{c,d,g}	Exercise may result in a reduction in depressive symptoms but the evidence is very uncertain.
Depressive symptoms (exercise vs control subgroup analysis: type of control - stretching, meditation or relaxation)	-	SMD 0.48 SD lower (0.8 lower to 0.15 lower)	-	385 (11 RCTs)	⊕⊕⊕⊕ Very low ^{c,d,h}	Exercise may result in a reduction in depressive symptoms but the evidence is very uncertain.
Depressive symptoms (exercise vs control subgroup analysis: type of control - Social/casual conversation)	-	SMD 0.48 SD lower (1.23 lower to 0.27 higher)	-	29 (1 RCT)	⊕⊕⊕⊕ Very low ^{c,d,i}	The evidence is very uncertain about the effect of exercise on depressive symptoms.

***The risk in the intervention group** (and its 95% confidence interval) is based on the assumed risk in the comparison group and the **relative effect** of the intervention (and its 95% CI).

CI: confidence interval; **SMD:** standardised mean difference

GRADE Working Group grades of evidence

High certainty: we are very confident that the true effect lies close to that of the estimate of the effect.

Moderate certainty: we are moderately confident in the effect estimate: the true effect is likely to be close to the estimate of the effect, but there is a possibility that it is substantially different.

Low certainty: our confidence in the effect estimate is limited: the true effect may be substantially different from the estimate of the effect.

Very low certainty: we have very little confidence in the effect estimate: the true effect is likely to be substantially different from the estimate of effect.

See interactive version of this table: https://gdt.gradepro.org/presentations/#/isof/isof_question_revman_web_453719213729092884.

^a Downgraded once because lack of blinding of participants and those delivering the intervention

^b Downgraded once because substantial borderline statistically significant heterogeneity of $I^2 = 66\%$.

^c Downgraded once because the lower bound of the 95% confidence interval for the effect estimate does not reach the point of clinical significance. Based upon a minimal clinically important difference (MCID) of reduction of 5 points on the Beck Depression Inventory, which is equivalent to a SMD of 0.45 (Button, 2015; Hengartner 2022).

^d Downgraded once because concerns about an unclear or high risk of bias in the areas of random sequence generation, allocation concealment, incomplete outcome data, and selective reporting.

^e Downgraded once because moderate statistically significant heterogeneity of $I^2 = 51\%$.

^f The lower bound of the 95% confidence interval for the effect estimate reaches the point of clinical significance. Based upon a MCID of reduction of 5 points on the Beck Depression Inventory, which is equivalent to a SMD of 0.45 (Button 2015 Hengartner 2022).

^g Downgraded once because substantial statistically significant heterogeneity of $I^2 = 68\%$.

^h Downgraded once because moderate statistically significant heterogeneity of $I^2 = 54\%$.

ⁱ Downgraded once for indirectness due to concerns about population applicability, as the study included only older adults with Alzheimer's disease

BACKGROUND

Description of the condition

Depression is the leading cause of ill health and disability in the world, affecting over 280 million people [1, 2]. Its effects are felt throughout all regions, irrespective of whether they consist of high-income or low-income and middle-income countries [3]. The prevalence of depression has continued to rise, increasing by more than 18% between 2005 and 2015, reflecting both population growth and an ageing population [3]. Although some studies suggest that the incidence of depression is becoming more stable globally [4], specific events such as the COVID-19 pandemic and 'lockdown' dramatically affected its prevalence [5]. Depression has a considerable effect on peoples' health, their functioning, disease outcomes, hospital-seeking behaviour and length of life [4, 6, 7]. Depression was the second main cause of non-fatal ill-health or disability globally in 2021, accounting for approximately 47 million years lived with disability (YLD) (6.1% of all YLD) [8]. In 2021, mortality associated with suicide accounted for 727,000 deaths worldwide [9]. Although it is a global concern, the effects of depression have been felt disproportionately in low- and middle-income countries, where over 80% of the world's non-fatal disease burden and 78% of suicides were reported in 2015 [3].

Depression encompasses a broad and varied range of mental health concerns. It is typified by persistent low mood; sadness; loss of pleasure in most activities and experiences; and is associated with emotional, cognitive, physical and behavioural symptoms [10]. People with depression may experience a loss of energy, a change in appetite, altered sleep patterns, anxiety, reduced concentration, indecisiveness, restlessness, feelings of worthlessness, guilt or hopelessness, and thoughts of self-harm or suicide [3]. It can be a long-lasting or recurring condition, affecting a person's ability to function and cope with daily life. The diagnosis and the severity of depression are determined by the extent of the symptoms incurred and the level of functional impairment. The fifth edition of the Manual of Mental Disorders (DSM-V) classifies major depressive disorders as the occurrence of at least five out of nine symptoms and a decline in a person's daily functioning [11]. The effects and severity of these symptoms have to be felt throughout each day and persist for at least two weeks [10, 11]. Importantly, at least one of the symptoms experienced has to be depressed mood or loss of interest or pleasure [11]. The DSM-V classifies severity of depression into three categories: mild (a few symptoms more than is required for diagnosis but these do not have a significant impact), moderate (number and intensity of symptoms between mild and severe) and severe (number of symptoms are substantially above what is required for diagnosis and subsequent symptoms are unmanageable and distressing, and interfere with social and occupational functioning) [11].

Description of the intervention and how it might work

Depression is treatable. It is treated with antidepressants, psychological therapies, alternative approaches or a combination of these options. Antidepressants are effective for the treatment of depression in primary care, especially for people with more than minor depression [12, 13]. In addition to benefiting the acutely ill, antidepressants can help reduce the risk of relapse into depression [10, 14, 15]. Recommendations for their use are often as part of combined treatment with high-intensity psychological interventions (cognitive behavioural therapy (CBT)

or interpersonal therapy (IPT)) [10, 15]. The effectiveness of antidepressants in managing depression may be tempered by several concerns, including: adverse side effects that impact on people's health and quality of life [16, 17, 18, 19]; a lag time between starting antidepressants and improvements in mood [20]; and poor adherence [21, 22].

Psychological treatments provide another option, particularly if antidepressants are not effective or are poorly tolerated [10, 15]. Their use may be as an adjunct to other interventions, particularly antidepressants, or as a standalone intervention [10, 15]. Increasingly, the use of psychological therapies is being recommended in different countries [10, 15, 23, 24, 25, 26, 27, 28, 29], reflecting their effectiveness irrespective of the severity of depression, the population being treated or the care setting [30, 31]. Despite the apparent benefits of psychological treatments, their effectiveness depends on people's expectations, perceived stigma, sustained motivation, degree of psychological mindedness and the influence of possible side effects (e.g. emotional discomfort, increased stress, resurfacing of memories, or physical problems (e.g. sleep problems, tiredness, headaches)) [32, 33, 34].

Other approaches have also been used as alternatives to, or to supplement, pharmacological and psychological treatments. These include music therapy, light therapy, acupuncture, family therapy, marital therapy, exercise, relaxation therapy and other complementary techniques [35, 36, 37, 38, 39]. The increasing use of alternative interventions for treating depression may reflect dissatisfaction with conventional treatments, people's beliefs and philosophies influencing their treatment choice, and the possibility that alternative treatments may be effective [35, 40].

Exercise has received growing attention as an approach to managing depression, whether as part of stepped-care programmes, combination therapy or monotherapy [15, 32, 41, 42, 43, 44, 45]. It is defined as the "planned, structured and repetitive bodily movement done to improve or maintain one or more components of physical fitness" [46]. The effect of exercise on depression has been the subject of research for several decades, with evidence continuing to emerge that assesses its effectiveness for different severities of depression (e.g. mild, moderate or severe), population groups (e.g. age, sex), management approaches (e.g. preventative, first-line or second-line) and against different comparators (e.g. usual care, waiting list, antidepressants, psychological treatments) [47, 48, 49, 50, 51, 52, 53, 54, 55, 56, 57].

The increasing consideration given to exercise as an approach to managing depression has been reflected in recent clinical guidelines that have emerged from national and professional policy bodies, although the definition of exercise and the extent of recommendations differ. In England in 2022, the National Institute for Health and Care Excellence (NICE) published a guideline on the treatment and management of depression in adults [15]. It advises that people doing any form of physical activity on a regular basis (e.g. walking, jogging, swimming, dance, gardening) could enhance their sense of wellbeing. Moreover, undertaking the activity outdoors can provide greater benefits. As a treatment option for both severe and mild-to-moderate depression, group physical activity interventions (usually for a group of eight participants) that are specifically designed for people with depression are recommended. These interventions should be provided by trained

practitioners, usually consist of one session per week for 10 weeks, include moderate-intensity aerobic exercise and not directly target thoughts and feelings [15]. For people with more severe depression, consideration should be given to other treatment options and more contact with therapists should be considered carefully before recommending group exercise [15]. The Royal Australian and New Zealand College of Psychiatrists (RANZCP) advises that exercise should be recommended to all patients due to the known health benefits [44]. Although RANZCP recommend aerobic exercise, they recognise that a combination of aerobic and resistance exercise is probably optimal. RANZCP note some uncertainty around the optimal amount and intensity, indicating that it should be set at a level that demands sustained concentration and effort [58]. The Canadian Network for Mood and Anxiety Treatments (CANMAT) recommends that supervised exercise of low to moderate intensity for 30 to 40 minutes per session, three to four times a week for at least nine weeks is effective as first-line monotherapy for mild depression and as second-line adjunctive treatment for moderate severity illness [29, 59, 60]. Portuguese and Brazilian clinical guidance emphasises the benefits of exercise delivered by trained professionals as an option for treating depression, though it notes difficulties in implementation [61]. In contrast, guidance in the USA is equivocal. Although the American Psychiatric Association makes no specific recommendation concerning the role of exercise for treating depression, other agencies identify benefits associated with exercise in treating depression [24]. The American Psychological Association recommends exercise as a monotherapy for adults with depression for whom psychotherapy or pharmacotherapy is ineffective or unacceptable [24].

The focus of this systematic review is on assessing the effectiveness of exercise for treating depression. The review does not consider the influence of any referral schemes or broader programmes of care. The review will use the American College of Sports Medicine (ACSM) criteria to define the concept of exercise. Given that there are many forms of bodily movement or physical activity that differ in their characteristics, it is important to provide some clarity regarding those that will be included in this review.

Depression and low levels of physical activity have been shown to be closely associated in a two-way relationship [62, 63]. While depression can lead to decreased levels of exercise through low motivation and energy, it is also evident that low levels of physical activity are a risk factor for people developing low mood and depression [62, 64, 65]. Although such an association does not necessarily imply causality, there are plausible reasons why physical activity and exercise may improve mood [32, 66]. Exercise may act as a diversion from negative thoughts, and the mastery of a new skill may be important [67]. Social contact may be part of the mechanism of action [32]. Self-efficacy may provide the mechanism by which exercise might have an antidepressant effect; people who experience an improvement in mood following exercise show higher self-efficacy levels at three weeks and nine weeks post-exercise [68]. Self-efficacy has been found to be intricately linked with self-esteem, which, in turn, is considered to be one of the strongest predictors of overall subjective well-being [69]. Low self-esteem is also considered to be closely related to mental illness [70]. It has been shown that exercise has a direct positive effect on levels of self-esteem, which occurred prior to improvements in depressive symptoms [71, 72, 73]. Physical activity may have physiological effects, such as changes in endorphin and monoamine levels [74], or reduction in the levels of

the stress hormone cortisol [75], all of which may improve mood. Exercise stimulates the growth of new nerve cells and the release of proteins known to improve the health and survival of nerve cells, for example, brain-derived growth neurotrophic factor [76, 77, 78, 79]. Low brain-derived neurotrophic factor levels can be seen in untreated patients suffering major depression compared to healthy controls [80, 81]. This positive effect of exercise on low brain-derived neurotrophic factor levels has been seen specifically in patients suffering from depression [82, 83].

Why it is important to do this review

Despite several new systematic reviews and meta-analyses on this topic [53, 52, 84, 51, 85, 49, 50, 54, 55, 86, 87, 88, 89, 90, 91, 92, 93, 94, 95, 96, 97], doubts remain about whether exercise is beneficial in treating depression when compared to other treatment options (and if it is, the extent of that benefit and what type of exercise benefits what type of person [43, 51, 84]). This uncertainty is reflected in differences in the pooled estimates of the comparative effects of exercise for treating depression, which range from statistically significant benefit (standardised mean difference (SMD) -1.13, 95% confidence interval (CI) -1.48 to -0.78) to no difference (SMD -0.14, 95% CI -0.16 to 0.12) [57].

Underlying the uncertainties and ongoing debate are differences in the scope of and methods used by systematic reviews and meta-analyses: elements that vary include type (e.g. aerobic only [50, 53, 93], aerobic or strength [49], inclusion of yoga, stretching, tai chi and qigong [50] or all types of exercise [51, 52, 54, 55, 84, 87]), treatment strategy used (e.g. alone [84], as an adjunct/combination therapy [84] or both/undefined [49, 50, 53, 54]) and definition (e.g. physical exercise [50, 51], physical activity [54]). In contrast, in this review, we have not defined the frequency, duration and intensity of treatment or limited inclusion by these factors.

Comparator interventions have also differed in different systematic reviews. They have included active pharmacological interventions (e.g. antidepressants) [50, 53, 84] and non-pharmacological interventions (e.g. psychotherapy, psychoeducation, physical activity/exercise) [50, 52, 53, 84], non-active interventions (e.g. placebo) [51, 55, 84], alternative therapies (e.g. meditation, relaxation, bright light therapy [50, 98, 99, 100, 101]), usual care [51, 52, 53, 55, 84], no treatment/waiting list controls [51, 55], or no comparator has been stated [49].

Despite these systematic reviews all having a focus on adults with major depressive disorders, the criteria used for diagnosing depression, and for establishing its severity, have varied. Importantly, these criteria were often the primary outcome measures for assessing the effectiveness of the interventions. Diagnosis has occurred through clinical interviews [53], occurrence of unspecified depression or depressive symptoms [55] or the use of different combinations of clinical interviews, diagnostic systems (DSM-IV, ICD-10, Research Diagnostic Criteria), self-reported depression scales using different thresholds (validated or unvalidated) (e.g. Beck Depression Inventory, Hamilton Rating Scale for Depression, exercise self-efficacy, depression coping self-efficacy) [49, 50, 51, 54, 55], with some variation in interpretation by researchers and/or clinicians. Depressive conditions also varied in the different reviews, whether by specifying levels of severity required for inclusion [49], excluding multiple diagnoses [55], focusing on unipolar depression only [52], or including unipolar, bipolar and dysthymia [51, 54]. In some instances, participants

were included who had depression diagnosed as a comorbidity, with another condition as the main diagnosis [100].

Evidence synthesised in the previous systematic reviews on this topic originated from studies that differ in their design and methodological rigour, raising concerns about risk of bias [55]. Many focused solely on randomised controlled trials (RCTs) [50, 51, 52, 53, 54, 84, 91, 98, 99, 100, 101, 102, 103], whereas others included a range of different randomised and non-randomised controlled studies and uncontrolled studies [49, 104, 105, 106, 107]. Inadequacies in allocation concealment, blinding, sample size and intention-to-treat analysis have been identified within studies included in the systematic reviews [55, 99, 100, 101, 102], raising the likelihood that results may have been affected [55]. Several meta-analyses have shown the influence of methodological rigour on estimates of pooled effect size, with statistically significant benefits from treatment with exercise for depression disappearing when focusing on subgroups of more rigorous studies [91, 99, 101].

Given the heterogeneous nature of the evidence base assessing exercise as a treatment for depression, it is important to be transparent and consistent about the different decisions made regarding the methods that have been adopted. Also, it is important to recognise that differences will exist between different systematic reviews and meta-analyses, interpreting the results within the context of the evidence presented. This review has continued to evolve, having originally been published in 2001 in the British Medical Journal [102], converted to a Cochrane review in 2008 [99] and updated in 2012 [32] and 2013 [108]. Although the current update broadly adheres to the protocol agreed with the Cochrane Depression, Anxiety and Neurosis Review Group (CCDAN) as outlined in previous versions, it has addressed issues raised since the previous version which necessitate consideration [43, 51, 55]. Our intention in this review is to update the evidence base and to address any issues that have arisen in the assessment of the effectiveness of exercise for treating depression. Changes made to the scope of the review and the methods used are described below under 'Differences between protocol and review'.

OBJECTIVES

To determine the effectiveness of exercise in the treatment of depression in adults compared with no intervention, waiting list control or placebo, or where exercise is used as an adjunct to an established treatment that is received by both exercising and non-exercising groups.

To determine the effectiveness of exercise compared with other active interventions for depression in adults (psychological therapies, pharmacological treatments or alternative interventions such as light therapy).

METHODS

We followed the Methodological Expectations of Cochrane Intervention Reviews (MECIR) [109] when conducting the review and reported the review using PRISMA 2020 [110].

Differences between protocol and review

Changes for this update (2025 version)

- We performed an additional sensitivity analysis to explore the effect of excluding those trials for which we used the arm

with the largest clinical effect rather than the largest 'dose' of exercise.

- We specified a new analysis that assessed 'other' interventions, which included bright light therapy, reflexology, occupational or art therapy, supplements and health education.
- Some studies with a control arm that had been specified as placebo in previous versions of the review have been reclassified as specific active therapies in this 2025 update.
- We modified our categorisation of the intensity of exercise in the studies to align with ACSM criteria.
- We undertook new subgroup analyses to look at the effects of exercise when given as an adjunct.

Criteria for considering studies for this review

Types of studies

We included randomised controlled trials (RCTs), including parallel, cluster, or individual, or the first phase of cross-over trials. We defined a trial as an RCT if the allocation of participants to intervention and comparison groups was described as randomised (including terms such as 'randomly', 'random' or 'randomisation').

Types of participants

Participant characteristics

Adult men and women aged 18 and over (with no upper age limit) in any setting, including inpatients. We included studies if the participants were defined by the author of the trial as having depression (by any method of diagnosis and of any severity). We excluded trials that randomised people both with and without depression, even if results from the subgroups of participants with depression were reported separately, in keeping with what was done in previous versions of this review [32, 99, 108].

Diagnosis

The effects of exercise on depressive symptoms in participants with emotional distress (but not fulfilling a diagnosis of depression) or those who are healthy were not included in this review. We acknowledge that it can sometimes be difficult to distinguish between depression and dysthymia in the 'real world', as there need be only two weeks of decreased interest and enjoyment to diagnose depression. However, we were primarily interested in the role of exercise in people with depression for whom there is substantial morbidity rather than in people with mild, transient episodes of low mood.

We did not include studies that investigated the effect of exercise on anxiety and neurotic disorders, dysthymia (i.e. low mood not fulfilling diagnostic criteria for depression) or postnatal depression.

Comorbidities

We did not exclude studies or participants based on comorbidities.

Setting

Any setting was eligible for inclusion in the review.

Types of interventions

We defined exercise as "planned, structured and repetitive bodily movement done to improve or maintain one or more components of physical fitness" [46]. This definition has been applied since

the previous update [32] and it was agreed to continue its use for future updates by the CCDAN Review Group editorial team. We also excluded studies that assessed the provision of advice on how to increase physical activity that does not fulfil the ACSM definition of exercise.

Experimental intervention

Any type of exercise (as defined above). We excluded studies that measured outcomes immediately before and after a single exercise session, and trials that provided less than a week of exercise.

Comparator intervention

We included studies in which exercise was compared to no intervention, a 'waiting list control' or placebo (defined by study authors); and those in which exercise was used as an adjunct to an established treatment (which was received (in an identical way) by participants in both the exercising and non-exercising group, e.g. exercise plus cognitive behavioural therapy (CBT) versus CBT alone).

We also included other types of active treatments, where the aim of the treatment was to improve mood. This included pharmacological treatments, psychological therapies or other alternative treatments. Where a study identified a comparator as placebo and stated it was an active treatment, whether pharmacological, psychological or other therapies, we classified it as the specified active therapy.

Although this strategy was the same as that included in the original review [102], some studies that were specified as placebo in previous versions have been reclassified as specific active therapies in this 2025 update. We made this change only when the interventions were clearly stated and the comparator was included in an analysis.

We excluded studies that compared two different types of exercise but had no non-exercising comparison group.

We excluded studies described by the authors as 'combination treatments', where exercise was one component of the combination, because we could not disentangle the effect of exercise from the effect of the other components of the intervention.

Outcome measures

Critical outcomes

- Our primary outcome was a measure of depression or mood, either continuous or dichotomous. Continuous measures of depression were reported using a variety of depression scales, the most common of which were the Beck Depression Inventory [111] and the Hamilton Rating Scale for Depression [112].

In previous versions of the review, where trials used a number of different tools to assess depression, we included only the main outcome measure in the meta-analysis. We defined the main outcome measure using a hierarchy of criteria: i) identified by the trial authors as the main outcome measure, ii) outcome reported in the abstract, iii) first outcome reported in the Results section.

Where trials used dichotomous data for primary outcomes, and also provided data on continuous outcome measures, we used

the data provided in the trial reports for the continuous outcome measure in our meta-analysis. This was because we knew from previous updates that most trials reported only continuous outcomes.

Timing of outcome assessment

We extracted data at the end of treatment and also at the end of any longer-term follow-up after the intervention had stopped.

Important outcomes

- Acceptability of treatment, assessed by a) attendance at exercise interventions, and b) the number of participants completing the interventions
- Quality of life (e.g. SF-36 Health Survey [113], health-related quality of life)
- Cost (e.g. average treatment cost, health service costs)
- Adverse events (e.g. musculoskeletal pain, fatigue)
- In order to better understand the generalisability of exercise for depression, we also extracted data on the number of people screened for inclusion and the number recruited (Table 1).

Search methods for identification of studies

Electronic searches

We carried out the following electronic searches (Supplementary material 1).

- The Cochrane Depression, Anxiety and Neurosis Review Group's Specialised Register (CCDANCTR) (February 2010 to February 2016)
- The Cochrane Central Register of Controlled Trials (CENTRAL), in the Cochrane Library (February 2010 to November 2023)
- MEDLINE Ovid (February 2010 to November 2023)
- EMBASE Ovid (February 2010 to November 2023)
- PsycINFO (February 2010 to November 2023)

The searches of MEDLINE, Embase, PsycINFO and CENTRAL between 2010 and 2016 were encompassed within our searches of CCDANCTR, which included these databases. Also, we searched the ISRCTN Registry (in March 2013, January 2017, February 2019, November 2020, July 2021 and September 2025) to identify any ongoing trials. We performed an electronic search of ClinicalTrials.gov and WHO International Clinical Trials Registry Platform (ICTRP) in March 2013, January 2017, February 2019, November 2020, July 2021 and September 2025. An RCT filter developed by CCDAN was used.

Because the searches for the CDCANCTR register were kept up-to-date and comprehensive, we were advised by the CCDAN editorial team that it was not necessary to search the other databases until the November 2019 update search.

We searched for errata and retractions for the included studies by searching for the papers on MEDLINE and Embase, and checking those records, updates on the publisher's website and the Retraction Watch database.

Previous updates of the review provide details of the search strategy prior to the searches conducted for the current update [32, 99, 108, 114].

Searching other resources

Due to limitations in available resources for this current update, we did not repeat handsearching conducted in previous updates. We contacted authors only to obtain any missing information about trial details (e.g. standard deviations).

We screened the bibliographies of all included articles for additional references.

Data collection and analysis

Selection of studies

We selected studies through a two-stage process. The full literature search results were independently screened by pairs of review authors (JH, DM, CL, AC, CH, CS, KD), identifying all those citations that may meet the selection criteria. Full manuscripts of all selected citations were then retrieved and assessed independently by the pairs of review authors against the study selection criteria. Any differences in opinion were resolved through consensus or, if necessary, by arbitration by a third review author who was independent of the original decision. For the searches of the CCDANCTR up to February 2020, their Trials Search Co-ordinator/Information Specialist checked abstracts, excluded obviously irrelevant ones, and then sent a list of the remaining citations to the review authors for scrutiny. We created a PRISMA flow diagram to outline the study selection process.

Data extraction and management

We extracted data, when available, at the end of treatment and at the end of follow-up. For this update, seven review authors (JH, DM, AC, CH, CS, CL, KD) independently extracted data for our primary and secondary outcomes for the new trials identified. All authors, working in pairs, verified extracted data from a subset of included studies. If differences in opinion occurred, these were resolved by discussion, or, if necessary, through recourse to a third review author. The data we extracted were participants, interventions, outcome measures, results, number of people screened, number randomised, number allocated to exercise, number who dropped out of the exercise arm (Table 1), secondary clinical outcomes, cost, adverse events and main conclusions. All the review authors used the same structured paper extraction form, which had been piloted.

In addition, we extracted data on the type of exercise from all the included trials, to enable a categorisation of the intensity of exercise according to ACSM criteria. This was undertaken by one author (JH) and checked by a second author (DM or AC). In categorising the intensity levels in this updated version of the review, we made a readjustment to maintain the five-intensity-level categorisation stated in the ACSM criteria [115], which are very light, light, moderate, vigorous and near maximum to maximum. Wherever possible, we used direct measures to estimate the intensity levels (percentage of heart rate reserve, percentage of oxygen uptake reserve, percentage of maximum heart rate and one-maximum-repetition percentage) [116]. Where this was not available, an indirect measure was utilised. If available, the classification from the author was used; if not, a classification of the type of exercise metabolic equivalent (METs) was derived from the Compendium of Physical Activities (<https://sites.google.com/site/compendiumofphysicalactivities/Activity>). All measurements of intensity were then cross-referenced to approximate the classification of exercise intensity ACSM criteria [115]. For the

MET values, when given, age was taken into consideration using the mean age (if reported). Wherever possible, we calculated 'triangulation of intensity levels'. If multiple intensity levels were used throughout the session/intervention, the programme intensity that represented most of the session/intervention programme was classified as the intervention intensity level. Where multiple direct and indirect conflicting measurements were given for relative intensity levels, we used the direct measure to classify the exercise intensity compared to indirect measures. Where a range of direct or indirect measures were given that crossed two levels of intensity, the intensity level was set where most of the overlap occurred. Where two intensity levels were described within the intervention, we selected the intensity level that represented most of the session or programme as the overall intensity level.

Main comparisons

We undertook the following analyses.

- Exercise versus 'control' (as defined above)
- Exercise versus psychological therapies
- Exercise versus pharmacological treatments
- Exercise versus 'other' interventions (including bright light therapy, reflexology, occupational or art therapy, supplements, health education)

Risk of bias assessment in included studies

We used the Cochrane Collaboration risk of bias tool to assess risks of bias, according to Chapter 8 of the *Cochrane Handbook for Systematic Reviews of Interventions* [117]. Two review authors (from JH, AC, DM, CH, CS, CL, KD) independently extracted data from each study on random sequence generation, allocation concealment, blinding of participants, blinding of those delivering the intervention, blinding of outcome assessors, incomplete outcome data, selective reporting and other potential biases. We categorised each of these domains as being at high risk of bias, unclear risk of bias or low risk of bias. All authors met in pairs to check decisions. We resolved any differences through discussion—or recourse to a third review author when necessary.

For concealment of allocation, we distinguished between trials that were adequately concealed (central randomisation at a site remote from the study; computerised allocation in which records are in a locked, unreadable file that can be assessed only after entering participant details; the drawing of opaque envelopes), inadequately concealed (open lists or tables of random numbers; open computer systems; drawing of non-opaque envelopes) and unclear (no information in report, and the authors either did not respond to requests for information or were unable to provide information).

Trials could only be defined as 'intention-to-treat' if participants were analysed according to the allocated treatment and all participants either completed allocated treatments or any missing outcome data were replaced using a recognised statistical method (e.g. last observation carried forward (LOCF)).

For blinding, we distinguished between trials in which the main outcome was measured by an assessor who was blind to treatment allocation (blind) and those in which the main outcome was measured either by the participants themselves (i.e. self report) or by a non-blinded assessor (not blind).

Measures of treatment effect

We undertook a narrative review of all studies and a meta-analysis of those studies with appropriate data. Where trials used a number of different tools to assess depression, we included the main outcome measure only in the meta-analysis. The main outcome measure was defined using a hierarchy of criteria as follows: identified by the authors as the main outcome measure; outcome reported in the abstract; first outcome reported in the Results section. For continuous data where different scales were used, the standardised mean difference (SMD) was calculated and reported with a 95% confidence interval (CI). For dichotomous data, we calculated the risk ratio and reported this with a 95% CI.

We interpreted the SMDs using the following rule of thumb: 0.2 represents a small effect, 0.5 a moderate effect and 0.8 a large effect [118]. We planned to pool follow-up data from those trials that reassessed participants at the end of the intervention and at subsequent time points (long-term follow-up). Long-term follow-up could mean an assessment at any point of time (after the initial assessment) after the intervention had been completed.

Unit of analysis issues

Studies with multiple treatment groups

Where studies with multiple treatment groups included a control arm, an exercise arm and an 'established treatment' arm (e.g. CBT, antidepressants), we extracted data on exercise versus control, and exercise versus established treatment. This meant that data from the exercise arm were included in two separate comparisons. If trials compared an established treatment (e.g. CBT, antidepressants) versus exercise versus both the established treatment and exercise, we made two comparisons: (i) established treatment plus exercise versus established treatment alone, and included this in the meta-analysis of treatment versus control; (ii) exercise versus established treatment (e.g. CBT, antidepressants). This means that data from the 'established treatment alone' arm was used in two separate comparisons. In the review versions in 2001 [102], 2009 [99] and 2012 [108], for trials that included more than one intensity of exercise, we used the exercise arm with the greatest clinical effect. Similarly, when trials provided more than one type of exercise, we used the type of exercise with the greatest clinical effect. However, because this may overestimate the effect of exercise, in this and the previous update of the review [32], the exercise arm that provides the biggest 'dose' of exercise was used, and we performed a sensitivity analysis to explore the effect of using the smallest dose.

Cross-over trials

For cross-over trials, we intended to use the first phase of the trial only due to the potential 'carry-over' effect of exercise. To date, we have not included any cross-over trials.

Cluster-randomised trials

If we identified cluster-randomised trials that were incorrectly analysed using individuals as the unit of analysis, we intended to make corrections using the intracluster correlation coefficient (ICC). If this had not been available, we would have imputed the ICC from similar studies. In fact, we did not find any cluster-RCTs to include.

Dealing with missing data

In two previous versions of this review [99, 102], we found current contact details of all authors through correspondence addresses in study reports and by searching websites. We contacted all authors by email or post (sending reminders to non-responders) to establish missing details from the methods and results sections of the written reports and to determine authors' knowledge of, or involvement in, any current work in the area. For the last two updates [32, 108] and this 2025 update, we contacted authors only if there were missing data items, or if we needed more detail to determine whether to include the study.

Some trials, in which participants dropped out, reported data from only the remaining participants, so we used these data in our meta-analyses. For trials that attempted to impute data from missing participants (e.g. LOCF for continuous data), we used the imputed values and categorised the trial as 'intention-to-treat'. When we could not obtain information either from the publication or from the authors, we classified the trial as 'not intention-to-treat', and used the data from the available cases in the meta-analysis.

Reporting bias assessment

We used a funnel plot to explore reporting biases when 10 or more studies were included in the meta-analysis. However, other reasons such as heterogeneity and small-study effects also cause asymmetry in funnel plots [117].

Synthesis methods

We used a random-effects model based on DerSimonian and Laird's method to calculate the pooled effect size [119], using a standardised mean difference where studies measured the same outcome but used different scales. Where studies used the same scales for the outcome or there was only one included study, we presented a mean difference. We synthesised data from trials where outcome data were collected as soon as the intervention ended, and performed a separate synthesis of data collected weeks or months after the intervention ended, to explore whether any benefits were retained after the intervention had been completed. With complex interventions, we had to decide whether the interventions were sufficiently similar to be combined into a meta-analysis. We included trials that fulfilled the ACSM definition of exercise [46], and we combined these data in a meta-analysis.

We used the χ^2 test, together with the I^2 statistic, to assess heterogeneity.

A P value of 0.1 or less indicates significant heterogeneity when considering χ^2 . The ranges for I^2 are:

- 0% to 40%: might not be important;
- 30% to 60%: may represent moderate heterogeneity;
- 50% to 90%: may represent substantial heterogeneity;
- 75% to 100%: considerable heterogeneity.

Note that the importance of the observed value of I^2 depends on (i) the magnitude and direction of effects and (ii) the strength of evidence for heterogeneity (e.g. P value from the χ^2 test, or a confidence interval for I^2) [117].

Investigation of heterogeneity and subgroup analysis

We conducted subgroup analyses on studies comparing exercise versus control, with all subgroup analyses prespecified in the protocol used for the current and previous versions of the review.

- Type of exercise
 - In this update, like the previous update, we explored the effect of different types of exercise (aerobic, resistance exercise or mixed aerobic and resistance) for those trials comparing exercise versus control on the primary outcome of reduction in depression symptoms.
- Intensity of exercise
 - In this update, like the previous update, we explored the effects of the intensity of exercise on the primary outcome of reduction in depression symptoms through a subgroup analysis that subdivided intensity of exercise into very light, light, moderate, vigorous, near maximum to maximum.
- Number of exercise sessions
 - In this update, like the previous update, we explored the effect of the number of exercise sessions on reduction in depression symptoms, by extracting data on the length of the exercise programme and the frequency of exercise sessions. We categorised studies by the total number of sessions and then grouped the total number as 0 to 12, 13 to 24, 25 to 36, at least 37.
- Diagnosis of depression
 - In this update, like the previous update, we explored how the diagnosis of depression at baseline (using a cut-point on a scale, or by psychiatric interview) influenced the effect of exercise on reduction in depressive symptoms at the end of treatment.
- Type of control
 - Given concerns raised about the potential additive effects of the use of exercise as an adjunct to an established treatment received in an identical way by participants in both exercising and non-exercising groups (e.g. exercise plus CBT versus CBT alone), we undertook new subgroup analyses to look at these effects separately on the reduction in depressive symptoms. This was part of a subgroup analysis that encompassed the different types of control.

Equity-related assessment

Equity issues were not formally assessed within the current or previous versions of the review.

Sensitivity analysis

We undertook sensitivity analyses to explore how much of the variation between studies comparing exercise to no exercise is explained by between-study differences of the following types.

- Publication type (peer-reviewed journal, conference abstract/proceedings, doctoral dissertation)
- Allocation concealment
- Intention-to-treat analysis (as defined above)
- Blinding
- Treatment dose

We then performed a sensitivity analysis, as we had done previously, including trials that were at low risk of bias for all three key criteria: allocation concealment, intention-to-treat, blinding.

We also performed a sensitivity analysis using those trials that had several arms, for which we had included the arm with the biggest 'dose' of exercise in the initial analysis. In this sensitivity analysis, we included the arm with the smallest dose.

Certainty of the evidence assessment

Summary of findings tables

We created summary of findings tables for the comparisons of exercise versus (1) all non-active control types; (2) psychological therapy; (3) pharmacological treatment and (4) specific non-active control types (subgroup analyses). We generated the tables in GRADEpro GDT software [120]. The outcomes presented were depressive symptoms (post-treatment and follow-up), treatment completion, quality of life and adverse events. We evaluated the certainty of the estimates in the summary of findings tables using the GRADE approach [121, 122]. We categorised the certainty of the evidence as high, moderate, low or very low, based on overall risk of bias, imprecision, inconsistency, indirectness and potential publication bias. Two review authors (JH, CH) independently assessed the certainty of the outcomes presented. In cases of disagreement, consensus was sought through discussion; if the disagreement was not resolved, a third review author (AC) provided arbitration.

Consumer involvement

No patients or members of the public were involved in the development and conduct of the original review or its updates.

RESULTS

Description of studies

Results of the search

The results of searches for the previous updates have been described in detail elsewhere [32, 99, 102, 108].

In the previous update [32], the review authors undertook the full literature search in March 2013. They repeated the searches again on 2 May 2013, listing any potentially eligible studies as 'studies awaiting classification', which were to be considered in a subsequent update. Of the seven studies identified by Cooney and colleagues [32] as 'studies awaiting classification' (Aghakhani 2011 [123, 124, 125, 126, 127, 128]; DEMO II 2012 [129, 130, 131]; Gotta 2012 [132]; Martiny 2013 [133, 134, 135, 136]; Murphy 2012 [137, 138, 139, 140, 141]; Pinniger 2012 [142]; Sturm 2012 [143, 144]), none met the inclusion criteria for this update. We excluded five of these studies because participants did not have to have depression to enter the trial (Aghakhani 2011; Gotta 2012; Murphy 2012; Pinniger 2012; Sturm 2012); we excluded one study because participants had multiple pathologies (Martiny 2013); and we excluded the final study because the control group received flexibility exercises (DEMO II 2012).

Cooney and colleagues also identified 23 trials that were ongoing [32]. Since then, 14 of these trials have been registered as completed. We excluded four of them from this update due to the participants not having depression on entering the trial (Kerse 2010 [145, 146] (ACTRN12605000475640); Rezaei 2015 [147] (IRCT201205159763); Pibernik-Okanovic 2015 [148] (ISRCTN05673017); Taylor-Young 2011 [149] (NCT00643695)). Another seven trials, although showing as having been completed

on trial registers, do not appear to have published their findings (NCT00931814 [150]; NCT01383811 [151]; NCT01464463 [152]; UMIN000001488 [153]; NCT01763983 [154]; ACTRN12612000094875 [155, 156]; CTR/2012/09/002985 [157]). Although we contacted all primary trial investigators on these trials to request additional information (where contact details were available), we received no responses. Due to this, we classified them as ongoing, and they will be checked at the next update for linked publications. We judged the remaining three completed trials to have met the selection criteria and included them in this update (Pfaff 2014 [158] (ACTRN12609000150246); Barclay 2014 [159] (NCT01787201); Kruisdijk 2018 [160, 161, 162] (NTR1894)). Of the remaining nine trials that were identified as ongoing by Cooney and colleagues [32], we excluded four trials as they did not meet the selection criteria: intervention groups in three trials used either yoga (Schneider 2016 [163] (NCT01024790)), self-help (Ström 2013 [164] (NCT01573130)) or a multimodal intervention with meditation as a warm-up routine (Sadeghi 2016 [165] (IRCT2012061910003N1)) and the control group in a fourth trial used exercise (Clark 2014 [166] (NCT01573728)). A further two trials have been terminated (NCT01401569 [167]; NCT01805479 [168]), and we classified two trials as still ongoing (NCT01619930 [169, 170, 171]; NCT01696201 [172]). We identified that another trial had been included in a previous update of the review (Krogh 2009 [173, 174, 175] (NCT00103415)).

In February 2016, February 2019, November 2020 and November 2023, we replicated the original search of the Cochrane Depression, Anxiety and Neurosis Group Clinical Trials Register (CCDANCTR). We identified 31,482 citations, with an additional citation identified

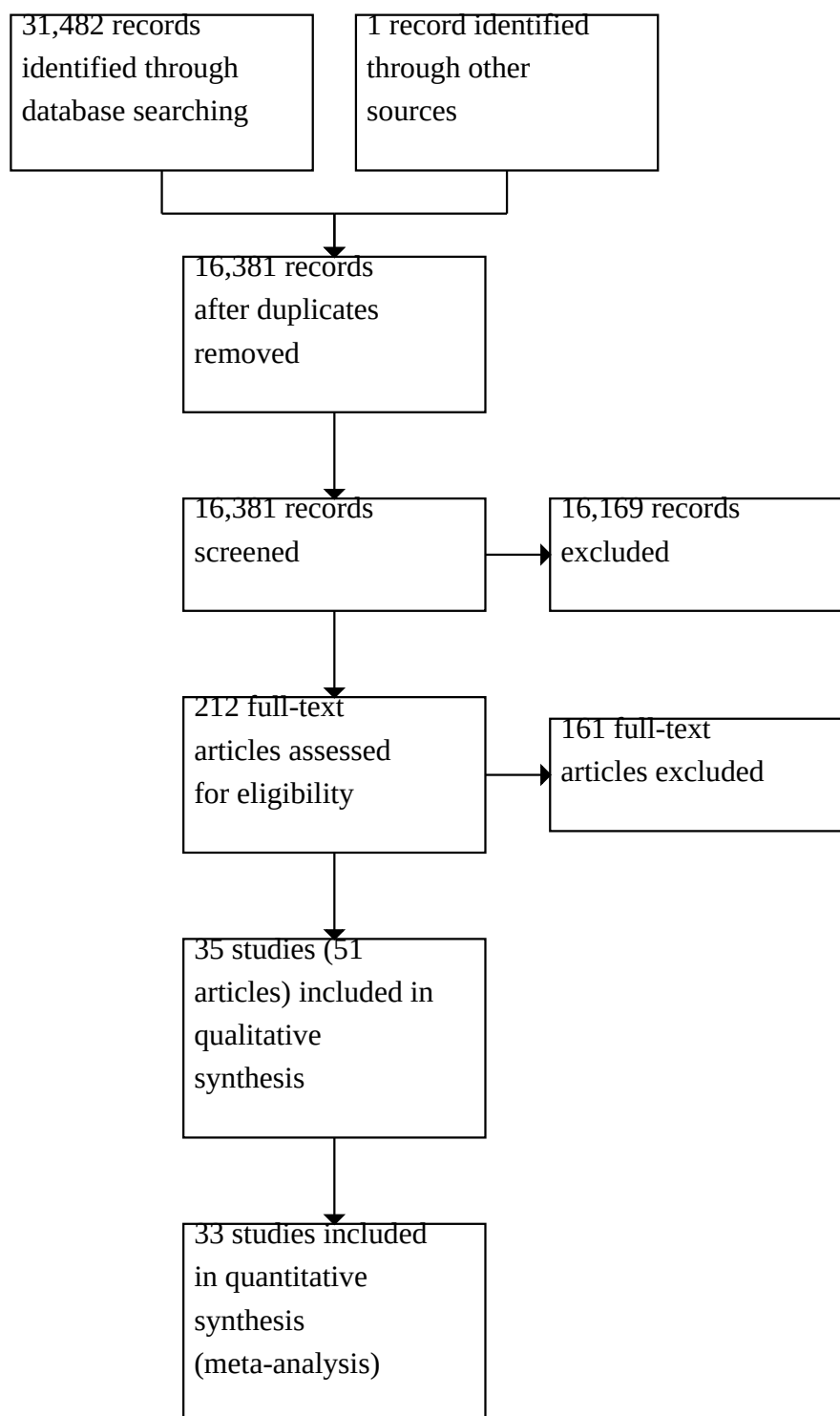
through other sources. Some 16,381 citations remained after deduplication of 15,102 citations. Of these, we excluded 16,169 citations and retrieved 212 full papers. We assessed these against the selection criteria and excluded a further 177 full-text articles.

We included 35 studies (Abdelbasset 2019 [176]; Abdollahi 2017 [177]; Bademli 2023 [178]; Barclay 2014; Belvederi Murri 2015 [179, 180, 181, 182, 183, 184]; Bose 2015 [185]; Brush 2022 [186]; Buschert 2019 [187]; Cheung 2018 [188]; De Groot 2019 [189, 190, 191]; Doose 2015 [192]; Eman 2017 [193]; Haller 2018 [194]; Heinzl 2022 [195]; Heissel 2015 [196]; Hidalgo 2021 [197]; Ho 2014 [198, 199]; Huang 2015 [200]; Karg 2020 [201, 202, 203, 204, 205]; Khorvash 2012 [206]; Khoshnab 2017 [207]; Kruisdijk 2018; La Rocque 2021 [208]; Legrand 2014 [209]; Legrand 2016 [210]; Olson 2017 [211]; Pfaff 2014; Prakhinkit 2014 [212]; Roy 2018 [213]; Salehi 2016 [214]; Schuch 2015 [215, 216, 217]; Siqueira 2016 [218]; Taheri 2018 [219]; Wang 2022 [220]; Yu 2023 [221]). We identified 16 additional papers that were linked to seven included studies and three existing linked papers, specifically Belvederi Murri 2015 [180, 181, 182, 183, 184], Blumenthal 2007 [222, 223, 224, 225] [223, 224, 225], De Groot 2019 [190, 191], Ho 2014 [199], Karg 2020 [202, 203, 204, 205], Kruisdijk 2018 [161, 162], and Schuch 2015 [216, 217].

Through the database searches, plus additional searching of Clinicaltrials.gov, the International Clinical Trials Registry Platform (ICTRP) and the ISRCTN registry in July 2021 and September 2025, we identified an additional 66 ongoing studies (see [Supplementary material 4](#)).

See the PRISMA flow diagram for details of the study selection process for the current update ([Figure 1](#)).

Figure 1. Study flow diagram showing the results of the searches for the current update



Included studies

With the inclusion of the 35 new trials in this update, there are now 73 trials included in the review (Abdelbasset 2019; Abdollahi 2017; Bademli 2023; Barclay 2014; Belvederi Murri 2015; Blumenthal 1999 [226, 227, 228, 229]; Blumenthal 2007; Blumenthal 2012a [230, 231]; Bonnet 2005 [232]; Bose 2015; Brenes 2007 [233]; Brush 2022; Buschert 2019; Cheung 2018; Chu 2008 [234, 235]; De Groot 2019; Doose 2015; Doyne 1987 [236, 237]; Dunn 2005 [238, 239, 240, 241]; Eman 2017; Epstein 1986 [242]; Fetsch 1979 [243]; Foley 2008 [244]; Fremont 1987 [245, 246]; Gary 2010 [247]; Greist 1979 [248, 249, 250]; Haller 2018; Heinzel 2022; Heissel 2015; Hemat-Far 2012 [251]; Hess-Homeier 1981 [252]; Hidalgo 2021; Ho 2014; Hoffman 2010 [253, 254]; Huang 2015; Karg 2020; Khorvash 2012; Khoshnab 2017; Klein 1985 [255]; Knubben 2007 [256]; Krogh 2009; Kruisdijk 2018; La Rocque 2021; Legrand 2014; Legrand 2016; Martinsen 1985 [257]; Mather 2002 [258]; McCann 1984 [259]; McNeil 1991 [260]; Mota-Pereira 2011 [261]; Mutrie 1988 [262, 263]; Nabkasorn 2005 [264]; Olson 2017; Orth 1979 [265]; Pfaff 2014; Pilu 2007 [266, 267]; Pinchasov 2000 [268]; Prakhinkit 2014; Reuter 1984 [269]; Roy 2018; Salehi 2016; Schuch 2015; Setaro 1985 [270]; ; Shahidi 2011 [271, 272]; Sims 2009 [273, 274]; Singh 1997 [275, 276]; Singh 2005 [277]; Siqueira 2016; Taheri 2018; Veale 1992 [278]; Wang 2022; Williams 2008 [279]; Yu 2023). The 73 trials recruited a total of at least 4985 participants at randomisation, and at least 3624 participants completed the trials. In two trials, data were not reported on the number of participants at randomisation or follow-up (Reuter 1984; Wang 2022), while eight trials did not report participant numbers at follow-up (Brenes 2007; Epstein 1986; Hess-Homeier 1981; Khorvash 2012; Khoshnab 2017; Pinchasov 2000; Reuter 1984; Wang 2022).

We contacted the authors of three studies for additional data and received one response (Schuch 2015).

See [Table 2](#) and [Supplementary material 7](#) for details about the characteristics of the included studies.

Excluded studies

In total, 326 studies have been excluded from the various versions of the review after full-text review; the reasons for exclusion are listed below. As with previous updates, we decided not to list reviews as excluded studies. Further details of excluded studies can be found in [Supplementary material 3](#).

In 195 studies, participants either did not have to have depression (as defined by the authors of the study) to be eligible for the study, had a condition that was excluded (e.g. dysthymia, bipolar, postnatal depression), or were a subgroup with another condition (e.g. people with obesity) (Abascal 2008 [280]; Abd El-Kadar 2016 [281]; Abt 2005 [282]; Aghakhani 2011; Aidar 2018 [283]; Akandere 2011 [284]; Alsaraireh 2017 [285]; Annesi 2010 [286]; Arcos-Carmona 2011 [287]; Armstrong 2003 [288]; Armstrong 2004 [289]; Asbury 2009 [290]; Aylin 2009 [291]; Badger 2007 [292]; Baker 2006 [293]; Bashir 2022 [294]; Berke 2007 [295]; Blumenthal 2012b [296]; Bosch 2009 [297]; Boyll 1986 [298]; Brittle 2009 [299]; Broberg 2021 [300]; Burton 2009 [301]; Carneiro 2015 [302]; Carneiro 2016 [303]; Carneiro 2017 [304]; Carney 1987 [305]; Chan 2013b [306]; Chang 2020 [307]; Chang 2021 [308]; Chapman 2017 [309]; Chen 2009 [310]; Christensen 2005 [311]; Christensen 2012 [312]; Clegg 2011 [313]; Courneya 2007 [314]; Courneya 2014 [315]; Dalton 1980 [316]; Danielsson 2020 [317]; Demiralp 2011 [318]; DeVaney 1991 [319]; DiLorenzo 1999 [320]; Dotson 2016 [321]; Eby 1985 [322]; Echeta

2018 [323]; Edwards 2013 [324]; Edwards 2018 [325]; Elavsky 2007 [326]; El-Rafie 2016 [327]; Emery 1990a [328]; Emery 1990b [329]; Ersek 2008 [330]; Fisher 2020 [331]; Fox 2007 [332]; Gao 2016 [333]; Gary 2007 [334]; Ghroubi 2009 [335]; Gotta 2012; Gottlieb 2009 [336]; Gusi 2008 [337]; Gutierrez 2012 [338]; Haffmans 2006 [339]; Hallgren 2015 [340, 341, 342, 343, 344, 345, 346]; Hallgren 2016 [347]; Hannaford 1988 [348]; Haugen 2007 [349]; Helgadottir 2016 [350]; Helgadottir 2017a [351]; Helgadottir 2017b [352]; Hembree 2000 [353]; Herrera 1994 [354]; Huang 2015a [355]; Hughes 1986 [356]; Imayama 2013 [357]; Imboden 2020 [358]; Jacobsen 2012 [359]; Johansson 2011 [360]; Justine 2010 [361]; Karlsson 2007 [362]; Katayama 2020 [363]; Kerr 2008 [364]; Kerse 2010; Kim 2004 [365]; Kim 2019 [366]; Kim 2023 [367]; Knapen 2003 [368]; Knapen 2006 [369]; Krogh 2013 [370]; Krogh 2014 [371]; Kubesh 2003 [372]; Kulcu 2007 [373]; Kupecz 2001 [374]; Lai 2006 [375]; Latimer 2004 [376]; Lautenschlager 2008 [377]; Leppämäki 2002 [378, 379]; Levendoglu 2004 [380]; Lever-van Milligen 2012 [381]; Levin 2016 [382]; Levin 2018 [383]; Lvinger 2011 [384]; Lin 2007 [385]; Littbrand 2011 [386]; Littlecott 2014 [387]; Lok 2017 [388]; Lolak 2008 [389]; Lovrenčić 2015 [390]; Luo 2023 [391]; Machado 2007 [392]; Mackintosh 2013 [393]; Mailey 2010 [394]; Majumder 2015 [395]; Makizako 2020 [396]; Martin 2009 [397]; Martiny 2013; Matthews 2011 [398]; Meyer 2022 [399]; Midtgaard 2011 [400]; Milani 2007 [401]; Morey 2003 [402]; Motl 2004 [403]; Mudge 2008 [404]; Murphy 2012; Mutrie 2007 [405]; NCT02885285 [406]; NCT03542396 [407]; NCT03833349 [408]; Neidig 1998 [409, 410, 411]; Netz 1994 [412]; Neuberger 2007 [413]; Neunhauserer 2013 [414]; Ng 2017 [415]; Nguyen 2001 [416]; Nitzsche 2015 [417]; Nyunt 2014 [418]; Oeland 2010 [419]; Oertel-Knochel 2014 [420]; Ouzouni 2009 [421]; Pakkala 2008 [422]; Palmer 2005 [423]; Pelham 1993 [424]; Penninx 2002 [425, 426]; Penttinen 2011 [427]; Perales 2015 [428]; Pereira 2013 [429]; Perna 2010 [430]; Peterson 2014 [431]; Pibernik-Okanovic 2015; Pinniger 2012; Poelke 2015 [432]; Rashidi 2017 [433]; Rawson 2015 [434]; Rebar 2016 [435]; Rezaei 2015; Rhodes 1980 [436]; Robledo Colonia 2012 [437]; Rogerson 2020 [438]; Ruunsunen 2012 [439]; Salminen 2005 [440]; Sarsan 2006 [441]; Sims 2006 [442]; Skrinar 2005 [443]; Smith 2008 [444]; Songoyard 2012 [445]; Soucy 2017 [446]; Soucy 2018 [447]; Stein 1992 [448]; Stener-Victorin 2013 [449]; Stern 1983 [450, 451]; Strömbeck 2007 [452]; Sturm 2012; Sung 2009 [453]; Szuhany 2018 [454]; Tapps 2009 [455]; Taylor 1986 [456]; Taylor-Young 2011; Tenorio 1986 [457]; Thomas 2013 [458]; Thomson 2010 [459]; Tomas-Carus 2008 [460]; Tsang 2003 [461]; Underwood 2013 [462, 463, 464, 465]; van Beljouw 2015 [466]; Van der Waerden 2013 [467]; Vankova 2014 [468]; Wadden 2014 [469]; Weinstein 2007 [470]; White 2007 [471]; Whitham 2011 [472]; Wilbur 2009 [473]; Wipfli 2008 [474]; Wipfli 2011 [475]; Yacaman-Mendez 2019 [476]; Yeh 2015 [477]; Zheng 2015 [478]).

We excluded eight studies involving adolescents (Beffert 1993 [479]; Brown 1992 [480]; Hughes 2009 [481]; MacMahon 1988 [482]; NCT00964054 [483]; Rofey 2008 [484]; Roshan 2011 [485]; Silverman 1998 [486]).

We excluded 26 studies that did not use a randomised trial design (Attia 2012 [487]; Biddle 1989 [488]; Bromby 2010 [489]; Broocks 1997 [490]; Burbach 1997 [491]; Chekroud 2019 [492]; Deslandes 2010 [493]; Hedayati 2012 [494]; Jaffery 2017 [495]; Kinser 2020 [496]; Knapen 2015 [497]; Korman 2020 [498]; Leibold 2010 [499]; Martinsen 1988a [500]; Martinsen 1989c [501]; Martinsen 1993 [502]; Munro 1997 [503]; NCT00416221 [504]; Pourmousavi 2016 [505]; Raglin 1990 [506]; Rao 2020 [507]; Salmon 2001 [508]; Silveira 2010 [509]; Tsang 2013 [510]; Van de Vliet 2003 [511]; Weiss 1989 [512]).

We excluded 52 studies on the basis of the intervention: they assessed an intervention that was not exercise (Alexopoulos 2014 [513]; Bowden 2012 [514]; Busch 2013 [515]; Chan 2011 [516]; Chu 2018 [517]; Ciocon 2003 [518]; Duarte 2015 [519]; Fitzsimmons 2001 [520]; Grosse 2023 [521]; He 2022 [522]; Keller-Varady 2023 [523]; Kleinstaub 2017 [524]; Nystrom 2017 [525]; Van der Merwe 2004 [526]; Walker 2014 [527]) or that did not fulfil the ACSM definition of exercise (tai chi (Chou 2004 [528]; Srinivasan 2014 [529]); qigong: (Chan 2013a [530]; Chow 2012 [531]; Tsang 2006 [532]); yoga (Immink 2011 [533]; Kahl 2016 [534]; Oretzky 2006 [535]; Schneider 2016) and pilates (Rashidi 2013 [536]), compared to waiting list controls; they involved only a single session of exercise (Bartholomew 2005 [537]; Bodin 2004 [538]; Gustafsson 2009 [539]; Legrand 2018 [540]) or exercise for only four days (Berlin 2003 [541]); they provided exercise counselling or psychological interventions rather than exercise (Chalder 2012 [542, 543, 544, 545, 546, 547, 548]; Kratz 2014 [549]; Piette 2011 [550]; Ström 2013; Vickers 2009 [551]); or they were multifaceted interventions (Chau 2020 [552]; Chau 2022 [553]; Danielsson 2014 [554]; Farrand 2014 [555]; Garcia-Toro 2010 [556]; Garcia-Toro 2014 [557]; Gyurak 2013 [558]; McClure 2008 [559, 560]; Meyer 2016 [561]; O'Neil 2011 [562]; Patten 2017 [563]; Shachar-Malach 2015 [564]; Szuhany 2020 [565, 566] Patten 2019 [567]; Sadeghi 2016; Sneider 2008 [568]; Verrall 2012 [569]).

In 31 trials, the comparator was inappropriate, often incorporating an exercise-based comparator intervention (Balchin 2016 [570]; Bosscher 1993 [571]; Clark 2014; DEMO II 2012; Gerber 2020 [572]; Hanssen 2017 [573, 574]; Hanssen 2018 [575]; Haussleiter 2020 [576]; Imboden 2016 [577, 578]; Imboden 2017 [579]; Judith 2006 [580]; Kerling 2015 [581, 582, 583]; Kerling 2017 [584]; Kerling 2018 [585]; Labbe 1988 [586]; Lacombe 1988 [587]; Lavebratt 2017 [588]; Lavretsky 2011 [589]; Legrand 2009 [590]; Martinsen 1988b [591]; Moraes 2020 [592]; NCT00546221 [593]; NCT01152086 [594]; Niederer 2017 [595]; Passmore 2006 [596]; Sexton 1989 [597]; TREAD 2003 [598, 599, 600, 601, 602, 603, 604, 605, 606]; Trivedi 2011 [607]; Uebelacker 2020 [608]; Wieman 1980 [609]; Williams 1992 [610]).

We excluded four trials as they did not assess the effects of the interventions on a measure of depression (Kratzer 2021 [611]; Patten 2016 [612]; Perri 1984 [613]; Roth 1987 [614]).

We excluded nine studies as either they were ongoing, had been terminated, had limited or no data available as the study had not been published or was available only as an abstract (Chau 2015 [615]; CTRI/2020/04/024632 [616]; NCT01401569; NCT01805479; NCT04337398 [617]; NCT04572035 [618]; Peacock 2006 [619]; Schwarz 2012 [620]; Skuhareuskaya 2020 [621]).

We excluded one trial as it appeared to be plagiarised from another included study with only the setting changed (Veloz 2022 [622]).

Ongoing studies

We identified 66 ongoing studies through our searches of CCDANCTR (23 studies: Heinzl 2018 [623]; Heissel 2020 [624]; Lopez-Torres 2019 [625]; Martland 2020 [626]; NCT02407704 [627]; NCT02679053 [628]; NCT03358433 [629]; NCT03056287 [630]; NCT03107039 [631]; NCT03731728 [632]; NCT04403373 [633]; NCT05703776 [634]; NCT05136027 [635]; NCT05313542 [636]; NCT05812001 [637]; NCT06387732 [638]; NCT06398496

[639]; NCT06450704 [640]; NCT06594588 [641]; Neves 2018 [642]; Safakar 2020 [643]; Schmitter 2020 [644]; UMIN000039519 [645]) and the Clinicaltrials.gov, ICTRP and ISRCTN trial registries (43 studies: ACTRN12620001372976 [646]; ACTRN12624001150538 [647]; ChiCTR-IOR-14005463 [648]; ChiCTR2200058744 [649]; ChiCTR2300074393 [650]; ChiCTR2300075559 [651]; ChiCTR2300077020 [652]; ChiCTR2400087257 [653]; ChiCTR2400087261 [654]; ChiCTR2500097477 [655]; ChiCTR2400083854 [656]; ChiCTR2400084484 [657]; ChiCTR2400091623 [658]; ChiCTR2500107313 [659]; CTRI/2016/06/007055 [660]; DRKS00005823 [661]; DRKS00008869 [662]; DRKS00023257 [663]; DRKS0008745 [664]; DRKS00026117 [665]; DRKS00032065 [666]; IRCT20130812014333N86 [667]; Irct2014020616501N [668]; Irct2015093024282N [669]; Irct2016082929598N [670]; Irct2016111630924N [671]; IRCT20190805044438N4 [672]; IRCT20210617051606N1 [673]; IRCT20211011052727N1 [674]; IRCT20241206063964N1 [675]; IRCT20250402065189N1 [676]; ISRCTN10469580 [677]; ISRCTN14914508 [678]; JPRN-UMIN000029117 [679]; JPRN-UMIN000036468 [680]; JPRN-UMIN000039519 [681]; NCT03403881 [682]; NCT04714528 [683]; NL8432 [684]; RBR-248x3v [685]; RBR-38nkj6 [686]; RBR-9zgxd [687]; RBR-4pgd3ct [688]). Added to the nine studies that remain ongoing from the 2013 update by Cooney and colleagues (ACTRN12612000094875; CTRI/2012/09/002985; NCT00931814; NCT01383811; NCT01464463; NCT01619930; NCT01696201; NCT01763983; UMIN000001488), there are currently 75 studies ongoing. See [Supplementary material 4](#).

New studies found for this update

We are including 35 additional trials, which recruited a total of 2526 participants (where reported) at randomisation. One trial did not state the number of participants randomised (Wang 2022). Of these, 1844 participants (where reported) remained in the trials by the time of final outcome analysis (Abdelbasset 2019; Abdollahi 2017; Bademli 2023; Barclay 2014; Belvederi Murri 2015; Bose 2015; Brush 2022; Buschert 2019; Cheung 2018; De Groot 2019; Doose 2015; Eman 2017; Haller 2018; Heinzl 2022; Heissel 2015; Hidalgo 2021; Ho 2014; Huang 2015; Karg 2020; Khorvash 2012; Khoshnab 2017; Kruisdijk 2018; La Rocque 2021; Legrand 2014; Legrand 2016; Olson 2017; Pfaff 2014; Prakhinkit 2014; Roy 2018; Salehi 2016; Schuch 2015; Siqueira 2016; Taheri 2018; Wang 2022; Yu 2023). Three trials did not provide participant numbers at follow-up (Khorvash 2012; Khoshnab 2017; Wang 2022). See [Supplementary material 2](#).

Risk of bias in included studies

Random sequence generation

We classified 30 trials as being at low risk of bias (Bademli 2023; Belvederi Murri 2015; Blumenthal 2007; Blumenthal 2012a; Brenes 2007; Brush 2022; Cheung 2018; De Groot 2019; Haller 2018; Heinzl 2022; Hidalgo 2021; Ho 2014; Hoffman 2010; Huang 2015; Karg 2020; Knubben 2007; Krogh 2009; La Rocque 2021; Legrand 2014; Legrand 2016; Martinsen 1985; Mather 2002; Olson 2017; Pfaff 2014; Prakhinkit 2014; Sims 2009; Singh 1997; Singh 2005; Siqueira 2016; Yu 2023), one as being at high risk of bias (Hemat-Far 2012), and 42 trials as being at unclear risk of bias. A graphical representation of the risk of bias assessment can be seen in [Figure 2](#) and [Figure 3](#). See [Supplementary material 2](#) for the full risk of bias assessment for each study.

Figure 2. 'Risk of bias' graph: review authors' judgements about each risk of bias item presented as percentages across all included studies

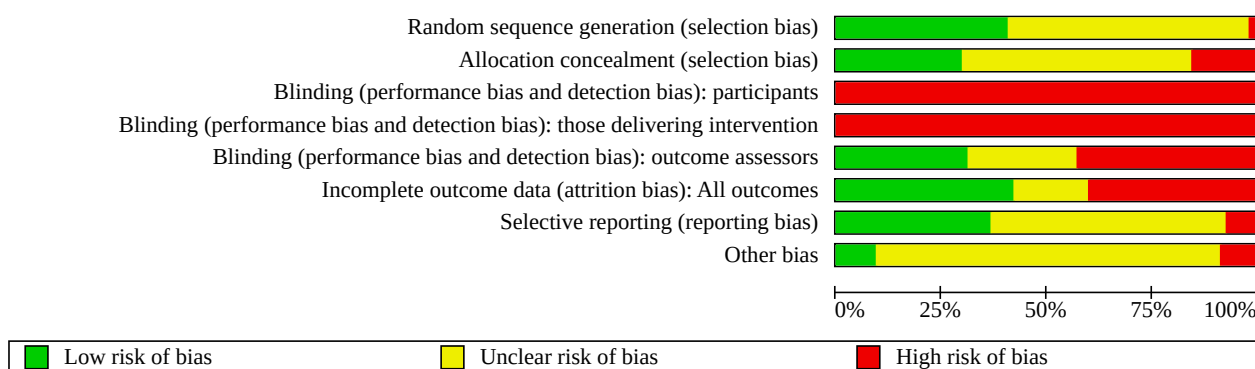


Figure 3. 'Risk of bias' summary: review authors' judgements about each risk of bias item for each included study

	Random sequence generation (selection bias)	Allocation concealment (selection bias)	Blinding (performance bias and detection bias): participants	Blinding (performance bias and detection bias): those delivering intervention	Blinding (performance bias and detection bias): outcome assessors	Incomplete outcome data (attrition bias): All outcomes	Selective reporting (reporting bias)	Other bias
Abdelbasset 2019	?	?	-	-	+	+	?	?
Abdollahi 2017	?	?	-	-	?	-	?	?
Bademli 2023	+	?	-	-	?	+	?	+
Barclay 2014	?	?	-	-	?	+	+	?
Belvederi Murri 2015	+	?	-	-	+	+	+	?
Blumenthal 1999	?	+	-	-	+	+	+	?
Blumenthal 2007	+	+	-	-	+	+	+	?
Blumenthal 2012a	+	+	-	-	+	+	+	?
Bonnet 2005	?	?	-	-	-	-	?	?
Bose 2015	?	?	-	-	+	-	?	?
Brenes 2007	+	?	-	-	+	?	?	?
Brush 2022	+	?	-	-	-	+	?	+
Buschert 2019	?	?	-	-	?	?	?	?
Cheung 2018	+	+	-	-	+	+	+	+
Chu 2008	?	?	-	-	-	-	?	?
De Groot 2019	+	+	-	-	+	+	+	?

Figure 3. (Continued)

	⬇	⬇	⬆	⬆	⬆	⬆	⬆	⬆
De Groot 2019	+	+	-	-	+	+	+	?
Doose 2015	?	?	-	-	-	+	+	?
Doyne 1987	?	-	-	-	-	-	?	?
Dunn 2005	?	+	-	-	+	+	+	?
Eman 2017	?	?	-	-	?	?	?	?
Epstein 1986	?	-	-	-	-	?	?	?
Fetsch 1979	?	?	-	-	-	-	?	?
Foley 2008	?	?	-	-	-	-	?	?
Fremont 1987	?	-	-	-	-	-	?	?
Gary 2010	?	?	-	-	+	-	+	+
Greist 1979	?	-	-	-	-	-	-	-
Haller 2018	+	?	-	-	?	-	-	-
Heinzel 2022	+	?	-	-	?	-	+	?
Heissel 2015	?	?	-	-	?	-	?	?
Hemat-Far 2012	-	?	-	-	-	?	+	-
Hess-Homeier 1981	?	-	-	-	-	-	?	?
Hidalgo 2021	+	?	-	-	?	+	-	?
Ho 2014	+	+	-	-	+	+	+	?
Hoffman 2010	+	+	-	-	-	+	+	?
Huang 2015	+	?	-	-	+	+	+	?
Karg 2020	+	+	-	-	+	+	+	+
Khorvash 2012	?	?	-	-	?	?	-	?
Khoshnab 2017	?	?	-	-	-	?	+	?
Klein 1985	?	-	-	-	-	-	?	?
Knubben 2007	+	+	-	-	+	?	?	?
Krogh 2009	+	+	-	-	+	+	+	?
Kruisdijk 2018	?	?	-	-	?	-	?	-
La Rocque 2021	+	+	-	-	+	+	+	?
Legrand 2014	+	?	-	-	-	-	+	?
Legrand 2016	+	+	-	-	-	+	+	?
Martinsen 1985	+	+	-	-	-	-	?	?
Mather 2002	+	+	-	-	+	+	?	?
McCann 1984	?	-	-	-	-	-	?	?
McNeil 1991	?	-	-	-	-	+	?	?
Mota-Pereira 2011	?	?	-	-	+	+	+	+
Mutrie 1988	?	-	-	-	-	+	?	?

Figure 3. (Continued)

Mota-Pereira 2011	?	?	+	+	+	+	+	+
Mutrie 1988	?	-	-	-	-	+	?	?
Nabkasorn 2005	?	?	-	-	-	-	?	?
Olson 2017	+	?	-	-	?	-	?	?
Orth 1979	?	?	-	-	-	+	?	?
Pfaff 2014	+	+	-	-	+	+	+	?
Pilu 2007	?	?	-	-	?	+	-	?
Pinchasov 2000	?	?	-	-	?	?	?	?
Prakhinkit 2014	+	-	-	-	-	?	+	?
Reuter 1984	?	-	-	-	-	-	?	?
Roy 2018	?	?	-	-	?	?	?	?
Salehi 2016	?	?	-	-	?	+	?	?
Schuch 2015	?	?	-	-	?	+	?	?
Setaro 1985	?	?	-	-	?	-	?	?
Shahidi 2011	?	?	-	-	-	-	+	?
Sims 2009	+	+	-	-	-	-	?	?
Singh 1997	+	+	-	-	-	+	?	?
Singh 2005	+	+	-	-	+	-	?	?
Siqueira 2016	+	+	-	-	-	+	+	+
Taheri 2018	?	?	-	-	-	?	?	-
Veale 1992	?	+	-	-	-	-	?	?
Wang 2022	?	?	-	-	?	?	?	-
Williams 2008	?	+	-	-	+	-	?	?
Yu 2023	+	?	-	-	+	-	+	?

Allocation concealment

Allocation concealment was adequate and therefore at low risk of bias in 22 trials (Blumenthal 1999; Blumenthal 2007; Blumenthal 2012a; Cheung 2018; De Groot 2019; Dunn 2005; Ho 2014; Hoffman 2010; Karg 2020; Knubben 2007; Krogh 2009; La Rocque 2021; Legrand 2016; Martinsen 1985; Mather 2002; Pfaff 2014; Sims 2009; Singh 1997; Singh 2005; Siqueira 2016; Veale 1992; Williams 2008). For the remaining trials, we judged the risk of bias as unclear or high (Abdelbasset 2019; Abdollahi 2017; Bademli 2023; Barclay 2014; Belvederi Murri 2015; Bonnet 2005; Bose 2015; Brenes 2007; Brush 2022; Buschert 2019; Chu 2008; Doose 2015; Doynne 1987; Eman 2017; Epstein 1986; Fetsch 1979; Foley 2008; Fremont 1987; Gary 2010; Greist 1979; Haller 2018; Heinzel 2022; Heissel 2015; Hemat-Far 2012; Hess-Homeier 1981; Hidalgo 2021; Huang 2015; Khorvash 2012; Khoshnab 2017; Klein 1985; Kruisdijk 2018; Legrand 2014; McCann 1984; McNeil 1991; Mota-Pereira 2011; Mutrie 1988; Nabkasorn 2005; Olson 2017; Orth 1979; Pilu 2007; Pinchasov 2000; Prakhinkit 2014; Reuter 1984; Roy 2018; Salehi 2016; Schuch 2015; Setaro 1985; Shahidi 2011; Taheri 2018; Wang 2022; Yu 2023).

Blinding of participants

In exercise trials, participants cannot be blinded to the treatment allocation. We were advised by Cochrane Editors to classify all trials as being at high risk of bias.

Blinding of those delivering the intervention

Similarly, for those trials where supervised exercise was provided, the person delivering the intervention could not be blinded, so we were advised by the Cochrane Editors to classify all trials as being at high risk of bias (note that not all studies reported whether exercise was performed with or without supervision).

Blinding of outcome assessors

Twenty-three trials included blinding of the outcome assessors, so we classified them as being at low risk of bias (Abdelbasset 2019; Belvederi Murri 2015; Blumenthal 1999; Blumenthal 2007; Blumenthal 2012a; Bose 2015; Brenes 2007; Cheung 2018; De Groot 2019; Dunn 2005; Gary 2010; Ho 2014; Huang 2015; Karg 2020; Knubben 2007; Krogh 2009; La Rocque 2021; Mather 2002; Mota-

Pereira 2011; Pfaff 2014; Singh 2005; Williams 2008; Yu 2023). We classified 19 trials as unclear (Abdollahi 2017; Bademli 2023; Barclay 2014; Buschert 2019; Eman 2017; Haller 2018; Heinzel 2022; Heissel 2015; Hidalgo 2021; Khorvash 2012; Kruisdijk 2018; Olson 2017; Piliu 2007; Pinchasov 2000; Roy 2018; Salehi 2016; Schuch 2015; Setaro 1985; Wang 2022) and 31 trials at high risk (Bonnet 2005; Brush 2022; Chu 2008; Dooze 2015; Doynne 1987; Epstein 1986; Fetsch 1979; Foley 2008; Fremont 1987; Greist 1979; Hemat-Far 2012; Hess-Homeier 1981; Hoffman 2010; Khoshnab 2017; Klein 1985; Legrand 2014; Legrand 2016; Martinsen 1985; McCann 1984; McNeil 1991; Mutrie 1988; Nabkasorn 2005; Orth 1979; Prakhinkit 2014; Reuter 1984; Shahidi 2011; Sims 2009; Singh 1997; Siqueira 2016; Taheri 2018; Veale 1992).

Incomplete outcome data

We classified 31 trials as low risk of bias because they performed 'intention-to-treat' (ITT) analysis (i.e. complete outcome data were reported) or, if there were missing outcome data, these were replaced using a recognised statistical method (e.g. last observation carried forward), and participants remained in the group to which they had been allocated (Abdelbasset 2019; Bademli 2023; Barclay 2014; Belvederi Murri 2015; Blumenthal 1999; Blumenthal 2007; Blumenthal 2012a; Brush 2022; Cheung 2018; De Groot 2019; Dooze 2015; Dunn 2005; Hidalgo 2021; Ho 2014; Hoffman 2010; Huang 2015; Karg 2020; Krogh 2009; La Rocque 2021; Legrand 2016; Mather 2002; McNeil 1991; Mota-Pereira 2011; Mutrie 1988; Orth 1979; Pfaff 2014; Piliu 2007; Salehi 2016; Schuch 2015; Singh 1997; Siqueira 2016).

One trial reported data for individual participants (Orth 1979), so by using 'last observation carried forward', we replaced data from the participants who did not complete the trial and included these data in the meta-analysis of ITT trials. One trial reported that the analysis was ITT, because it used a 'worse-case' scenario assumption to replace data from participants who did not complete the trial, but they included only 38 of the 39 randomised participants in the analyses, so we classified it as 'not ITT' (Knubben 2007). The remaining studies were classified as being at unclear or high risk of bias. Most trials did not report data from participants who dropped out.

Selective reporting

We attempted to identify whether a protocol was available by screening the reference list of the publications. We identified protocols for eight trials and checked that all prespecified outcome events were reported, and rated these as being at low risk of bias (Barclay 2014; Blumenthal 2012a; Dunn 2005; Heinzel 2022; Karg 2020; Krogh 2009; La Rocque 2021; Yu 2023).

We classified 19 other trials as being at low risk of bias as we judged that there was sufficient information in the methods to be sure that the trials had reported all their planned outcomes (Belvederi Murri 2015; Blumenthal 1999; Blumenthal 2007; Cheung 2018; De Groot 2019; Dooze 2015; Gary 2010; Hemat-Far 2012; Hoffman 2010; Ho 2014; Huang 2015; Khoshnab 2017; Legrand 2014; Legrand 2016; Pfaff 2014; Prakhinkit 2014; Mota-Pereira 2011; Shahidi 2011; Siqueira 2016).

We classified five trials at high risk of selective reporting bias. One trial did not have a methods section and only reported outcomes

within the results (Greist 1979), while a second trial stated that there was an eight-month interview follow-up, but these data were not reported (Piliu 2007). A third trial did not report any measure of variation for all its outcome measures (Khorvash 2012). One trial did not report outcomes at follow-up as was indicated it would do in the protocol (Hidalgo 2021). One trial reported data only for the intervention before and after its use rather than in comparison to the control intervention (Haller 2018).

Other biases

We considered seven trials to be at low risk of bias in terms of other potential biases (Bademli 2023; Brush 2022; Cheung 2018; Gary 2010; Karg 2020; Mota-Pereira 2011; Siqueira 2016), with six trials at high risk of bias (Greist 1979; Haller 2018; Hemat-Far 2012; Kruisdijk 2018; Taheri 2018; Wang 2022).

For 60 trials, the risk of bias was unclear (Abdelbasset 2019; Abdollahi 2017; Barclay 2014; Belvederi Murri 2015; Blumenthal 1999; Blumenthal 2007; Blumenthal 2012a; Bonnet 2005; Bose 2015; Brenes 2007; Buschert 2019; Chu 2008; De Groot 2019; Dooze 2015; Doynne 1987; Dunn 2005; Eman 2017; Epstein 1986; Fetsch 1979; Foley 2008; Fremont 1987; Heinzel 2022; Heissel 2015; Hess-Homeier 1981; Hidalgo 2021; Ho 2014; Hoffman 2010; Huang 2015; Khorvash 2012; Khoshnab 2017; Klein 1985; Knubben 2007; Krogh 2009; La Rocque 2021; Legrand 2014; Legrand 2016; Martinsen 1985; Mather 2002; McCann 1984; McNeil 1991; Mutrie 1988; Nabkasorn 2005; Olson 2017; Orth 1979; Pfaff 2014; Piliu 2007; Pinchasov 2000; Prakhinkit 2014; Reuter 1984; Roy 2018; Salehi 2016; Schuch 2015; Setaro 1985; Shahidi 2011; Sims 2009; Singh 1997; Singh 2005; Veale 1992; Williams 2008; Yu 2023).

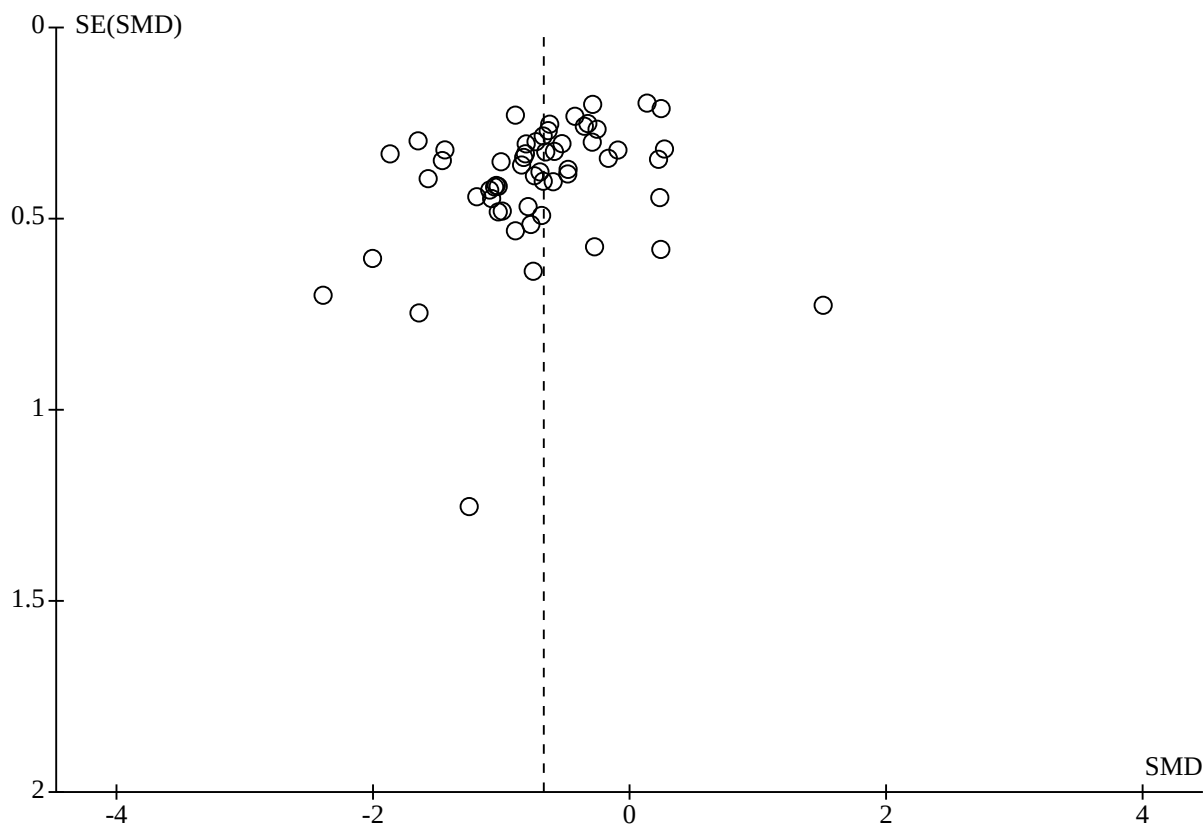
On the basis of the first 50 participants, one study changed the sample size calculation based on the observed standard deviation (Krogh 2009; their initial calculation was that a minimum of 186 participants (standard deviation (SD) = 6) were needed, but for the first 50 participants, the SD was 3.9, so they adjusted their sample size calculation to 135 participants). Hemat-Far 2012 told the control group to reduce their activity.

We decided to include data on continuous depression scores in our meta-analysis, rather than depression measured as a dichotomous outcome. This is because we knew from previous updates that very few trials had reported depression as a dichotomous outcome, and we wished to include as many trials as possible in our meta-analysis. For future updates, we will consider whether to perform a separate meta-analysis for trials that measured depression as a dichotomous outcome.

Publication bias

Visually, our funnel plot appeared to suggest some missing studies that may be due to a degree of publication bias, outcome reporting bias or heterogeneity (Egger regression intercept P value was 0.00265) [689] (funnel plot for exercise versus control - analysis 1.1 (Figure 4)). Use of Duval and Tweedie's trim and fill suggests that 15 studies may have been trimmed from the right of the mean (i.e. small studies favouring control intervention), and the adjusted SMD was -0.45 (95% CI -0.62 to -0.28), representing a moderate benefit from exercise [690].

Figure 4. Funnel plot of comparison: 1 Exercise versus control, outcome: 1.1 Reduction in depression symptoms post-treatment



Synthesis of results

Comparison 1: exercise versus control

Sixty-one trials (2576 participants) included a comparison of exercise with a control intervention, although only 57 trials (2189 participants) reported data that could be meta-analysed (see [Supplementary material 5](#)).

Primary outcome measure

1.1 Reduction in depressive symptom severity

Post-treatment

The pooled standardised mean difference (SMD) for the 57 trials (2189 participants), which was calculated using a random-effects model, was -0.67 (95% CI -0.82 to -0.52 ; $I^2 = 61$, $P < 0.001$; GRADE: low), showing that exercise may result in a reduction in depressive symptoms (Analysis 1.1 in [Supplementary material 5](#) and [Figure 5](#)).

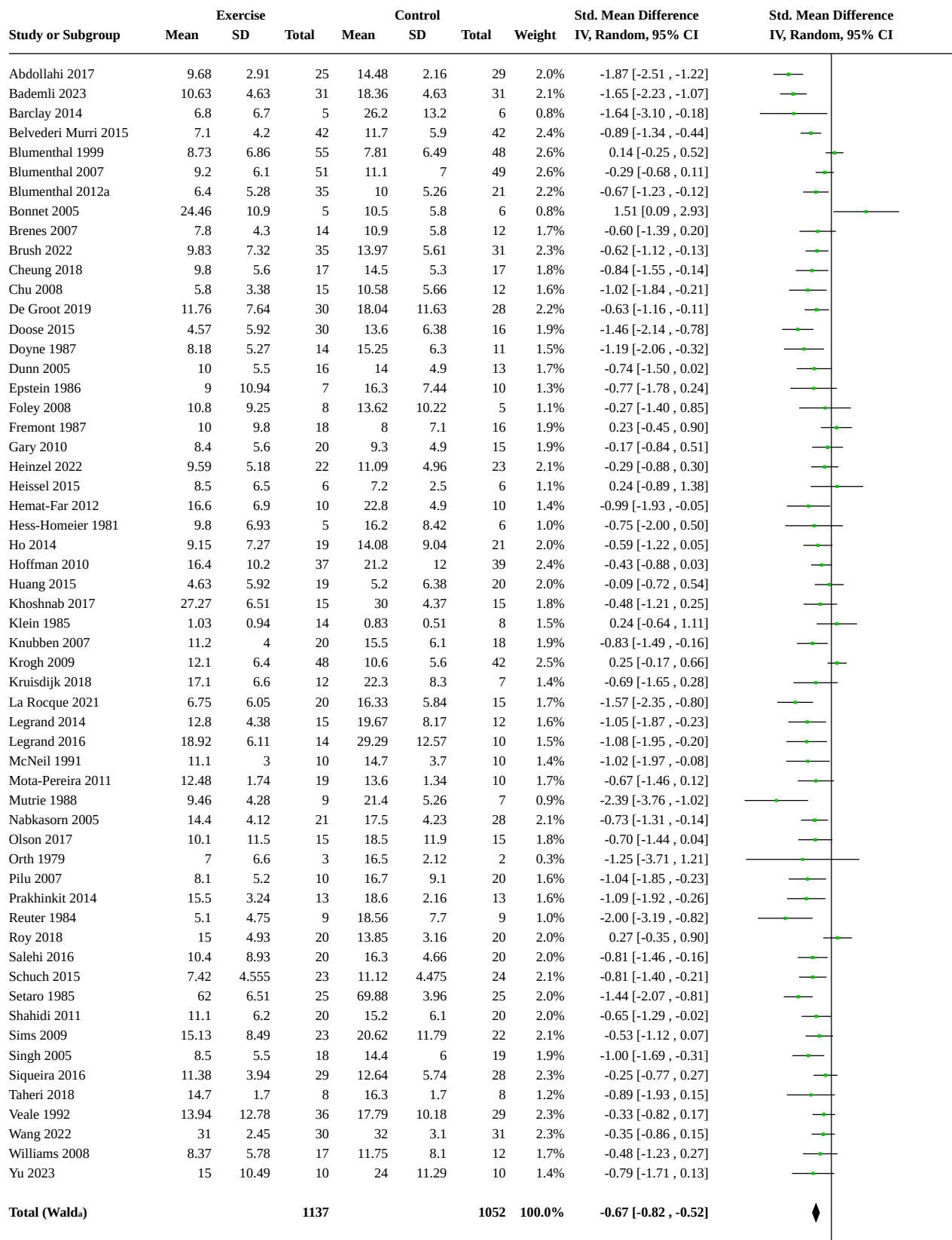
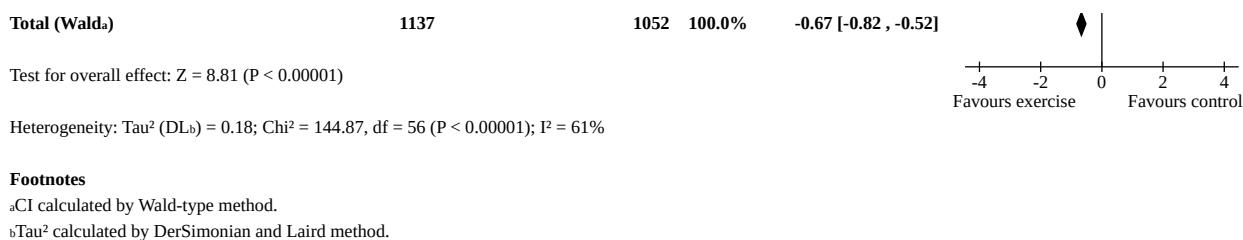
Figure 5. Forest plot of comparison: 1 Exercise versus 'control', outcome 1.1 Reduction in depression symptoms post-treatment

Figure 5. (Continued)

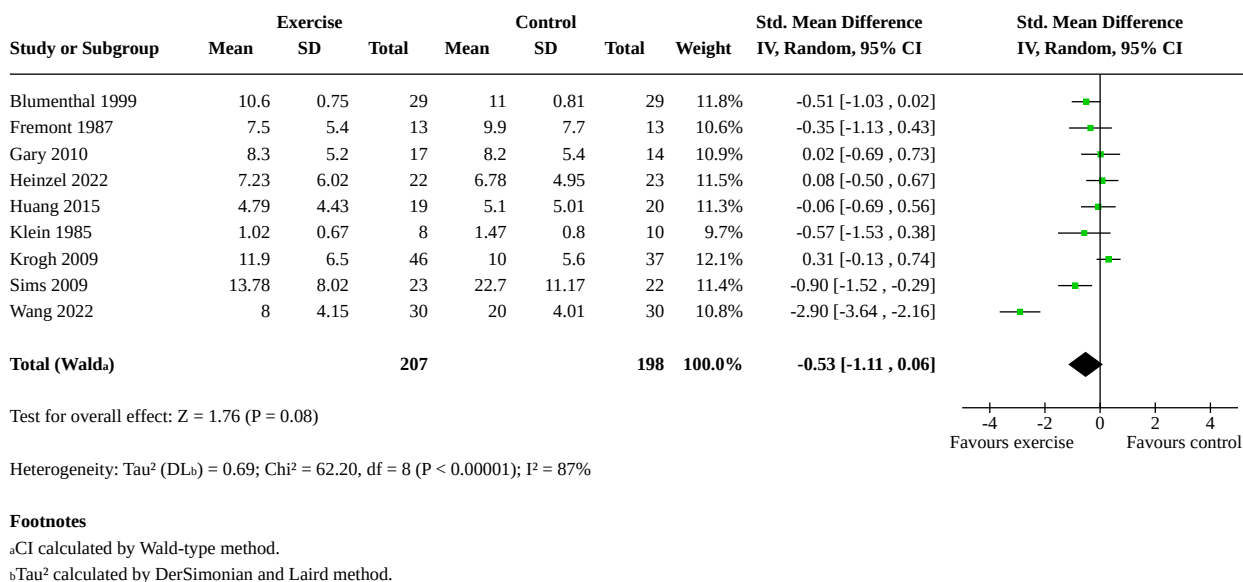


End of long-term follow-up

We were able to pool nine trials (405 participants) that provided long-term follow-up data, ranging from 8 weeks to 12 months of follow-up (Blumenthal 1999; Fremont 1987; Gary 2010; Heinzel 2022; Huang 2015; Klein 1985; Krogh 2009; Sims 2009; Wang 2022). The pooled SMD (random-effects) showed that the evidence is very

uncertain about the effect of exercise on depressive symptoms (SMD -0.53, 95% CI -1.11 to 0.06; $I^2 = 87$, $P < 0.001$; GRADE: very low) (Analysis 1.2; Figure 6). Follow-up data from Blumenthal 2007 were reported according to the proportion who had fully or partially remitted from depression, but continuous mood scores were not reported, so we could not include these data in the meta-analysis.

Figure 6. Forest plot of comparison: 1 Exercise versus 'control', outcome: 1.2 Reduction in depression symptoms follow-up



Secondary outcome measures

1.2a Acceptability of treatment: attendance at exercise

Twenty-six trials reported attendance rates for exercise. These were 50.6% for aerobic and 56.2% for strength exercise groups (Krogh 2009), 58% (Doose 2015; Pfaff 2014), 66.9% for exercise and 68.8% for yoga (La Rocque 2021), 68% (Legrand 2014), 70% (Doyle 1987), 72% (Dunn 2005), 73% (Ho 2014), 75% for progressive aerobic plus sertraline and 71% for non-progressive exercise plus sertraline (Belvederi Murri 2015), 78% (Nabkasorn 2005), 82% (Gary 2010), 84% for endurance and 90% for strength exercise (Haller 2018), 91% (Mota-Pereira 2011; Schuch 2015), 92% (Blumenthal 1999; Blumenthal 2007), 92.7% for moderate and 95% for vigorous exercise (Yu 2023), 92.9% and 81.8% attended ≥ 8 sessions, respectively (Legrand 2016), 94% (Blumenthal 2012a), 97.3% for high-intensity and 99.1% for low-intensity (Chu 2008), 95% to 100% (Singh 2005) and 100% (Huang 2015). One trial reported

the mean number of exercise sessions attended as 5.88 (Hoffman 2010). Another trial reported that 1.71 and 1.60 exercise sessions were attended for high- and low-intensity exercise, respectively (Heinzel 2022). One trial rescheduled missed visits (McNeil 1991) so participants attended the full course of exercise. One trial reported high adherence (no data reported) (Heissel 2015). As with exercise intensity, it is difficult to attribute any differences in outcome to differences in attendance rates because there were other sources of variation in the type of interventions (e.g. duration of intervention, type of exercise) and differences in the risk of bias between trials that might account for differences in outcome.

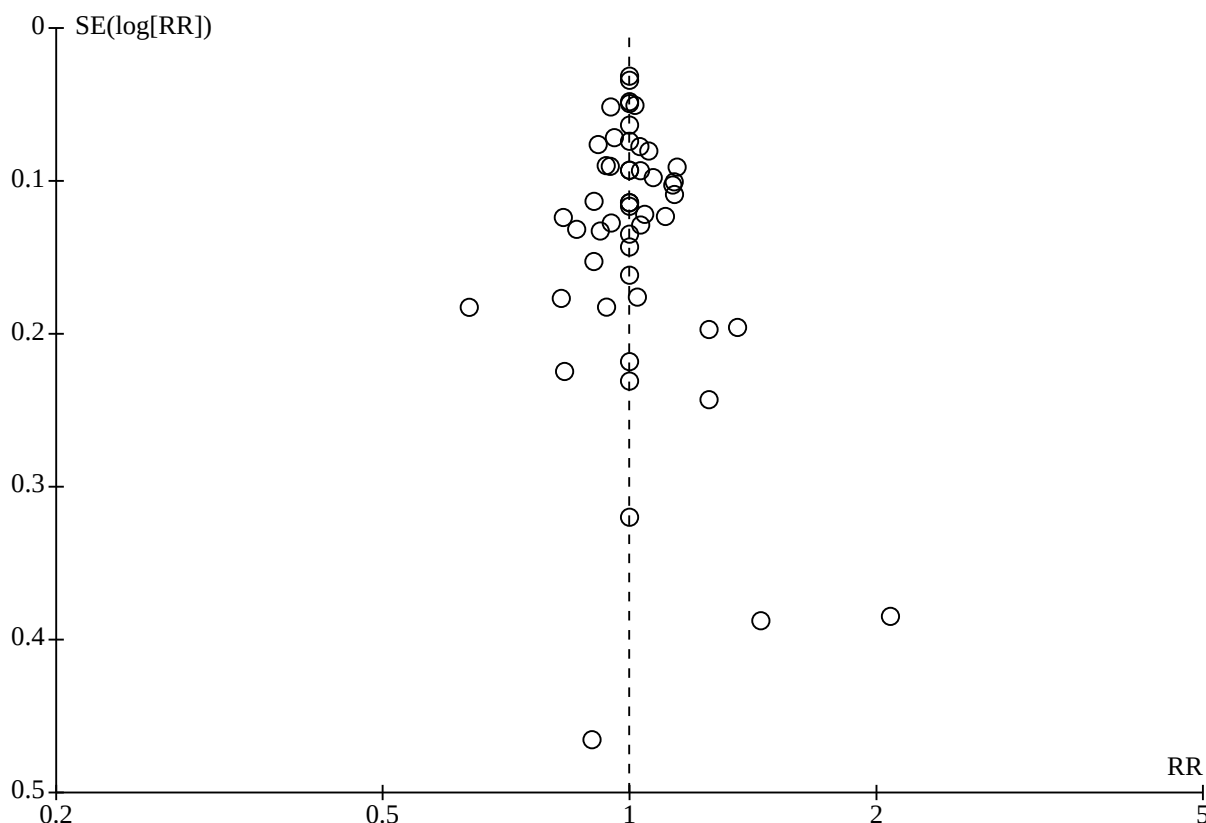
1.2b Acceptability of treatment: completing the intervention or control

We extracted data on the number of participants who were randomised in each trial and the number who completed each

trial (Table 1). This ranged from 100% completion (Bademli 2023; Barclay 2014; Hemat-Far 2012; Huang 2015; Khoshnab 2017; McNeil 1991; Mutrie 1988; Orth 1979; Pilu 2007; Salehi 2016; Siqueira 2016; Taheri 2018) to 46% completion (Hidalgo 2021). For the exercise intervention, this ranged from 100% completion (Bademli 2023; Barclay 2014; Cheung 2018; Doose 2015; Gary 2010; Hemat-Far 2012; Huang 2015; Khoshnab 2017; McNeil 1991; Mutrie 1988; Orth

1979; Pilu 2007; Salehi 2016; Siqueira 2016; Taheri 2018) to 38% completion (Hidalgo 2021). Fifty-one studies (2311 participants) reported how many participants completed the exercise and control arms (Analysis 1.3; Figure 7), producing a risk ratio of (RR) 1.0 (95% CI 0.98 to 1.02; GRADE: moderate certainty), showing that exercise results in little or no difference in completion compared to control.

Figure 7. Funnel plot of comparison: 1 Exercise versus 'control', outcome: 1.3 Completed intervention or control



1.3 Quality of life

Seven studies reported measures of quality of life at the end of the study (Brenes 2007; Gary 2010; Huang 2015; Pilu 2007; Schuch 2015; Sims 2009; Singh 2005), comparing interventions on different domains of quality-of-life questionnaires (Analysis 1.4 in Supplementary material 5). One trial reported changes in quality of life without presenting the necessary data to include in the meta-analysis (Singh 2005), and another reported quality of life at baseline but not at follow-up (Sims 2009).

The evidence was very uncertain for the quality of life mental health domain (SMD -0.05, 95% CI -0.85 to 0.76; $I^2 = 74\%$, $P = 0.02$; 3 studies, 98 participants; GRADE: very low) and psychological domain (SMD 0.35, 95% CI -1.22 to 1.92; $I^2 = 90\%$, $P = 0.001$; 2 studies, 80 participants; GRADE: very low). For social domains, exercise may result in little to no difference following exercise compared to control (SMD 0.37, 95% CI -0.18 to 0.92; $I^2 = 29\%$, $P = 0.24$; 2 studies, 80 participants; GRADE: low). In contrast, exercise may result in an increase in the environment domain (SMD 0.51, 95% CI 0.05 to 0.96; $I^2 = 0\%$, $P = 0.82$; 2 studies, 80 participants; GRADE: low) and physical domain (SMD 0.57, 95% CI 0.15 to 0.98;

$I^2 = 44\%$, $P = 0.13$; 5 studies, 178 participants; GRADE: very low) of quality of life, although the evidence is very uncertain for the physical domain. One study (39 participants) reported on a quality of life domain, where the evidence on the effect of exercise on quality of life was very uncertain (SMD 0.56, 95% CI -0.08 to 1.20; GRADE: very low).

1.4 Cost

No trial reported the cost or cost-effectiveness of exercise compared to another intervention.

1.5 Adverse events

Eleven trials reported data on adverse events between the exercise and the control groups (Belvederi Murri 2015; Blumenthal 1999; Blumenthal 2007; Blumenthal 2012a; Doose 2015; Dunn 2005; Knubben 2007; Legrand 2016; Prakhinkit 2014; Sims 2009; Singh 2005), showing that exercise is likely to result in little to no difference in adverse events compared to control. Dunn 2005 reported increased severity of depressive symptoms ($n = 1$), chest pain ($n = 1$) and joint pain/swelling ($n = 1$); all of these participants

discontinued exercise. Singh 2005 reported adverse events in detail (visits to a health professional, minor illness, muscular pain, chest pain, injuries requiring training adjustment, falls, deaths and hospital days) and found no difference between the groups. Knubben 2007 reported “no negative effects of exercise (muscle pain, tightness or fatigue)” in the course of the programme; two weeks after the training had finished, one person in the placebo group required gastric lavage and one person in the exercise group inflicted a superficial cut in her arm. Sims 2009 reported no adverse events or falls in either the exercise or control groups. Blumenthal 1999 reported that five participants in the exercise plus sertraline group dropped out due to adverse events associated with sertraline and four participants due to musculoskeletal injury. Five participants from the sertraline-only group dropped out owing to medication adverse events (Blumenthal 1999). Blumenthal 2007 reported side effects using a participant rating of a 36-item checklist, with the only statistically significant difference ($P = 0.03$) reported for participants experiencing post-treatment diarrhoea and loose stools (sertraline 31%; home-based exercise 21%; supervised exercise 10%; placebo 12%). Blumenthal 2012a reported more fatigue and sexual dysfunction in the sertraline group (20% and 26%, respectively) than in the exercise group (2.4% for both side effects). Although Prakhinkit 2014 reported that five participants withdrew either due to loss to follow-up, physical discomfort or lack of compliance, no details were provided regarding which interventions they received. Belvederi Murri 2015 reported two participants engaged with physical exercise plus sertraline left the study, one due to having ankle distortion and another through worsening musculoskeletal pain. Doose 2015 found that seven participants in the exercise group reported adverse events, including three who were unable to continue due to worsening of pre-existing orthopaedic injuries, two who were admitted to psychiatric wards with major depression and two who withdrew for personal reasons. Four in the control group also reported adverse events: one was admitted to hospital, one felt physically incapable of participating, one did not want to participate due to worsened pre-existing chronic neck pain and one withdrew for personal reasons (Doose 2015). Adverse events were reported in the exercise intervention group by Legrand 2016, including transient muscular/joint soreness ($n = 3$), headaches ($n = 1$) and fatigue ($n = 2$). Due to the diversity of different adverse events reported, we decided not to meta-analyse these data.

Comparison 2: exercise versus psychological therapies

Eleven trials (918 participants) compared exercise with psychological therapies (De Groot 2019; Epstein 1986; Fetsch 1979; Fremont 1987; Gary 2010; Greist 1979; Hess-Homeier 1981; Huang 2015; Karg 2020; Klein 1985; Setaro 1985).

Primary outcome

2.1 Reduction in depressive symptom severity

Post-treatment

We included data from 10 trials (414 participants) comparing exercise with psychological therapies in a meta-analysis (De Groot 2019; Epstein 1986; Fetsch 1979; Fremont 1987; Gary 2010; Hess-Homeier 1981; Huang 2015; Karg 2020; Klein 1985; Setaro 1985). When data were pooled, the SMD was 0.03 (95% CI -0.16 to 0.23), showing that exercise results in little to no difference compared to psychological therapies in changing depressive symptoms (Analysis 2.1 in [Supplementary material 5](#)). The analysis

provides moderate-certainty evidence, with heterogeneity that was considered unimportant ($I^2 = 0\%$, $P = 0.85$).

End of long-term follow-up

Long-term follow-up was reported by four studies (114 participants) ranging from 4 to 9 months (Fremont 1987; Gary 2010; Huang 2015; Klein 1985), which we judged as low-certainty evidence. When pooled in a random-effects meta-analysis (Analysis 2.2 in [Supplementary material 5](#)), the SMD of -0.11 (95% CI -0.48 to 0.26) showed that exercise may result in little to no difference compared to psychological therapies in changing depressive symptoms. No statistical heterogeneity was found ($I^2 = 0\%$, $P = 0.47$).

Secondary outcomes

2.2a Acceptability of treatment: attendance at exercise sessions

Two trials reported attendance rates (Gary 2010; Huang 2015). One trial reported adherence scores that were calculated based on the number of sessions attended of those prescribed. This trial reported that the adherence rates were 82% for exercise and 72% for CBT (Gary 2010). A second trial reported that the exercise group achieved 100% at 3-month follow-up, declining to 47% at 6- and 9-month follow-up (Huang 2015). Adherence rates for CBT were not reported (Huang 2015).

2.2b Acceptability of treatment: completing the intervention or psychological therapies

Seven trials (433 participants) reported the number of participants completing either the intervention trial or psychological therapies (Analysis 2.3 in [Supplementary material 5](#)) (De Groot 2019; Fetsch 1979; Fremont 1987; Gary 2010; Huang 2015; Karg 2020; Klein 1985). The risk ratio for treatment acceptability was 1.06 (95% CI 0.99 to 1.13; GRADE: moderate), showing that exercise probably results in little to no difference in completion of the intervention compared to psychological therapies.

2.3. Quality of life

Two trials (72 participants) reported changes in quality of life measures, specifically the Minnesota Living with Heart Failure Questionnaire (Gary 2010) and the SF-36 (Huang 2015) (Analysis 2.4 in [Supplementary material 5](#)).

When pooled through meta-analysis, it was shown that exercise may result in a slight increase in the physical domain of quality of life compared to psychological therapy (SMD 0.25, 95% CI -0.21 to 0.72; GRADE: low). In contrast, there was little to no difference between exercise and psychological therapies for the mental domain (SMD 0.05, 95% CI -0.41 to 0.52; GRADE: low). No statistical heterogeneity was evident for either physical ($I^2 = 0\%$, $P = 0.32$) or mental ($I^2 = 0\%$, $P = 0.81$) domains. Only one trial (37 participants) reported a summary quality of life score (Huang 2015), finding that exercise may result in a slight increase in quality of life compared to psychological therapy, although the evidence was very uncertain (SMD 0.37, 95% CI -0.28 to 1.02; GRADE: very low).

2.4 Cost

No trial reported the cost or cost-effectiveness of exercise compared to another intervention.

2.5 Adverse events

One trial by Karg 2020 reported that eight participants dropped out of the exercise intervention group, with one participant reporting physical complaints. Although 12 participants dropped out of the comparator group (CBT), none were identified as an adverse event associated with the comparator intervention (versus CBT).

Comparison 3: exercise versus pharmacological treatments

Six trials (677 participants) assessed the effectiveness of exercise compared to pharmacological treatments (Blumenthal 1999; Blumenthal 2007; Blumenthal 2012a; Brenes 2007; Hidalgo 2021; Khoshnab 2017).

Primary outcome

3.1 Reduction in depressive symptom severity

Post-treatment

For the five trials (330 participants) that compared exercise with pharmacological treatments (Blumenthal 1999; Blumenthal 2007; Blumenthal 2012a; Brenes 2007; Khoshnab 2017) the SMD was -0.11 (95% CI -0.33 to 0.10) (Analysis 3.1 in [Supplementary material 5](#)), indicating that exercise results in little to difference in depressive symptoms compared to pharmacological treatments. No statistical heterogeneity was indicated ($I^2 = 0\%$, $P = 0.69$).

End of long-term follow-up

One study (58 participants) reported long-term follow-up at 6 months (Blumenthal 1999), suggesting that exercise may result in a slight reduction in depressive symptoms compared to pharmacological treatment, although the evidence is very uncertain (MD -0.40 , 95% CI -0.80 to 0.00) (Analysis 3.2 in [Supplementary material 5](#)).

Secondary outcomes

3.2a Acceptability of treatment: attendance at exercise

Four trials reported on attendance rates (Blumenthal 1999; Blumenthal 2007; Blumenthal 2012a; Hidalgo 2021). Three trials reported attendance rates as 82% (Blumenthal 2007), 89.6% (Blumenthal 1999) and 94% (Blumenthal 2012a) for participants receiving exercise alone. Adherence to pharmacological treatment differed among the trials, ranging from 83% (Blumenthal 2007) to 100% achieving acceptable adherence (Blumenthal 1999; Blumenthal 2012a). A fourth trial reported that the mean number of sessions of physical activity attended was 18.9 (Hidalgo 2021).

3.2b Acceptability of treatment: completing the intervention or pharmacological intervention

For participant completion of the trial, there were data from five trials (655 participants) (Blumenthal 1999; Blumenthal 2007; Blumenthal 2012a; Hidalgo 2021; Khoshnab 2017). The risk ratio was 0.92 (95% CI 0.77 to 1.10; GRADE: low) (Analysis 3.3 in [Supplementary material 5](#)).

3.3. Quality of life

One trial (25 participants) by Brenes 2007 showed that the evidence is very uncertain about the benefits of exercise compared to pharmacological treatments on the SF-36 mental health domain (MD -11.90 , 95% CI -24.04 to 0.24 ; GRADE: very low) and physical health domain (MD 1.30 , 95% CI -0.67 to 3.27 ; GRADE: very low) (Analysis 3.4 in [Supplementary material 5](#)).

3.4 Cost

No trial reported the cost or cost-effectiveness of exercise compared to another intervention.

3.5 Adverse events

Blumenthal 1999 reported that three of 53 participants in the exercise group suffered musculoskeletal injuries; five participants from the medication group experienced medication adverse events. Blumenthal 2007 collected data on side effects by asking participants to rate a 36-item somatic symptom checklist and reported that "a few patients reported worsening of symptoms". Of the 36 side effects assessed, only one showed a statistically significant group difference ($P = 0.03$), with the sertraline group reporting worse post-treatment diarrhoea and loose stools (31%) compared to home-based exercise (21%) and supervised exercise (10%). Blumenthal 2012a also assessed the 36 side effects following the approach by Blumenthal 2007. Only two side effects differed significantly between the groups, with the sertraline group having worse post-treatment fatigue (20%) and increased sexual problems (26%) than the exercise group (2.4% for both side effects). Hidalgo 2021 reported that 8.9% of people undertaking the exercise intervention reported adverse events, including osteomuscular pain, mild contusions or dizziness. In the comparator group receiving antidepressant therapy, 22.5% reported adverse events, including dizziness (6.5%), epigastric pain (5.2%) or daytime drowsiness (3.2%) (Hidalgo 2021). The evidence indicates that exercise may result in fewer adverse events than pharmacological treatments.

Comparison 4: exercise versus 'other' comparators

Nine studies (365 participants) compared exercise with one of five other comparator interventions, which were bright light therapy (Pinchasov 2000), reflexology (Eman 2017), occupational or art therapy (Buschert 2019; Martinsen 1985), supplements (Taheri 2018) and health education (Abdelbasset 2019; Bose 2015; Mather 2002; Singh 1997).

Primary outcome

4.1 Reduction in depressive symptom severity

Post-treatment

Separate studies comparing exercise to bright light therapy (SMD -1.48 , 95% CI -2.56 to -0.41 ; 18 participants) (Pinchasov 2000), reflexology (SMD -1.32 , 95% CI -2.01 to -0.63 ; 40 participants) (Eman 2017) and supplements (SMD -1.41 , 95% CI -2.54 to -0.28 ; 16 participants) (Taheri 2018) found that exercise may be more effective than the different comparators (Analysis 4.1 in [Supplementary material 5](#)). Also, exercise may be more effective than health education (SMD -1.16 , 95% CI -1.99 to -0.33 ; 218 participants), although there was considerable statistical heterogeneity ($I^2 = 87\%$, $P < 0.001$) (Abdelbasset 2019; Bose 2015; Mather 2002; Singh 1997). For the two trials (73 participants) that compared exercise with occupational or art therapy (Buschert 2019; Martinsen 1985), the SMD was -0.51 (95% CI -1.75 to 0.73), with considerable statistical heterogeneity ($I^2 = 85\%$, $P = 0.01$), indicating that there may be little difference between the two interventions. In all comparisons, the limited evidence makes the findings very uncertain.

End of long-term follow-up

Three studies (170 participants) reported long-term follow-up for the comparison of exercise with health education ranging from 8 weeks to 26 months (Bose 2015; Mather 2002; Singh 1997), reporting that exercise may provide a comparative benefit (SMD -1.22 , 95% CI -2.63 to 0.19 ; $I^2 = 93\%$, $P < 0.001$) (Analysis 4.2 in [Supplementary material 5](#)). No studies reported long-term follow-up in comparisons with bright light therapy, reflexology, occupational or art therapy, or supplements.

Secondary outcomes

4.2a Acceptability of treatment: attendance at exercise

None of the trials of bright light therapy, reflexology, occupational or art therapy, or supplements reported this outcome. Two of the four trials assessing health education reported attendance rates, varying from 59% for aerobic exercise (Mather 2002) to 93% for high-intensity exercise (Singh 1997).

4.2b Acceptability of treatment: completing the intervention or control

All participants in the trials comparing exercise with reflexology (Eman 2017) and with supplements (Taheri 2018) completed the trial. The trial of bright light therapy did not report on this outcome (Pinchasov 2000). Two trials of occupational or art therapy reported on the acceptability of treatment (Buschert 2019; Martinsen 1985), with a risk ratio of 0.99 (95% CI 0.82 to 1.20) (Analysis 4.3 in [Supplementary material 5](#)). All four trials comparing exercise with health education reported the number of participants randomised and completing the trial (Abdelbasset 2019; Bose 2015; Mather 2002; Singh 1997), reporting a risk ratio of 1.01 (95% CI 0.96 to 1.05) (Analysis 4.3 in [Supplementary material 5](#)).

4.3. Quality of life

None of the trials reported on this outcome.

4.4 Cost

No trial reported the cost or cost-effectiveness of exercise compared to another intervention.

4.5 Adverse events

Only the trial by Singh 1997 comparing exercise with health education reported on adverse events, with one participant undertaking exercise being referred to a psychologist at six weeks due to increasing suicidality, and musculoskeletal symptoms in two participants necessitated adjustment of the training regime.

Subgroup analysis for reduction in depressive symptom severity

Type of exercise

We explored the influence of the type of exercise (aerobic, mixed or resistance) on outcomes for those trials comparing exercise versus control (Analysis 5.1 in [Supplementary material 5](#)). All types of exercise appeared beneficial compared to control, with no evidence of a difference between the subgroups ($P = 0.60$). Forty-eight studies (1937 participants) assessing aerobic exercise found a moderate to large benefit from exercise (SMD -0.64 , 95% CI -0.80 to -0.48), although there was substantial heterogeneity ($I^2 = 62\%$, $P < 0.001$). Mixed exercise was assessed by six studies (139 participants), which when pooled showed a large clinical effect

from exercise compared to control (SMD -0.92 , 95% CI -1.66 to -0.17), with substantial heterogeneity ($I^2 = 73\%$, $P = 0.002$). The three studies (112 participants) of resistance exercise appear to show a comparative benefit from exercise (SMD -0.80 , 95% CI -1.20 to -0.41), although confidence intervals were fairly wide. The effects of heterogeneity were considered not to be important ($I^2 = 0\%$, $P = 0.48$).

Intensity and sessions of exercise

We explored the influence of intensity of exercise (very light, light, moderate, vigorous, near maximum to maximum) on the reduction of depression for those trials comparing exercise versus control (Analysis 5.2 in [Supplementary material 5](#)), finding that there was no difference between the subgroups ($P = 0.72$). Fifty-seven studies making 59 different comparisons at four different levels of intensity (i.e. very light, light, moderate, vigorous), were included in the analyses. One study employed very light intensity exercise, finding limited difference between exercise and control (SMD -0.31 , 95% CI -1.13 to 0.50). Eight studies (357 participants) carried out light-intensity exercise, which had a moderate clinical effect (SMD -0.66 , 95% CI -0.94 to -0.38), with heterogeneity we considered to be moderate ($I^2 = 32\%$, $P = 0.17$). Thirty-one studies (1020 participants) carried out moderate-intensity exercise resulting in a moderate to large clinical effect (SMD -0.71 , 95% CI -0.92 to -0.49); however, the analysis was affected by moderate to substantial heterogeneity ($I^2 = 59\%$, $P < 0.001$). In the 19 studies (819 participants) that carried out vigorous intensity exercise, there was a moderate effect (SMD -0.57 , 95% CI -0.82 to -0.31), with substantial heterogeneity ($I^2 = 65\%$, $P < 0.001$).

We explored the influence of the total number of prescribed exercise sessions (0 to 12, 13 to 24, 25 to 36, 37+ sessions) on the reduction in depressive symptoms for those trials comparing exercise versus control (Analysis 5.3 in [Supplementary material 5](#)). Although exercise provided greater benefit compared to the control interventions, the extent of the difference in effect did not vary significantly between the subgroups ($P = 0.32$). Ten studies (373 participants) providing up to 12 sessions of exercise found a moderate beneficial effect compared to control (SMD -0.52 , 95% CI -1.09 to 0.04), with the meta-analysis affected by substantial statistically significant heterogeneity ($I^2 = 84\%$, $P < 0.001$). Increasing the number of sessions to between 13 and 36 sessions increased the benefit from exercise when compared to control. In the 20 studies (741 participants) where participants received 13 to 24 sessions (SMD -0.76 , 95% CI -0.96 to -0.56) and the 11 studies (363 participants) where participants received 25 to 36 sessions (SMD -0.72 , 95% CI -1.14 to -0.30), there appeared to be large benefits from exercise compared to control. In both cases, the meta-analyses were affected by heterogeneity, although this was moderate for 13 to 24 sessions ($I^2 = 39\%$, $P = 0.04$) and substantial for 25 to 36 sessions ($I^2 = 72\%$, $P < 0.001$). People receiving 37 or more sessions were assessed in 12 studies (559 participants), and a moderate benefit from exercise compared to control was found (SMD -0.48 , 95% CI -0.68 to -0.27), with levels of heterogeneity that we considered unimportant ($I^2 = 25\%$, $P = 0.20$). In four studies (152 participants), the number of sessions provided was unclear. When these four studies were pooled, benefit was found from exercise compared to control (SMD -0.77 , 95% CI -1.11 to -0.44), with unimportant levels of heterogeneity ($I^2 = 0\%$, $P = 0.88$).

Type of diagnosis

We performed subgroup analyses according to how the diagnosis of depression was made, whether it was a clinical diagnosis or categorised according to cut points on a scale or where it was unclear (Analysis 5.4 in [Supplementary material 5](#)). We found that there was no difference between the subgroups ($P = 0.72$). In the meta-analysis of the 35 trials (1449 participants) where depression was diagnosed clinically, a moderate-to-large benefit was found for exercise versus control (SMD -0.63 , 95% CI -0.81 to -0.44), although the analysis was affected by substantial heterogeneity ($I^2 = 62\%$, $P < 0.001$). Where trials used cut-points on a scale (18 trials, 616 participants), the pooled effect showed a moderate-to-large benefit favouring exercise relative to control (SMD -0.73 , 95% CI -0.97 to -0.49); the meta-analysis was characterised by moderate heterogeneity ($I^2 = 48\%$, $P = 0.01$). The method of diagnosis was unclear in four trials (123 participants). When pooled, the four trials showed uncertain effect (SMD -0.88 , 95% CI -1.88 to 0.12), with the analysis affected by considerable heterogeneity ($I^2 = 84\%$, $P < 0.001$).

Type of control

We categorised the type of control as either: placebo; no treatment, waiting list, usual care or self-monitoring; exercise plus treatment versus treatment alone; stretching, meditation or relaxation; or, social/casual conversation (Analysis 5.5 in [Supplementary material 5](#)). In the previous update, the categorisations were made by one author (KD) on the basis of data extracted at the initial stage of data extraction, and were checked by a second author (GM). In the current update, all the additional studies were classified by JH, AC, DM, CH, CS, CL. Although exercise was beneficial in reducing depressive symptoms compared to each of the different types of control, the difference between the subgroups was not significant ($P = 0.13$). When compared with the 'no treatment, waiting list, usual care or self-monitoring' comparator in 27 studies (897 participants), it was shown that exercise may result in a large reduction in depressive symptoms (SMD -0.85 , 95% CI -1.06 to -0.64 ; GRADE: low), with the meta-analysis affected by heterogeneity that we considered moderate ($I^2 = 51\%$, $P = 0.001$). The meta-analyses of 19 studies (812 participants) comparing exercise plus treatment versus treatment only (SMD -0.57 , 95% CI -0.83 to -0.30 ; GRADE: very low) and the 11 studies (385 participants) comparing exercise with stretching, meditation or relaxation (SMD -0.48 , 95% CI -0.80 to -0.15 ; GRADE: very low) found that exercise may result in a reduction in depressive symptoms, but the evidence is very uncertain. In both instances, the meta-analyses were affected by moderate to substantial heterogeneity (exercise plus treatment with treatment only $I^2 = 68\%$, $P < 0.001$; exercise with stretching, meditation or relaxation $I^2 = 54\%$, $P = 0.02$). Comparison of exercise with 'social/casual conversation' in one study (29 participants) (SMD -0.48 , 95% CI -1.23 to 0.27 ; GRADE: very low) and with 'placebo' in three studies (172 participants) (SMD -0.21 , 95% CI -0.81 to 0.40 ; GRADE: very low) showed that exercise may result in a reduction in depressive symptoms, but the evidence is very uncertain, with the comparison with placebo affected by substantial heterogeneity ($I^2 = 66\%$, $P = 0.05$). We considered whether to perform additional subgroup analyses according to supervised versus unsupervised exercise, indoor versus outdoor exercise and individual versus group exercise, and according to the type of control group (in the no-treatment comparison), but we wanted to focus on the main

subgroups of interest. Multiple subgroup analyses are generally considered inadvisable.

Sensitivity analysis

We conducted sensitivity analyses to explore the impact of study quality factors on the pooled effect size, comparing findings with the outcome from the meta-analysis of all 57 studies (Analysis 1.1 in [Supplementary material 5](#)). The characteristics of interest included the type of publication, allocation concealment, use of intention-to-treat analysis and blinding of outcomes assessment.

Peer-reviewed journal publications

For the 56 trials (2169 participants) that were reported in peer-reviewed journal publications or doctoral theses, the SMD was -0.65 (95% CI -0.80 to -0.51), showing a moderate clinical effect in favour of exercise, with substantial heterogeneity ($I^2 = 60\%$, $P < 0.001$) (Analysis 6.1 in [Supplementary material 5](#)).

Published as abstracts/conference proceedings only

Only one study (18 participants) was published as a conference abstract. It showed benefit from exercise (MD -13.46 , 95% CI -19.37 to -7.55) (Analysis 6.2 in [Supplementary material 5](#)).

Allocation concealment

For the 17 trials (916 participants) with adequate allocation concealment and therefore at low risk of bias, the SMD was -0.51 (95% CI -0.72 to -0.30) (Analysis 6.3 in [Supplementary material 5](#)), showing a moderate clinical effect in favour of exercise. Although this did not show a large difference from the pooled SMD for all 57 trials of exercise versus control (SMD -0.67 , 95% CI -0.82 to -0.52) (Analysis 1.1 in [Supplementary material 5](#)), it was lower. The meta-analysis of the 17 trials with adequate concealment was affected by substantial heterogeneity ($I^2 = 57\%$, $P = 0.002$).

Use of intention-to-treat analysis

For the 21 trials (994 participants) undertaking an intention-to-treat analysis, the pooled SMD of -0.77 (95% CI -1.05 to -0.50) showed some benefit in favour of exercise (Analysis 6.4 in [Supplementary material 5](#)), which was larger than that shown in the analysis of all 57 trials (Analysis 1.1 in [Supplementary material 5](#)). Again, the meta-analysis was affected by substantial heterogeneity ($I^2 = 74\%$, $P < 0.001$).

Blinded outcome assessment

For the 21 trials (998 participants) with blinded outcome assessments, the SMD was -0.53 (95% CI -0.73 to -0.33) (Analysis 6.5 in [Supplementary material 5](#)), showing a moderate effect in favour of exercise. The SMD was smaller than the pooled SMD for all 57 trials. It was affected by a substantial level of heterogeneity ($I^2 = 54\%$, $P = 0.002$).

Allocation concealment, intention-to-treat analysis and blinded outcome assessment

For the seven trials (447 participants) with adequate allocation concealment and intention-to-treat analysis and blinded outcome assessment, and therefore at low risk of bias, the SMD was -0.46 (95% CI -0.88 to -0.04) (Analysis 6.6 in [Supplementary material 5](#)), with substantial heterogeneity ($I^2 = 78\%$, $P < 0.001$). This moderate effect in favour of exercise was smaller than that shown in the

pooled SMD for all 57 trials (Analysis 1.1 in [Supplementary material 5](#): SMD -0.67, 95% CI -0.82 to -0.52).

Lowest dose

We assessed the effects of the dose of exercise on the outcome of reduction in depressive symptoms. Where trials included multiple arms with different doses, we included the arm with the lowest dose of exercise.

Thirteen trials included more than one dose of exercise (Belvederi Murri 2015; Blumenthal 2007; Chu 2008; Doyne 1987; Dunn 2005;

Heinzel 2022; Krogh 2009; Mutrie 1988; Orth 1979; Setaro 1985; Singh 2005; Williams 2008; Yu 2023), with the largest dose replaced with the lowest dose for this sensitivity analysis (Analysis 6.7; [Figure 8](#); [Supplementary material 5](#)). The effect was to slightly reduce the SMD from -0.67 (95% CI -0.82 to -0.52) to -0.60 (95% CI -0.74 to -0.46), showing a similar moderate benefit from exercise. The analysis was affected by moderate heterogeneity ($I^2 = 55\%$, $P < 0.001$). It is likely that this reflects the limited number of studies in which a lower dose was compared, as well as the difference in the type of exercise and extent of difference in the dose.

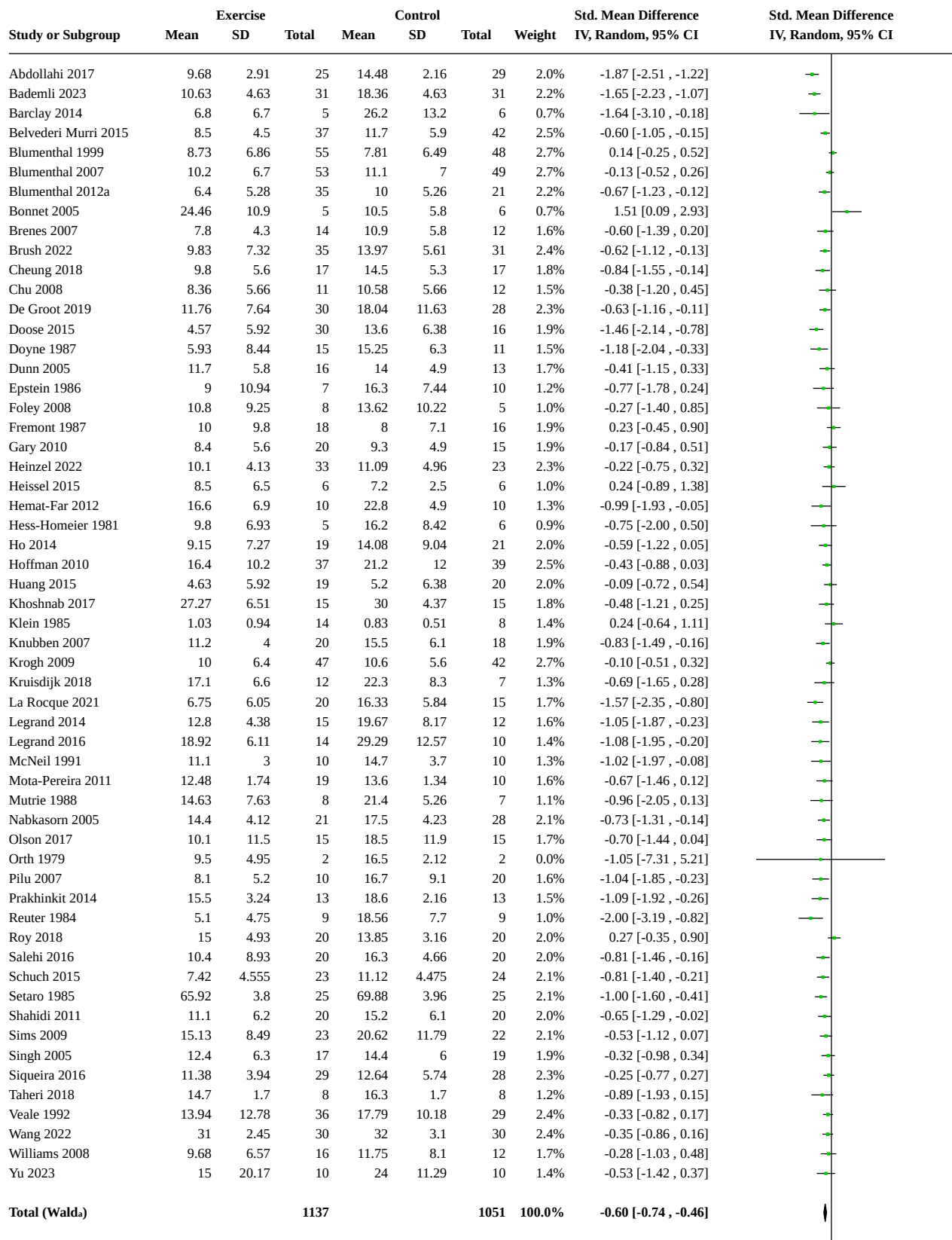
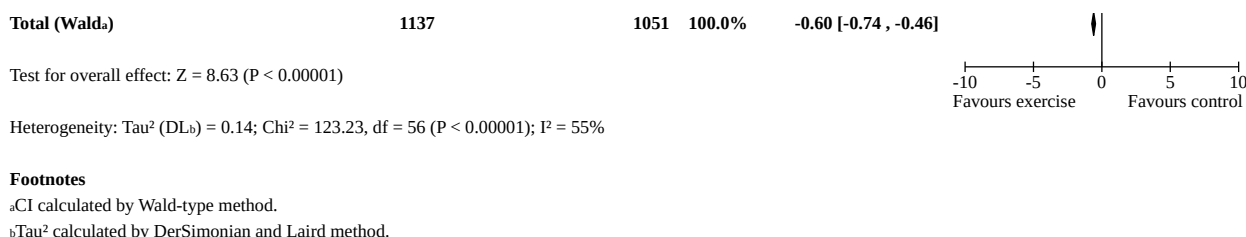
Figure 8. Forest plot of comparison: 9 Exercise versus control: sensitivity analyses, outcome: 9.7 Reduction in depression symptoms post-treatment: lowest dose of exercise

Figure 8. (Continued)



Recruitment and retention of participants

Table 1 presents data about the feasibility of recruiting and retaining participants both in the trial as a whole and in the exercise intervention in particular. We extracted data, when available, about the number of participants who were considered for inclusion in each trial, although this information was not available for all trials. The trials that did provide these data used different recruitment techniques (ranging from screening of people responding to advertisements to inclusion of those patients who were considered eligible by a referring doctor).

Equity assessment

We did not undertake a formal equity assessment.

Reporting biases

A degree of publication bias was thought to be evident from a visual inspection of the funnel plot (Egger regression intercept $P = 0.00365$) [689] (funnel plot for Analysis 1.1 - exercise versus control (Figure 4; Supplementary material 5)). A comparison of the estimates from the meta-analysis in Analysis 1.1 (SMD -0.67 (95% CI -0.82 to -0.52)) with that from using Duval and Tweedie's trim and fill (SMD -0.45 , 95% CI -0.62 to -0.28) indicates that small studies favouring control intervention may be missing, and, when adjusted, reduces the benefit from exercise [690]. Data for the review can be found in Supplementary material 6.

DISCUSSION

Summary of main results

This updated review includes 35 additional trials (2526 additional participants), with conclusions that are similar to the previous version of the review [108]. The pooled standardised mean difference (SMD) for depression (measured by continuous variable), at the end of treatment, represented a moderate clinical effect, suggesting that exercise may result in a reduction in depressive symptoms. We assessed the certainty of the evidence contributing to this outcome as low. The evidence was very uncertain for results at long-term follow-up. The evidence suggested there was little to no difference between exercise and psychological therapy or pharmacological treatment in the reduction of depressive symptoms.

There was some variation between studies with respect to attendance rates for exercise as an intervention, suggesting that there may be factors that influence the acceptability of exercise amongst participants. Treatment acceptability seemed similar between interventions when measured by participants' completion of the studies.

For the secondary outcome of quality of life, the evidence was more limited, restricting the conclusions that could be drawn. Likewise, the evidence was limited for the adverse events of exercise though it suggested they were experienced by only a small number of participants. Ideally, all the risks and benefits of exercise for depression should be evaluated in future trials.

There were no data on costs or cost-effectiveness associated with exercise as an intervention, so we are unable to comment on the cost-effectiveness of exercise for depression.

Some uncertainties remain regarding how effective exercise is for improving mood in people with depression, primarily due to methodological shortcomings in the studies in this review (see Limitations of the evidence included in the review below). Although it appears that exercise may result in a reduction in depressive symptoms, determining the optimum type, frequency and duration of exercise, and whether it should be performed supervised or unsupervised, indoors or outdoors, or in a group or alone is less certain.

Despite a degree of uncertainty, some patterns were evident. People doing moderate-intensity exercise reported larger benefits when compared with those doing either light or vigorous exercise. The greatest benefit was shown for people who received between 13 and 36 sessions of exercise; this benefit was reduced to a moderate effect for those who undertook fewer sessions or more sessions. Mixed or resistance exercise appeared more effective than aerobic exercise.

Importantly, both the interventions and comparators may not be as clearly defined in the included studies as they should be, particularly in terms of what other active interventions that might be received, whether as usual care or part of the intervention or comparator. Although their provision represents a pragmatic approach to treating participants, it may influence the comparative effectiveness of the comparisons.

Limitations of the evidence included in the review

Although the amount of evidence assessing the effectiveness of exercise compared with other interventions has increased since the last version of the review, most of the trials continue to include small samples and to be affected by a number of methodological weaknesses that may influence the interpretation of the evidence.

Study limitations/risk of bias

Our assessment of the risk of bias in the included studies indicated that most are at unclear or high risk of bias across many domains. We were directed by CCDAN that all studies should be classified as

high risk of bias for blinding of participants and those delivering the interventions, reflecting that it is almost impossible to blind to treatment allocation, meaning that all studies were at high risk of performance and detection bias. Where self-reported outcomes or clinician-rated outcomes are used (e.g. Hamilton Rating Scale for Depression), they may overestimate the effect sizes. Blinding of outcome assessors was also unclear or at high risk of bias for two-thirds of the trials. Selection bias was a concern, with random sequence generation and allocation concealment either unclear or at high risk of bias in 58% and 73% of included trials, respectively. Forty-two percent of trials were at low risk of attrition bias due to incomplete outcome data. Reporting bias was a concern due to selective reporting of outcomes in 63% of trials. Some 89% of trials were either at unclear or high risk of other biases.

With between 58% and 89% trials being at unclear or high risk of the different biases, caution should be taken when interpreting the results. We explicitly aimed to determine the influence of study quality, in particular allocation concealment, blinding and intention-to-treat analysis on effect sizes, as we had done in previous versions of the review [32, 99, 102, 108]. When we included only those trials with adequate allocation concealment and intention-to-treat analysis and blinding of outcome assessors, the effect size was reduced from moderate (Figure 5) to small (see analysis 6.6 in [Supplementary material 5](#)). There was substantial heterogeneity, which might be explained by a number of factors, including variation in the control interventions.

Consistency of effect

Despite most meta-analyses showing that exercise was as effective or more effective than the different comparators for treating depression, the results were often affected by substantial heterogeneity (i.e. $I^2 = 50\%$ to 90%), affecting the certainty of the results. Subgroup analysis and sensitivity analysis rarely indicated causes for such variability. It is evident that the inclusion criteria for our review were broad, allowing studies with diverse participants, interventions, comparators and sources of potential bias to be included. This may cause some uncertainty around applying the results in other situations.

Imprecision

As the amount of evidence evaluating the effectiveness of exercise for treating depression has increased, estimates have tended to become more precise. Imprecision was evident for the outcomes associated with quality of life, long-term follow-up (pharmacological treatments) and in subgroup analysis by exercise type, number of sessions and type of control, where there were fewer studies, many of small size, and where confidence intervals were wide. Limited evidence of the benefits of exercise compared to bright light therapy (Analysis 4.1), reflexology (Analysis 4.1), occupational or art therapy (Analysis 4.1), supplements (Analysis 4.1), and health education (Analysis 4.3), rendered these comparisons inconclusive (see [Supplementary material 5](#)). Additional trials would be required to provide any certainty around the relative effects.

Although subgroup analyses showed that effect sizes were higher for mixed exercise and resistance exercise than for aerobic exercise alone (Analysis 5.1), the limited evidence base (small number of studies and small sample sizes) for mixed exercise (6 trials, 139 people) and resistance exercise (3 trials, 112 people) resulted in

wide confidence intervals compared to that for aerobic exercise (48 trials, 1937 people). In addition, the mixed exercise group effect size did not reach the minimal clinically important difference. It was evident that light-intensity and moderate-intensity exercise were more effective than vigorous exercise, although only a small number of trials evaluated light-intensity exercise (8 trials, 357 people) (Analysis 5.2). Despite the limited difference in the benefits from different numbers of exercise sessions, it appeared that between 13 and 36 sessions was optimal. Some uncertainty was evident due to the wide confidence intervals and considerable heterogeneity that affected the analysis of 0 to 12 sessions and 25 to 36 sessions (Analysis 5.3). See [Supplementary material 5](#). Issues around imprecision reduced our confidence when interpreting the results.

Indirectness

Our review had a broad focus on any adult classified by the trialists as having depression through any method of diagnosis (either by a cut-off score on a depression scale or by having a clinical diagnosis of depression) and of any severity, who were willing to participate in a programme of regular physical exercise that fulfilled the American College of Sports Medicine (ACSM) definition. Although the focus of the research question was broad, which may have resulted in issues of heterogeneity, some areas were excluded that could be considered. The effects of exercise on anxiety, neurotic disorders, dysthymia and postnatal depression were not included; these topics should be the focus of other reviews. Dysthymia (i.e. low mood not fulfilling the diagnostic criteria for depression) may be more difficult to differentiate from depression itself in real-world situations, meaning that some loss of evidence may have occurred. The data we extracted on aspects of feasibility (see [Table 1](#)) suggest that a large number of people need to be screened to identify suitable participants, unless recruitment is from a clinical population (e.g. inpatients with depression). There was a wide range in the proportion of those screened who were subsequently randomised; this may be a function of the sampling frame (which may include a range of specifically screened or non-screened potential participants), and interest in being a research participant at a time of low mood, as much as whether potential participants are interested in exercise as a therapy. A substantial number of people dropped out of both the exercise and control programmes, and even those who remained in the trial until the outcome assessments were not able to attend all exercise sessions. As such, it may be that only the most motivated of individuals were included in these trials. In previous versions of the review, the review authors decided to exclude trials that included people both with and without depression, even if they reported data from a subgroup with depression, which may be perceived to influence the findings of the review. As a consequence, we excluded a large, high-quality, cluster-randomised trial recruiting 891 residents from 78 nursing homes (Underwood 2013), of whom 375 had baseline Geriatric Depression Scores suggesting depression. At the end of the treatment, there was no difference between the intervention and control group, for people both with and without depression at baseline. It was evident that the included studies recruited more women than men and there was a wide range in mean ages, influencing the generalisability of the findings.

Interventions were clearly defined as 'planned, structured and repetitive movement done to improve or maintain one or more components of physical fitness'. Again, this is a relatively broad definition, although we excluded some forms of 'exercise' such

as yoga, qigong and stretching due to the fact that they were interpreted as having other components (e.g., mental processes) that may contribute additional or synergistic effects to those provided by exercise itself. Importantly, these other forms of activity do not meet the ACSM definition of exercise. It may be argued that these are artificial divides and such forms of exercise should have been incorporated. Moreover, we did not include trials in which advice was given to increase activity or on exercise referral schemes for depression [691, 692, 693]. As a result, we excluded a large, high-quality trial ($n = 361$) in which people with depression in primary care were randomised to usual care or to usual care plus advice from a physical activity facilitator to increase activity (Chalder 2012), which showed no effect of the intervention on mood.

The primary outcome used in the studies, specifically measures of depression or mood, seemed to be well-defined and of direct relevance. We did not attempt to take into account the effects of exercise when the experience is pleasurable and self-determined, though this would have been difficult as such data were not reported in the trials. Although we assessed the effects of exercise on outcomes such as quality of life, adverse events or cost-effectiveness, the effects remain uncertain as most trials did not systematically report the information.

With most of the evidence considering the effectiveness of evidence versus 'control' and a more limited number of trials assessing exercise versus psychological or pharmacological treatments, it may be helpful to produce a network meta-analysis to allow direct and indirect comparisons to be incorporated, although the nature of the evidence base may lead to issues around heterogeneity and incoherence.

Arguably, the review could be broader in terms of the people, interventions and outcomes included, but we have elected to keep it more focused, partly to ensure that it remains feasible to update the review on a regular basis, with the resources we have available. The original review questions were conceived more than 20 years ago [102], and although they are still relevant today, it would be of value to broaden the research questions to include evidence for other modes of physical activity in different people. This could be through a series of related Cochrane reviews.

Publication bias

For this update, we searched several different sources for relevant studies. Initially, CCDAN Group's trial register (CCDANCTR) was searched in November 2013. CCDANCTR was a comprehensive source of trials up until it was last updated in April 2013; when it was no longer updated, we searched MEDLINE, Embase and PsycINFO up to November 2023. We also searched Current Controlled Trials, ClinicalTrials.gov and the WHO International Clinical Trials Registry Platform (ICTRP) up to September 2025 to identify ongoing trials. We scrutinised reference lists of the new trials identified. Ideally, we would have performed citation reference searches of all included studies, but with the large number of trials now in this review, this was no longer practical. As a consequence, it is possible that we may have missed some relevant trials. It is evident from the asymmetrical funnel plot suggesting small-study bias that some studies may have been missed or not published (Figure 4).

Exercise for depression remains a topic of considerable interest to researchers, with new trials and systematic reviews continuing to

emerge. It also remains important to policymakers, who would like clarity around its effectiveness and about whether specific types, durations, intensities or doses are most effective. The increase in evidence presented in the current version of the review highlights that additional evidence has added little to the overall conclusion of the previous review. If further research is to be undertaken, it should focus more on the specific aspects of exercise (i.e. types, durations, intensities, doses) and for whom it is most effective for managing depression. If this review is updated again, it should concentrate on those aspects.

Limitations of the review processes

The current version of the review has adhered to the research protocol agreed for previous versions of the review with the CCDAN group, with no opportunity to amend. The protocol specified that activities, such as yoga, stretching, relaxation and qigong, were not forms of exercise as they were either combined with other interventions (i.e. incorporating physical, mental and spiritual elements) or involved too low a dose of exercise. As such, these were considered as comparators. This may be a limitation in this review, given it could be argued that these are, in fact, exercise and may impact the comparison between exercise and non-exercise interventions. A similar limitation of the original protocol was the use of a comparator (i.e. 'control' group) that encompassed people who may have received an active intervention (e.g. usual care) and those who were inactive (e.g. no treatment), which may have affected the estimate of effect.

We attempted to avoid bias by ensuring that we had identified all relevant studies through comprehensive systematic searching of the literature and checking the reference lists of relevant studies. However, we accept that some publication bias is likely and this is indicated by the asymmetrical funnel plot (Figure 4). This is likely to lead to an overestimate of effect sizes, because positive trials are more likely to be published than negative trials. The searches for this and previous updates may have been less extensive than for the initial review in 2001 [102]. As the CCDAN register of trials was updated regularly from many different sources, and we translated searches into a broad range of databases, we think it is unlikely that we have missed relevant trials.

Although there is considerable interest in the continued development of a robust and accurate evidence base in this field to guide practice and healthcare investment, the addition of 35 trials in this updated version of the review has had a limited effect on the estimate of effectiveness compared to control (SMD -0.67 , 95% CI -0.82 to -0.52 ; $I^2 = 61\%$) compared to the previous version (SMD -0.62 , 95% CI -0.81 to -0.42 ; $I^2 = 63\%$). When compared with other interventions (i.e. pharmacological treatments, psychological therapies), additional trials have had a similarly limited effect on the estimates of effectiveness in previous versions of the review, bringing into question the value of further trials.

For a previous version of this review [99], we made post hoc decisions to exclude trials defined as 'combination' interventions, a trial in which the exercise intervention lasted only four days (Berlin 2003), and trials of postnatal depression (Armstrong 2003; Armstrong 2004). For the update in 2012 [108], we specified in advance that we would exclude trials that did not fulfil the ACSM criteria for exercise [46]; this meant that we excluded two studies that had previously been included (Chou 2004; Tsang 2006).

In previous versions of the review, we used data from the trial arm with the largest clinical effect; this approach could have biased the results in favour of exercise. For this update, we used the largest 'dose' of exercise and performed a sensitivity analysis to determine the effect of using the smaller 'dose' (Figure 8; Supplementary material 5). This showed that the effect size was very similar for both the lower and higher doses (-0.60 for the lower dose and -0.67 for the higher dose). We performed several subgroup analyses, which, by their nature, are simply observational. In some instances, there was little evidence evaluating a particular subgroup (e.g. exercise type: mixed exercise, resistance exercise; control: placebo), which may cause some uncertainty as to the outcome. We classified heterogeneity as substantial in many of our main analyses (see Supplementary material 5 - analyses 1.1, 1.2, 1.4, 3.3, 4.1, 4.2, 5.1 to 5.5 and 6.1 to 6.7). This may reflect the diverse participants, interventions, comparators and risks of bias, resulting in further uncertainty about these analyses.

Agreements and disagreements with other studies or reviews

A recent umbrella review and three recent systematic reviews have similarly concluded that exercise is an effective treatment for depression with a moderate size of effect [87, 92, 694, 695]. A fourth review found a larger effect, but the authors reduced this to a moderate effect after conducting sensitivity analyses excluding studies with a high risk of bias [89].

Two systematic reviews with network meta-analyses (one comparing exercise with conventional psychotherapies [696] and another comparing exercise with antidepressants [96]) concluded that exercise has a moderate positive effect compared to no treatment or treatment-as-usual, but that treatment effects are similar and not superior to behavioural activation, cognitive behavioural therapy or antidepressants. A similar conclusion was reached in a narrative review that suggests that exercise as a stand-alone treatment is effective in reducing symptoms of depression compared to control groups but is not more effective than active treatments such as antidepressants or cognitive behavioural therapy [697]. A meta-analysis that only included RCTs of participants with a diagnosis of major depressive disorder who were not receiving any other treatment did not find a statistically significant difference in effect between exercise interventions and various comparators [698].

A network meta-analysis comparing different types of exercise found that walking or jogging, yoga, strength training, and dancing were more effective than other exercises and that the benefits increased with exercise intensity [94]. Another network meta-analysis comparing aerobic, resistance, mixed exercise or mind-body exercise found that all categories of exercise were effective in reducing depressive symptoms, but aerobic and mind-body exercises were slightly more beneficial than mixed or resistance exercise [699]. A systematic review of strength training interventions found a positive moderate-size effect on symptoms of depression [700]. However, a review of resistance training exercise was inconclusive and suggested that more well-designed trials of this type of exercise are needed to assess its efficacy [701].

AUTHORS' CONCLUSIONS

Implications for practice

Like previous systematic reviews and meta-analyses, our review found that exercise may result in a reduction in depressive symptoms. Our review also found that people undertaking light or moderate intensity exercise may have a larger benefit than those undertaking vigorous exercise, and that between 13 and 36 sessions of exercise may be more beneficial than either fewer or more sessions. We are less certain which type of exercise is most beneficial. Long-term follow-up was lacking in most of the studies, which means that the duration of any benefits is unclear, although those reported appeared to decline over time following completion of the exercise programme.

The evidence also suggests that exercise is probably as effective as psychological therapies and may be as effective as pharmacological treatments, but the number of trials reporting these comparisons was small, as was the number of participants randomised.

Equity-related implications for practice

The review does not directly address equity issues, partly due to the initial framing of the review in its first and subsequent versions, but also due to the lack of consideration of this aspect within the included trials. This is an issue that is starting to be addressed in trials and this should come through within systematic reviews. Most trials in the review focused on participants with depression, without ensuring that there was representation of different groups within the population (other than specific age ranges or sex groups), which makes it difficult to make any meaningful assessment of equity issues.

Implications for research

This review would be strengthened by additional large-scale high-quality studies where all participants at the time of recruitment were diagnosed as having depression through clinical interview, and where they adhered closely to an exercise regimen as a sole intervention and were further assessed through diagnostic clinical interview post-intervention. Participants should come from a broad range of backgrounds to ensure studies are not selective and findings are generalisable. Studies should explore which types, intensities and number of exercise sessions are effective for different types of participants. It would also be worth considering whether any long-lasting effects of exercise correlate with sustained increases in physical activity over time. Given the use of psychological therapies and pharmacological treatments, further research should be undertaken comparing these interventions with exercise. In addition, economic evaluations should routinely be undertaken as part of trials to assess cost-effectiveness. Although further high-quality focused research would improve the evidence base, it is not clear if such RCTs would be commissioned. This concern is based on the lack of improvement in the quality of the RCTs found for this review over the last 12 years and the limited change in the estimates of effect that has come from doubling the number of trials included in this systematic review.

A future update of the current review will only be valuable if large-scale and high-quality trials are conducted, and if these trials give consideration to equity issues.

Equity-related implications for research

Equity should be a focus of any future update of this review. It will necessitate a re-examination of the included trials to identify any consideration of equity in them. Given the limited consideration of equity in trials to date, any future trials should address equity issues and funders should make this a key part of commissioning research.

SUPPLEMENTARY MATERIALS

Supplementary materials are available with the online version of this article: [10.1002/14651858.CD004366.pub7](https://doi.org/10.1002/14651858.CD004366.pub7).

Supplementary material 1 Search strategies

Supplementary material 2 Characteristics of included studies

Supplementary material 3 Characteristics of excluded studies

Supplementary material 4 Characteristics of ongoing studies

Supplementary material 5 Analyses

Supplementary material 6 Data package

Supplementary material 7 Characteristics of included studies

ADDITIONAL INFORMATION

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Professor Gillian E Mead: developed a course on exercise after stroke, which is licensed to Later Life Training. She receives royalty payments from Later Life Training, which are paid into an account at the University of Edinburgh to support further research. She personally receives royalties from a book about Exercise and

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Data, code and other materials

As part of the published Cochrane review, the following are made available for download for users of the Cochrane Library (see [Supplementary material 6](#)): full search strategies for each database; full citations of each unique report for all studies (included, ongoing or awaiting classification, or excluded at the full text screen) in the final review; study data, including study information, study arms, and study results or test data; consensus risk of bias assessments; and analysis data, including overall estimates and settings, subgroup estimates, and individual data rows. Appropriate permissions have been obtained for such use. Analyses and data management were conducted within Cochrane's authoring tool, RevMan, using the inbuilt computation methods.

Date	Event	Description
8 January 2026	New search has been performed	A new authorship team using the same methods as previous versions has updated the review. The new searches were conducted up to November 2023. We included 35 additional studies (2526 people) in the review, bringing the total to 73 studies (4985 people).
8 January 2026	New citation required but conclusions have not changed	Despite the new evidence included in the review, there was limited change in the outcomes, and the review's conclusion that exercise provides a moderate beneficial effect compared to control remains the same.

History

Protocol first published: Issue 3, 2003

Review first published: Issue 4, 2008

Date	Event	Description
2 May 2013	New citation required but conclusions have not changed	New studies incorporated
2 May 2013	New search has been performed	New search performed
12 November 2009	Feedback has been incorporated	Feedback received from a trialist received 3 November 2009 concerning the DOSE 2002 study was addressed on 10 November 2009
13 May 2009	New citation required but conclusions have not changed	Incorrect 'date assessed as up to date' changed from February 2000 to March 2007 (date of last electronic searches). Abstract corrected to reflect true history of searches

Exercise for depression (Review)

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Date	Event	Description
13 August 2008	New search has been performed	This is an updated version of a previous review published in the <i>BMJ</i> in 2001 and includes several new trials
30 July 2008	Amended	Converted to new review format
21 February 2000	New citation required and conclusions have changed	Substantive amendment

REFERENCES

1. World Health Organisation. Depressive disorder (depression). <https://www.who.int/news-room/fact-sheets/detail/depression> (accessed 1 November 2023).
2. GBD 2019 Mental Disorders Collaborators. Global, regional, and national burden of 12 mental disorders in 204 countries and territories, 1990-2019: a systematic analysis for the Global Burden of Disease Study 2019. *Lancet. Psychiatry* 2022;**9**(2):137-50. [DOI: [10.1016/S2215-0366\(21\)00395-3](https://doi.org/10.1016/S2215-0366(21)00395-3)]
3. World Health Organization. Depression and other common mental disorders. Global health estimates. https://www.who.int/mental_health/management/depression/prevalence_global_health_estimates/en/ 2017.
4. Zhang Y, Jia X, Yang Y, Sun N, Shi S, Wang W. Change in the global burden of depression from 1990-2019 and its prediction for 2030. *Journal of Psychiatric Research* 2024;**178**:16-22. [DOI: [10.1016/j.jpsychires.2024.07.054](https://doi.org/10.1016/j.jpsychires.2024.07.054)]
5. World Health Organisation. COVID-19 pandemic triggers 25% increase in prevalence of anxiety and depression worldwide. <https://www.who.int/news/item/02-03-2022-covid-19-pandemic-triggers-25-increase-in-prevalence-of-anxiety-and-depression-worldwide> 2022.
6. Moussavi S, Chatterji S, Verdes E, Tandon A, Patel V, Ustun B. Depression, chronic diseases, and decrements in health: results from the World Health Surveys. *Lancet* 2007;**370**(9590):808-9.
7. Seymour J, Benning TB. Depression, cardiac mortality and all-cause mortality. *Advances in Psychiatric Treatment* 2009;**15**(2):107-13.
8. Agudelo-Botero M, Giraldo-Rodríguez L, Rojas-Russell M, González-Robledo MC, Balderas-Miranda JT, Castillo-Rangel D, et al. Prevalence, incidence and years of life adjusted for disability due to depressive disorders in Mexico: results of the Global Burden of Disease Study 2019. *Journal of Affective Disorders Reports* 2021;**6**:-. [DOI: [10.1016/j.jadr.2021.100206](https://doi.org/10.1016/j.jadr.2021.100206)]
9. World Health Organisation. Suicide worldwide in 2021. Global health estimates. <https://www.who.int/publications/i/item/9789240110069> 2021.
10. National Institute for Health and Clinical Excellence. Depression in adults: recognition and management. <http://www.nice.org.uk/guidance/CG90> 2009.
11. American Psychiatric Association. Diagnostic and Statistical Manual of Mental Disorders, Fifth Edition. Arlington: Arlington, 2013. [DOI: [10.1176/appi.books.9780890425596](https://doi.org/10.1176/appi.books.9780890425596)]
12. Arroll B, Elley CR, Fishman T, Goodyear-Smith FA, Kenealy T, Blashki G, et al. Antidepressants versus placebo for depression in primary care. *Cochrane Database of Systematic Reviews* 2009, Issue 3. Art. No: CD007954. [DOI: [10.1002/14651858.CD007954](https://doi.org/10.1002/14651858.CD007954)]
13. Linde K, Kriston L, Rucker G, Jamil S, Schumann I, Meissner K, et al. Efficacy and acceptability of pharmacological treatments for depressive disorders in primary care: systematic review and network meta-analysis. *Annals of Family Medicine* 2015;**13**(1):69-79.
14. Geddes JR, Carney SM, Davies C, Furukawa TA, Kupfer DJ, Frank E, et al. Relapse prevention with antidepressant drug treatment in depressive disorders: a systematic review. *Lancet* 2003;**361**(9358):653-61.
15. NICE. Depression in adults: treatment and management. <https://www.nice.org.uk/guidance/ng222> 2022.
16. Ferguson JM. SSRI antidepressant medications: adverse effects and tolerability. *Primary Care Companion to the Journal of Clinical Psychiatry* 2001;**3**(1):22-7.
17. Kelly K, Posternak M, Alpert JE. Toward achieving optimal response: understanding and managing antidepressant side effects. *Dialogues in Clinical Neuroscience* 2008;**10**(4):409-18.
18. Andrews PW, Thomson JA, Amstadter A, Neale MC. Primum non Nocere: an evolutionary analysis of whether antidepressants do more harm than good. *Frontiers in Psychology* 2012;**24**(3):1-19.
19. Oliva V, Lippi M, Paci R, Del Fabro L, Delvecchio G, Brambilla P, et al. Gastrointestinal side effects associated with antidepressant treatments in patients with major depressive disorder: a systematic review and meta-analysis. *Progress in Neuro-Psychopharmacology & Biological Psychiatry* 2021;**109**(110266):- . [DOI: [10.1016/j.pnpbp.2021.110266](https://doi.org/10.1016/j.pnpbp.2021.110266)]
20. Machado-Vieira R, Salvatore G, Luckenbaugh DA, Manji HK, Zarate CA. Rapid onset of antidepressant action: a new paradigm in the research and treatment of major depressive disorder. *Clinical Psychiatry* 2008;**69**(6):946-58.
21. Tamburrino MB, Nagel RW, Chahal MK, Lynch DJ. Antidepressant medication adherence: a study of primary care patients. *Primary Care Companion to the Journal of Clinical Psychiatry* 2009;**11**(5):205-11.
22. Sansone RA, Sansone LA. Antidepressant adherence: are patients taking their medications? *Innovations in Clinical Neuroscience* 2012;**9**(5):41-6.
23. Scottish Intercollegiate Guidelines Network. Non-pharmaceutical management of depression in adults. <http://www.sign.ac.uk/pdf/sign114.pdf> 2010.
24. American Psychological Association. APA clinical practice guideline for the treatment of depression across three age cohorts. Available at <https://www.apa.org/depression-guideline/guideline.pdf> 2019.
25. American College of Physicians 2016. Nonpharmacologic versus pharmacologic treatment of adult patients with major depressive disorder: a clinical practice guideline from the American College of Physicians. *Annals of Internal Medicine* 2016;**164**(5):350-9.
26. Parikh SV, Quilty LC, Ravitz P, Rosenbluth M, Pavlova B, Grigoriadis S, et al; Uher R; CANMAT Depression Work Group.

CANMAT depression work group. Canadian network for mood and anxiety treatments (CANMAT) 2016 clinical guidelines for the management of adults with major depressive disorder: section 2. Psychological treatments. *Canadian Journal of Psychiatry* 2016;**61**(9):524-39.

27. Royal Australian and New Zealand College of Psychiatrists. Australian and New Zealand clinical practice guidelines for the treatment of depression. *Australian and New Zealand Journal of Psychiatry* 2004;**38**(6):389-407.

28. Wolf NJ, Hopko DR. Psychosocial and pharmacological interventions for depressed adults in primary care: a critical review. *Clinical Psychology Review* 2008;**28**(1):131-61.

29. Parikh SV, Segal ZV, Grigoriadis S, Ravindran AV, Kennedy SH, Lam RW, et al. Canadian Network for Mood and Anxiety Treatments (CANMAT) clinical guidelines for the management of major depressive disorder in adults. II. Psychotherapy alone or in combination with antidepressant medication. *Journal of Affective Disorders* 2009;**117**(1):S15-S25.

30. Cuijpers P, Andersson G, Donker T, Van Straten A. Psychological treatment of depression: results of a series of meta-analyses. *Nordic Journal of Psychiatry* 2011;**65**(6):354-64.

31. Orgeta V, Qazi A, Spector A, Orrell M. Psychological treatments for depression and anxiety in dementia and mild cognitive impairment: systematic review and meta-analysis. *British Journal of Psychiatry* 2015;**207**(4):293-8.

32. Cooney GM, Dwan K, Greig CA, Lawlor DA, Rimer J, Waugh FR, et al. Exercise for depression. *Cochrane Database of Systematic Reviews* 2013, Issue 9. Art. No: CD004366. [DOI: [10.1002/14651858.CD004366.pub6](https://doi.org/10.1002/14651858.CD004366.pub6)]

33. Barlow DH. Negative effects from psychological treatments: a perspective. *American Psychologist* 2010;**65**(1):13-20.

34. Britt TW. Perceived stigma and barriers to care for psychological treatment: implications for reactions to stressors in different contexts. *Journal of Social and Clinical Psychology* 2008;**27**(4):317-35. [DOI: [10.1521/jscp.2008.27.4.317](https://doi.org/10.1521/jscp.2008.27.4.317)]

35. Solomon D, Adams J. The use of complementary and alternative medicine in adults with depressive disorders. A critical integrative review. *Journal of Affective Disorders* 2015;**179**:101-13.

36. Meekums B, Karkou V, Nelson EA. Dance movement therapy for depression. *Cochrane Database of Systematic Reviews* 2015, Issue 2. Art. No: CD009895. [DOI: [10.1002/14651858.CD009895.pub2](https://doi.org/10.1002/14651858.CD009895.pub2)]

37. Joyce J, Herbison GP. Reiki for depression and anxiety. *Cochrane Database of Systematic Reviews* 2015, Issue 4. Art. No: CD006833. [DOI: [10.1002/14651858.CD006833.pub2](https://doi.org/10.1002/14651858.CD006833.pub2)]

38. Aalbers S, Fusar-Poli L, Freeman RE, Spreen M, Ket JC, Vink AC, et al. Music therapy for depression. *Cochrane Database of Systematic Reviews* 2017, Issue 11. Art. No: CD004517. [DOI: [10.1002/14651858.CD004517.pub3](https://doi.org/10.1002/14651858.CD004517.pub3)]

39. Tao L, Jiang R, Zhang K, Qian Z, Chen P, Lv Y, et al. Light therapy in non-seasonal depression: an update meta-analysis. *Psychiatry Research* 2020;**291**(113247):- . [DOI: [10.1016/j.psychres.2020.113247](https://doi.org/10.1016/j.psychres.2020.113247)]

40. Astin JA. Why patients use alternative medicine: results of a study. *JAMA* 1998;**279**(19):1548-53.

41. Blumenthal JA, Smith PJ, Hoffman BM. Is exercise a viable treatment for depression? *ACSM's Health & Fitness Journal* 2012;**16**(4):14-21.

42. Craft LL, Perna FM. The benefits of exercise for the clinically depressed. *Primary Care Companion to the Journal of Clinical Psychiatry* 2004;**6**(3):104-11.

43. Ekkekakis P. Honey, I shrunk the pooled SMD! Guide to critical appraisal of systematic reviews and meta-analyses using the Cochrane review on exercise for depression as example. *Mental Health and Physical Activity* 2015;**8**:21-36.

44. Malhi GS, Bell E, Bassett D, Boyce P, Bryant R, Hazell P, et al. The 2020 Royal Australian and New Zealand College of Psychiatrists clinical practice guidelines for mood disorders. *Australian and New Zealand Journal of Psychiatry* 2021;**55**(1):7-117. [DOI: [10.1177/0004867420979353](https://doi.org/10.1177/0004867420979353)]

45. Stubbs B, Vancampfort D, Hallgren M, Firth J, Veronese N, Solmi M, et al. EPA guidance on physical activity as a treatment for severe mental illness: a meta-review of the evidence and Position Statement from the European Psychiatric Association (EPA), supported by the International Organization of Physical Therapists in Mental Health (IOPTMH). *European Psychiatry* 2018;**54**:124-44. [DOI: [10.1016/j.eurpsy.2018.07.004](https://doi.org/10.1016/j.eurpsy.2018.07.004)]

46. American College of Sports Medicine. ACSM's Resource Manual for Guidelines for Exercise Testing and Prescription. 4th edition. Lippincott, Williams and Wilkins, 2001.

47. Beesley S, Mutrie N. Exercise is beneficial adjunctive treatment in depression. *BMJ (Clinical Research Ed.)* 1997;**315**(7121):1542-3.

48. Andel V, Glen E, Austin D. Physical fitness and mental health: a review of the literature. *Adapted Physical Activity Quarterly* 1984;**3**(1):207.

49. Silveira H, Moraes H, Oliveira N, Coutinho ES, Laks J, Deslandes A. Physical exercise and clinically depressed patients: a systematic review and meta-analysis. *Neuropsychobiology* 2013;**67**(2):61-8.

50. De Souza Moura AM, Lamego MK, Paes F, Ferreira Rocha NB, Simoes-Silva V, Rocha SA, et al. Comparison among aerobic exercise and other types of interventions to treat depression: a systematic review. *CNS & Neurological Disorders - Drug Targets* 2015;**14**(9):1171-83.

51. Schuch FB, Vancampfort D, Richards J, Rosenbaum S, Ward PB, Stubbs B. Exercise as a treatment for depression: a meta-analysis adjusting for publication bias. *Journal of Psychiatric Research* 2016;**77**:42-51.

52. Farah WH, Alsawas M, Mainou M, Alahdab F, Farah MH, Ahmed AT, et al. Non-pharmacological treatment of depression: a systematic review and evidence map. *Evidence-based Medicine* 2016;**21**(6):214-21.
53. Danielsson L, Noras AM, Waern M, Carlsson J. Exercise in the treatment of major depression: a systematic review grading the quality of evidence. *Physiotherapy Theory and Practice* 2013;**29**(8):573-85.
54. Stubbs B, Vancampfort D, Rosenbaum S, Ward PB, Richards J, Ussher M, et al. Challenges establishing the efficacy of exercise as an antidepressant treatment: a systematic review and meta-analysis of control group responses in exercise randomised controlled trials. *Sports Medicine* 2017;**46**(5):699-713.
55. Josefsson T, Lindwall M, Archer T. Physical exercise intervention in depressive disorders: meta-analysis and systematic review. *Scandinavian Journal of Medicine & Science in Sports* 2014;**24**(2):259-72.
56. Banyard H, Edward KL, Garvey L, Stephenson J, Azevedo L, Benson AC. The effects of aerobic and resistance exercise on depression and anxiety: systematic review with meta-analysis. *International Journal of Mental Health Nursing* 2025;**34**(3):e70054. [DOI: [10.1111/inm.70054](https://doi.org/10.1111/inm.70054)]
57. Budde H, Dolz N, Mueller-Alcazar A, Schacht F, Velasques B, Ribeiro P, et al. A 10 years update of effects of exercise on depression disorders-in otherwise healthy adults: a systematic review of meta-analyses and neurobiological mechanisms. *PLoS One* 2025;**20**(5):e0317610. [DOI: [10.1371/journal.pone.0317610](https://doi.org/10.1371/journal.pone.0317610)]
58. Malhi GS, Bell E, Bassett D, Boyce P, Bryant R, Hazell P, et al. The 2020 Royal Australian and New Zealand College of Psychiatrists clinical practice guidelines for mood disorders. *Australian and New Zealand Journal of Psychiatry* 2021;**55**(1):7-117.
59. Ravindran AV, Balneaves LG, Faulkner G, Ortiz A, McIntosh D, Morehouse RL, et al. Canadian network for mood and anxiety treatments (CANMAT) 2016 clinical guidelines for the management of adults with major depressive disorder: section 5. Complementary and alternative medicine treatments. *Canadian Journal of Psychiatry* 2016;**61**(9):576-87.
60. Lam RW, Kennedy SH, Adams C, Bahji A, Beaulieu S, Bhat V, et al. Canadian network for mood and anxiety treatments (CANMAT) 2023 update on clinical guidelines for management of major depressive disorder in adults [Réseau canadien pour les traitements de l'humeur et de l'anxiété (CANMAT) 2023 : mise à jour des lignes directrices cliniques pour la prise en charge du trouble dépressif majeur chez les adultes]. *Canadian Journal of Psychiatry. Revue Canadienne de Psychiatrie* 2024;**69**(9):641-87.
61. Carneiro LF, Mota MP, Schuch F, Deslandes A, Vasconcelos-Raposo J. Portuguese and Brazilian guidelines for the treatment of depression: exercise as medicine. *Brazilian Journal of Psychiatry* 2017;**40**(2):210-1.
62. Roshanaei-Moghaddam B, Katon WJ, Russo J. The longitudinal effects of depression on physical activity. *General Hospital Psychiatry* 2009;**31**(4):306-15.
63. Dubash S. The interplay of depression symptoms and physical activity: bidirectional insights from 25-years of the Americans' changing lives panel. *Mental Health and Physical Activity* 2024;**26**:100599. [DOI: [10.1016/j.mhpa.2024.100599](https://doi.org/10.1016/j.mhpa.2024.100599)]
64. Smith PJ, Blumenthal JA. Exercise and physical activity in the prevention and treatment of depression. In: Handbook of Physical Activity and Mental Health. London: Routledge, 2013:145-60.
65. Teychenne M, Ball K, Salmon J. Physical activity and likelihood of depression in adults: a review. *Preventive Medicine* 2008;**46**(5):397-411.
66. Wanjau MN, Möller H, Haigh F, Milat A, Hayek R, Lucas P, et al. Physical activity and depression and anxiety disorders: a systematic review of reviews and assessment of causality. *AJPM Focus* 2023;**2**(2):100074. [DOI: [10.1016/j.focus.2023.100074](https://doi.org/10.1016/j.focus.2023.100074)]
67. LePore SJ. Expressive writing moderates the relation between intrusive thoughts and depressive symptoms. *Journal of Personality and Social Psychology* 1997;**73**(5):1030-7.
68. Craft LL. Exercise and clinical depression: examining two psychological mechanisms. *Psychology of Sport and Exercise* 2005;**6**:151-71.
69. Diener E. Subjective well-being. *Psychological Bulletin* 1984;**95**:542-75.
70. Fox KR. The effects of exercise on self-perceptions and self-esteem. In: Biddle SJ, Fox KR, editor(s). Physical Activity and Psychological Well-Being. London: Routledge, 2000:88-117.
71. Park SH, Han KS, Kang CB. Effects of exercise programs on depressive symptoms, quality of life, and self-esteem in older people: a systematic review of randomized controlled trials. *Applied Nursing Research* 2014;**27**(4):219-26.
72. Yiğitler K. The effects of participation in regular exercise on self-esteem and hopelessness of female university students. *Social Behavior and Personality: an International Journal* 2014;**42**(2):1233-43.
73. Legrand FD. Effects of exercise on physical self-concept, global self-esteem, and depression in women of low socioeconomic status with elevated depressive symptoms. *Journal of Sport & Exercise Psychology* 2014;**36**(4):357-65.
74. Gomez-Merino D, Béquet F, Berthelot M, Chennaoui M, Guezennec CY. Site-dependent effects of an acute intensive exercise on extracellular 5-HT and 5-HIAA levels in rat brain. *Neuroscience Letters* 2001;**301**(2):143-6.
75. Chen MJ. The neurobiology of depression and physical exercise. In: Handbook of Physical Activity and Mental Health. London: Routledge, 2013:169-84.
76. Cotman CW, Berchtold NC. Exercise: a behavioural intervention to enhance brain health and plasticity. *Trends in Neurosciences* 2002;**25**(6):295-301.

77. Ernst C, Olson AK, Pinel JP, Lam RW, Christie BR. Antidepressant effects of exercise: evidence for an adult neurogenesis hypothesis? *Journal of Psychiatry and Neurosciences* 2006;**31**(2):84-92.
78. Lu J, Xu Y, Hu W, Gao Y, Ni X, Sheng H, et al. Exercise ameliorates depression-like behavior and increases hippocampal BDNF level in ovariectomized rats. *Neuroscience Letters* 2014;**573**:13-18.
79. Yau SY, Lau BW, Tong JB, Wong R, Ching YP, Qiu G, et al. Hippocampal neurogenesis and dendritic plasticity support running-improved spatial learning and depression-like behaviour in stressed rats. *PLoS One* 2011;**6**(9):e24263. [DOI: [10.1371/journal.pone.0024263](https://doi.org/10.1371/journal.pone.0024263)]
80. Ristevska-Dimitrovska G, Shishkov R, Gerazova VP, Vujovic V, Stefanovski B, Novotni A, et al. Different serum BDNF levels in depression: results from BDNF studies in FYR Macedonia and Bulgaria. *Psychiatria Danubina* 2013;**25**(2):123-7.
81. Sen S, Duman R, Sanacora G. Serum brain-derived neurotrophic factor, depression, and antidepressant medications: meta-analyses and implications. *Biological Psychiatry* 2008;**64**(4):527-32.
82. Heyman E, Gamelin FX, Goekint M, Piscitelli F, Roelands B, Leclair E, et al. Intense exercise increases circulating endocannabinoid and BDNF levels in humans--possible implications for reward and depression. *Psychoneuroendocrinology* 2012;**37**(6):844-51.
83. Laske C, Banschbach S, Stransky E, Bosch S, Straten G, Machann J, et al. Exercise-induced normalization of decreased BDNF serum concentration in elderly women with remitted major depression. *International Journal of Neuropsychopharmacology* 2010;**13**(5):595-602.
84. Kvam S, Kleppe CL, Nordhus IH, Hovland A. Exercise as a treatment for depression: a meta-analysis. *Journal of Affective Disorders* 2016;**202**:67-86.
85. Schuch FB, Dunn AL, Kanitz AC, Delevatti RS, Fleck MP. Moderators of response in exercise treatment for depression: a systematic review. *Journal of Affective Disorders* 2016;**195**:40-9.
86. Bailey AP, Hetrick SE, Rosenbaum S, Purcell R, Parker AG. Treating depression with physical activity in adolescents and young adults: a systematic review and meta-analysis of randomised controlled trials. *Psychological Medicine* 2018;**48**(7):1068-83. [DOI: [10.1017/S0033291717002653](https://doi.org/10.1017/S0033291717002653)]
87. Correia ÉM, Monteiro D, Bento T, Rodrigues F, Cid L, Vitorino A, et al. Analysis of the effect of different physical exercise protocols on depression in adults: systematic review and meta-analysis of randomized controlled trials. *Sports Health* 2024;**16**(2):285-94.
88. Guo S, Liu F, Shen J, Wei M, Yang Y. Comparative efficacy of seven exercise interventions for symptoms of depression in college students: a network of meta-analysis. *Medicine* 2020;**99**(47):e23058. [DOI: [10.1097/MD.00000000000023058](https://doi.org/10.1097/MD.00000000000023058)]
89. Heissel A, Heinen D, Brokmeier LL, Skarabis N, Kangas M, Vancampfort D, et al. Exercise as medicine for depressive symptoms? A systematic review and meta-analysis with meta-regression. *British Journal of Sports Medicine* 2023;**57**(16):1049-57.
90. Jazayeri D, Zanjani Z, Joeekar S, Alvani SR, Molaei A. Comparison study of the effects of resistance and endurance exercise interventions for the treatment of depression: a meta-analytical review. *Journal of Contemporary Psychotherapy* 2022;**52**:237-46. [DOI: [10.1007/s10879-022-09537-8](https://doi.org/10.1007/s10879-022-09537-8)]
91. Krogh J, Hjorthøj C, Speyer H, Gluud C, Nordentoft M. Exercise for patients with major depression: a systematic review with meta-analysis and trial sequential analysis. *BMJ Open* 2017;**7**(9):e014820. [DOI: [10.1136/bmjopen-2016-014820](https://doi.org/10.1136/bmjopen-2016-014820)]
92. Lee J, Gierc M, Vila-Rodriguez F, Puterman E, Faulkner G. Efficacy of exercise combined with standard treatment for depression compared to standard treatment alone: a systematic review and meta-analysis of randomized controlled trials. *Journal of Affective Disorders* 2021;**295**:1494-511.
93. Morres ID, Hatzigeorgiadis A, Stathi A, Comoutos N, Arpin-Cribbie C, Krommidas C, et al. Aerobic exercise for adult patients with major depressive disorder in mental health services: a systematic review and meta-analysis. *Depression and Anxiety* 2019;**36**:39-53. [DOI: [10.1002/da.22842](https://doi.org/10.1002/da.22842)]
94. Noetel M, Sanders T, Gallardo-Gómez D, Taylor P, Del Pozo Cruz B, Van den Hoek D, et al. Effect of exercise for depression: systematic review and network meta-analysis of randomised controlled trials. *BMJ (Clinical Research Ed.)* 2024;**384**:e075847. [DOI: [10.1136/bmj-2023-075847](https://doi.org/10.1136/bmj-2023-075847)]
95. Pérez-López FR, Martínez-Domínguez SJ, Lajusticia H, Chedraui P. Effects of programmed exercise on depressive symptoms in midlife and older women: a meta-analysis of randomized controlled trials. *Maturitas* 2017;**106**:38-47.
96. Recchia F, Leung CK, Chin EC, Fong DY, Montero D, Cheng CP, et al. Comparative effectiveness of exercise, antidepressants and their combination in treating non-severe depression: a systematic review and network meta-analysis of randomised controlled trials. *British Journal of Sports Medicine* 2022;**56**(23):1375-80.
97. Zhang Z, Liu JY, Zhu KT, Huo GQ. A META analysis and systematic review of the effects of exercise interventions on middle-aged and elderly patients with depression. *PLoS One* 2025;**20**(1):e0303594. [DOI: [10.1371/journal.pone.0303594](https://doi.org/10.1371/journal.pone.0303594)]
98. Stathopoulou G, Powers MB, Berry AC, Smits JA, Otto MW. Exercise interventions for mental health: a quantitative and qualitative review. *Clinical Psychology: Science and Practice* 2006;**13**:179-93.
99. Mead GE, Morley W, Campbell P, Greig CA, McMurdo M, Lawlor DA. Exercise for depression. *Cochrane Database of Systematic Reviews* 2009, Issue 3. Art. No: CD004366. [DOI: [10.1002/14651858.CD004366.pub4](https://doi.org/10.1002/14651858.CD004366.pub4)]

- 100.** Rethorst CD, Wipfli BM, Landers DM. The antidepressive effects of exercise. *Sports Medicine (Auckland, N.Z.)* 2009;**39**(6):491-511.
- 101.** Krogh J, Nordentoft M, Sterne J, Lawlor DA. The effect of exercise in clinical depressed adults: systematic review and meta-analysis of randomised controlled trials. *Journal of Clinical Psychiatry* 2011;**72**(4):529-38.
- 102.** Lawlor DA, Hopker SW. The effectiveness of exercise as an intervention in the management of depression: systematic review and meta-regression analysis of randomised controlled trials. *BMJ (Clinical Research Ed.)* 2001;**322**(7289):763-7.
- 103.** Sjosten N, Kivela SL. The effects of physical exercise on depressive symptoms among the aged: a systematic review. *International Journal of Geriatric Psychiatry* 2006;**21**(5):410-8.
- 104.** North TC, McCullagh P, Tran ZV. Effect of exercise on depression. *Exercise and Sport Sciences Reviews* 1990;**18**:379-415.
- 105.** Piquart M, Duberstein PR, Lyness JM. Effects of psychotherapy and other behavioral interventions on clinical depressed older adults: a meta-analysis. *Aging & Mental Health* 2007;**11**(6):645-57.
- 106.** Blake H. How effective are physical activity interventions for alleviating depressive symptoms in older people? A systematic review. *Clinical Rehabilitation* 2009;**23**:873-87.
- 107.** Craft LL, Landers DM. The effect of exercise on clinical depression and depression resulting from mental illness: a meta-analysis. *Journal of Sport & Exercise Psychology* 1998;**20**(4):339-57.
- 108.** Rimer J, Dwan K, Lawlor DA, Greig CA, McMurdo M, Morley W, et al. Exercise for depression. *Cochrane Database of Systematic Reviews* 2012, Issue 7. Art. No: CD004366. [DOI: [10.1002/14651858.CD004366.pub5](https://doi.org/10.1002/14651858.CD004366.pub5)]
- 109.** Higgins J, Lasserson T, Thomas J, Flemyng E, Churchill R. Standards for the conduct of new Cochrane Intervention Reviews, and the planning and conduct of updates. Methodological Expectations of Cochrane Intervention Reviews (MECIR). Available at <https://www.cochrane.org/authors/handbooks-and-manuals/mecir-manual> 2023.
- 110.** Page MJ, McKenzie JE, Bossuyt PM, Boutron I, Hoffmann TC, Mulrow CD et al. The PRISMA 2020 statement: an updated guideline for reporting systematic reviews. *BMJ* 2021;**372**:n71. [DOI: [10.1136/bmj.n71](https://doi.org/10.1136/bmj.n71)]
- 111.** Beck AT, Ward CH, Mendelson M, Mock J, Erbaugh J. An inventory for measuring depression. *Archives of General Psychiatry* 1961;**4**(6):561-71.
- 112.** Hamilton M. A rating scale for depression. *Journal of Neurology, Neurosurgery and Psychiatry* 1960;**23**:56-62.
- 113.** Ware JE Jr, Sherbourne CD. The MOS 36-item short-form health survey (SF-36). I. Conceptual framework and item selection. *Medical Care* 1992;**30**(6):473-83.
- 114.** Mead GE, Morley W, Campbell P, Greig CA, McMurdo M, Lawlor DA. Exercise for depression. *Cochrane Database of Systematic Reviews* 2008, Issue 4. Art. No: CD004366. [DOI: [10.1002/14651858.CD004366.pub3](https://doi.org/10.1002/14651858.CD004366.pub3)]
- 115.** Garber CE, Blissmer B, Deschenes MR, Franklin BA, Lamonte MJ, Lee IM, et al. American College of Sports Medicine position stand. Quantity and quality of exercise for developing and maintaining cardiorespiratory, musculoskeletal, and neuromotor fitness in apparently healthy adults: guidance for prescribing exercise. *Medicine and Science in Sports and Exercise* 2011;**43**(7):1334-59.
- 116.** Howley ET. Type of activity: resistance, aerobic and leisure versus occupational physical activity. *Medicine & Science in Sports & Exercise* 2001;**33**(6):s419-20. [DOI: [10.1097/00005768-200106001-00005](https://doi.org/10.1097/00005768-200106001-00005)]
- 117.** Higgins JP, Thomas J, Chandler J, Cumpston M, Li T, Page MJ, et al (editors). *Cochrane Handbook for Systematic Reviews of Interventions* version 6.0 (updated July 2019). Available from <https://www.cochrane.org/handbook>.
- 118.** Schünemann HJ, Oxman AD, Vist GE, Higgins JP, Deeks JJ, Glasziou P, et al. Chapter 12: Interpreting results and drawing conclusions. In: Higgins JP, Green S. *Cochrane Handbook for Systematic Reviews of Interventions* Version 5.1.0 (updated March 2011). The Cochrane Collaboration, 2011. Available from <https://www.cochrane.handbook.org>.
- 119.** DerSimonian R, Laird N. Meta-analysis in clinical trials. *Controlled Clinical Trials* 1986;**7**(3):177-88.
- 120.** Schünemann H, Brożek J, Guyatt G, Oxman A, editors. GRADE handbook for grading quality of evidence and strength of recommendations. Updated October 2013. The GRADE Working Group. Available from <https://guidelinedevelopment.org/handbook> 2013.
- 121.** GRADE Working Group. GRADE Handbook for Grading Quality of Evidence and Strength of Recommendations (updated 2013). GRADE Working Group, 2013. Available from <https://book.grade.pro/>.
- 122.** Guyatt GH, Oxman AD, Vist GE, Kunz R, Falck-Ytter Y, Alonso-Coello P, et al; GRADE Working Group. GRADE: an emerging consensus on rating quality of evidence and strength of recommendations. *BMJ (Clinical Research Ed.)* 2008;**336**(7650):924-6.
- 123.** Aghakhani N, Sharif F, Khademvatan K, Rahbar N, Eghtedar S, Shojaei Motlagh V. The reduction in anxiety and depression by education of patients with myocardial infarction. *International Cardiovascular Research Journal* 2011;**5**(2):66-8.
- 124.** Aghakhani N, Khademvatan K, & Dehghani MR. The effect of written material and verbal method education on anxiety and depression in patients with myocardial infarction in selected hospitals in Iran. *Journal of Advances in Medical Education & Professionalism* 2014;**2**(4):165-9.
- 125.** Aghakhani N, Sharif F, Khademvatan K, Rahbar N, Eghtedar S, Shojaei Motlagh V. The reduction in anxiety and depression by education of patients with myocardial infarction

[abstract 279]. *European Psychiatry [abstracts of the 21st European Congress of Psychiatry, EPA; 2013 Apr 6-9; Nice France]* 2013;**28**(Suppl 1):..

126. Aghakhani N. Effect of education on anxiety and depression in hospitalized patients with myocardial infarction in Urmia, Iran [poster]. *European Psychiatry [abstracts of the 20th European Congress of Psychiatry, EPA. 2012 Mar 3-6; Prague Czech Republic]* 2012;**27**(Suppl 1):P-1005.

127. Aghakhani N, Rahbar N, Khademvatan K, Jasemi M, Eghtedar S. The effect of education on anxiety and depression in patients with myocardial infarction (MI) in selected hospitals, Iran [abstract]. In: *Asian Journal of Psychiatry [abstracts from the 3rd Congress of Asian Psychiatry, Wcap; 31 Jul - 4 Aug 2011; Melbourne, VIC Australia]*. 2011.

128. Aghakhani N. Effect of education on anxiety and depression in patients with myocardial infarction [conference abstract]. In: *European Psychiatry [abstracts from the 19th European Congress of Psychiatry, EPA 2011 Mar 12-15; Vienna, Austria]*. 2011.

129. Krogh J, Videbech P, Thomsen C, Gluud C, Nordentoft M DEMO-II Trial. Aerobic exercise versus stretching exercise in patients with major depression - a randomised clinical trial. *PloS One* 2012;**7**(10):e48316.

130. Krogh J, Gotze JP, Jorgensen MB, Kristensen LO, Kistorp C, Nordentoft M. Copeptin during rest and exercise in major depression. *Journal of Affective Disorders* 2013;**151**(1):284-90. [PMID: 23856279]

131. Krogh J, Rostrup E, Thomsen C, Elfving B, Videbech P, Nordentoft M. The effect of exercise on hippocampal volume and neurotrophines in patients with major depression-A randomized clinical trial. *Journal of Affective Disorders* 2014;**165**:24-30.

132. Gotta G. Sex differences and the effects of exercise on depression and executive dysfunctioning in older adults (thesis). *Dissertation Abstracts International: Section B: The Sciences and Engineering* 2012;**72**(10-B):6385.

133. Martiny K, Refsgaard E, Lund V, Lunde M, Sørensen L, Thougard B, et al. The day-to-day acute effect of wake therapy in patients with major depression using the HAM-D6 as primary outcome measure: results from a randomised controlled trial. *PloS One* 2013;**8**(6):e67264.

134. Martiny K, Refsgaard E, Lund V, Lunde M, Thougard B, Lindberg L, et al. Maintained superiority of chronotherapeutics vs. exercise in a 20-week randomized follow-up trial in major depression. *Acta Psychiatrica Scandinavica* 2015;**131**(6):446-57. [PMID: 25689725]

135. Martiny K, Refsgaard E, Lund V, Lunde M, Sorensen L, Thougard B, et al. A 9-week randomised trial comparing a chronotherapeutic intervention (wake and light therapy) to exercise in major depressive disorder patients treated with duloxetine. *Journal of Clinical Psychiatry* 2012;**73**(9):1234-42.

136. Martiny K, Refsgaard E, Lund V, Lunde M, Bech P. Wake therapy yields sustained response and remission. A rater-

blind 9 weeks controlled study using a chronotherapeutic intervention compared to exercise [conference abstract S06-03]. *European Psychiatry [abstracts from the 19th European Congress of Psychiatry, EPA 2011 Mar 12-15; Vienna, Austria]* 2011;**26**:2031. [DOI: [10.1016/S0924-9338\(11\)73734-8](https://doi.org/10.1016/S0924-9338(11)73734-8)]

137. Murphy SM, Edwards RT, Williams N, Raisanen L, Moore G, Linck P, et al. An evaluation of the effectiveness and cost effectiveness of the National Exercise Referral Scheme in Wales, UK: a randomised controlled trial of a public health policy initiative. *Journal of Epidemiology and Community Health* 2012;**66**(8):745-53.

138. Littlecott HJ, Moore GF, Moore L, Murphy S. Psychosocial mediators of change in physical activity in the Welsh national exercise referral scheme: secondary analysis of a randomised controlled trial. *International Journal of Behavioral Nutrition and Physical Activity* 2014;**11**:109. [PMID: 25209188]

139. Edwards RT, Linck P, Hounsborne N, Raisanen L, Williams N, Moore L, et al. Cost-effectiveness of a national exercise referral programme for primary care patients in Wales: results of a randomised controlled trial. *BMC Public Health* 2013;**13**:1021. [DOI: [10.1186/1471-2458-13-1021](https://doi.org/10.1186/1471-2458-13-1021)]

140. Murphy S, Raisanen L, Moore G, Edwards RT, Linck P, Williams N, et al. A pragmatic randomised controlled trial of the Welsh National Exercise Referral Scheme: protocol for trial and integrated economic and process evaluation. *BMC Public Health* 2010;**10**:352. [DOI: [10.1186/1471-2458-10-352](https://doi.org/10.1186/1471-2458-10-352)]

141. Littlecott HJ, Moore GF, Moore L, Murphy S. Psychosocial mediators of change in physical activity in the Welsh national exercise referral scheme: secondary analysis of a randomised controlled trial. *International Journal of Behavioral Nutrition and Physical Activity* 2014;**11**:109. [DOI: [10.1186/s12966-014-0109-9](https://doi.org/10.1186/s12966-014-0109-9)]

142. Pinniger R, Brown RF, Thorsteinssona EB, McKinley P. Argentine tango dance compared to mindfulness meditation and a waiting-list control: a randomised trial for treating depression. *Complementary Therapies in Medicine* 2012;**20**(6):377-84.

143. Sturm J, Ploderl M, Fartacek C, Kralovec K, Neunhauserer D, Niederseer D, et al. Physical exercise through mountain hiking in high-risk suicide patients. A randomized crossover trial. *Acta Psychiatrica Scandinavica* 2012;**126**(6):467-75.

144. Neunhauserer D, Sturm J, Baumgartlinger MM, Niederseer D, Ledl-Kurkowski E, Steidle E, et al. Hiking in suicidal patients: Neutral effects on markers of suicidality. *American Journal of Medicine* 2013;**126**(10):927-30.

145. Kerse N, Falloon K, Moyes SA, Hayman KJ, Dowell T, Kolt GS, et al. DeLLITE Depression in late life: An intervention trial of exercise. Design and recruitment of a randomised controlled trial [ACTRN12605000475640]. *BMC Geriatrics* 2008;**8**:ArtID 12.

146. Kerse N, Hayman KJ, Moyes SA, Peri K, Robinson E, Dowell A, et al. Home-based activity program for older people with depressive symptoms: DeLLITE - a randomized controlled trial. *Annals of Family Medicine* 2010;**8**:214-23.

- 147.** Rezaei J, Abdi A, Rezaei M, Heydarnezhadian J, Jalali R. Effect of regular exercise program on depression in hemodialysis patients. *International Scholarly Research Notices* 2015;**2015**:182030. [PMID: 27347502]
- 148.** Pibernik-Okanovic M, Hermanns N, Ajdukovic D, Kos J, Prasek M, Sekerija M, et al. Does treatment of subsyndromal depression improve depression-related and diabetes-related outcomes? A randomised controlled comparison of psychoeducation, physical exercise and enhanced treatment as usual. *Trials* 2015;**16**:305. [PMID: 26174334]
- 149.** Taylor-Young P, Hauser P, Schuldheis S, Winters-Stone K. Efficacy of an exercise intervention to decrease depressive symptoms in depressed veterans with hepatitis c. In: <https://clinicaltrials.gov/study/NCT00643695>. Kissimmee, Florida: University of South Florida College of Nursing, 2011.
- 150.** NCT00931814. Effects of exercise on depression symptoms, physical function, and quality of life in community-dwelling elderly. [ClinicalTrials.gov/show/NCT00931814](https://clinicaltrials.gov/show/NCT00931814) (accessed 1 March 2013).
- 151.** NCT01383811. Clinical and neuroendocrine/metabolic benefits of exercise in treatment resistant depression (TRD): a feasibility study. [ClinicalTrials.gov/ct2/show/NCT01383811](https://clinicaltrials.gov/ct2/show/NCT01383811) (accessed 1 March 2013).
- 152.** NCT01464463. The impact of psychological interventions (with and without exercise) on psychometric and immunological measures in patients With major depression. [ClinicalTrials.gov/show/NCT01464463](https://clinicaltrials.gov/show/NCT01464463) (accessed 1 March 2013).
- 153.** UMIN000001488. A randomised controlled trial of exercise class for older persons with mild depression. www.umin.ac.jp/ctr/index.htm (accessed 1 March 2013).
- 154.** NCT01763983. Effects of cognitive behaviour therapy and exercise on depression and cognitive deficits in multiple sclerosis. [ClinicalTrials.gov/show/NCT01763983](https://clinicaltrials.gov/show/NCT01763983) (accessed 1 March 2013).
- 155.** Mackintosh S. Improving mood through physical activity for carers and care recipients (IMPACCT): protocol for a randomised trial. *Journal of Physiotherapy* 2013;**59**(2):125.
- 156.** ACTRN12612000094875. A randomised controlled trial to improve depression in family carers through a physical activity intervention: IMPACCT Study. www.anzctr.org.au/ACTRN12612000094875.aspx (accessed 1 March 2013).
- 157.** CTR/2012/09/002985. Effect of sprint interval training on depression: a randomized controlled trial [Clinical Trials Registry - India]. www.ctri.nic.in/Clinicaltrials/pmaindet2.php?trialid=5224 (accessed 1 March 2013).
- 158.** Pfaff JJ, Alfonso H, Newton RU, Sim M, Flicker L, Almeida OP. ACTIVEDEP: a randomised, controlled trial of a home-based exercise intervention to alleviate depression in middle-aged and older adults. *British Journal of Sports Medicine* 2014;**48**(3):226-32. [PMID: 23833045]
- 159.** Barclay T, Richards S, Schoffstall J, Magnuson C, McPhee C, Price J, et al. A pilot study on the effects of exercise on depression symptoms using levels of neurotransmitters and EEG as markers. *European Journal of Psychology and Educational Studies* 2014;**1**(1):30-5.
- 160.** Kruisdijk F, Hendriksen I, Tak E, Beekman A-J, Hopman-Rock M. EFFORT-D study process evaluation: challenges in conducting a trial into the effects of running therapy in patients with major depressive disorder. *Annals of General Psychiatry* 2018;**17**(10):1-10.
- 161.** Kruisdijk F, Hopman-Rock M, Beekman AT, Hendriksen IJ. Personality traits as predictors of exercise treatment adherence in major depressive disorder: lessons from a randomised clinical trial. *International Journal of Psychiatry in Clinical Practice* 2020;**24**(4):380-6. [DOI: [10.1080/13651501.2020.1787452](https://doi.org/10.1080/13651501.2020.1787452)]
- 162.** Kruisdijk F, Hopman-Rock M, Beekman AT, Hendriksen I. EFFORT-D: results of a randomised controlled trial testing the EFFECT of running therapy on depression. *BMC Psychiatry* 2019;**19**(170):1-14.
- 163.** Schneider KL, Panza E, Handschin B, Ma Y, Busch AM, Waring ME, et al. Feasibility of pairing behavioral activation with exercise for women with type 2 diabetes and depression: the get it study pilot randomized controlled trial. *Behavior Therapy* 2016;**47**(2):198-212.
- 164.** Ström M, Uckelstam C-J, Andersson G, Hassmén P, Umefjord G, Carlbring P. Internet-delivered therapist-guided physical activity for mild to moderate depression: a randomized controlled trial. *Black K, Ed. PeerJ* 2013;**1**(178):..
- 165.** Sadeghi K, Ahmadi SM, Ahmadi SM, Rezaei M, Miri J, Abdi A, et al. A comparative study of the efficacy of cognitive group therapy and aerobic exercise in the treatment of depression among the students. *Global Journal of Health Science* 2016;**8**(10):54171. [PMID: 27302433]
- 166.** Clark D. The role of exercise in depression among middle and older aged adults. Indiana University 2014.
- 167.** NCT01401569. Efficacy of exercise and counseling intervention on relapse in smoker with depressive disorders (STOB-ACTIV). <https://clinicaltrials.gov/ct2/show/NCT01401569>.
- 168.** NCT01805479. Exercise training in depressed traumatic brain injury survivors. <https://clinicaltrials.gov/ct2/show/NCT01805479>.
- 169.** Carlbring P, Lindner P, Martell C, Hassmén P, Forsberg L, Ström L, et al. The effects on depression of Internet-administered behavioural activation and physical exercise with treatment rationale and relapse prevention: study protocol for a randomised controlled trial. *Trials* 2013;**14**:35.
- 170.** Svartvatten N, Segerlund M, Dennhag I, Andersson G, Carlbring P. A content analysis of client e-mails in guided internet-based cognitive behavior therapy for depression. *Internet Interventions* 2015;**2**(2):121-7.
- 171.** NCT01619930. The effects of behavioural activation and physical exercise on depression. [ClinicalTrials.gov/show/NCT01619930](https://clinicaltrials.gov/show/NCT01619930) (accessed 1 March 2013).

- 172.** NCT01696201. Effect of a supervised exercise program during whole pregnancy on outcomes and level of depression. *ClinicalTrials.gov/show/NCT01696201* (accessed 1 March 2013).
- 173.** Krogh J, Saltin B, Gluud C, Nordentoft M. The DEMO Trial: a randomized, parallel-group, observer-blinded clinical trial of strength versus aerobic versus relaxation training for patients with mild to moderate depression. *Journal of Clinical Psychiatry* 2009;**70**(6):790-800.
- 174.** Krogh J, Nordentoft M, Mohammad-Nezhad M, Westrin A. Growth hormone, prolactin and cortisol response to exercise in patients with depression. *Journal of Affective Disorders* 2010;**125**:189-97.
- 175.** Krogh J, Petersen L, Timmermann M, Saltin B, Nordentoft M. Design paper: the DEMO trial: a randomized, parallel-group, observer-blinded clinical trial of aerobic versus non-aerobic versus relaxation training for patients with light to moderate depression [NCT00103415]. *Contemporary Clinical Trials* 2007;**28**(1):79-89.
- 176.** Abdelbasset WK, Alqahtani BA. A randomized controlled trial on the impact of moderate-intensity continuous aerobic exercise on the depression status of middle-aged patients with congestive heart failure. *Medicine* 2019;**98**(17):e15344. [DOI: [10.1097/MD.00000000000015344](https://doi.org/10.1097/MD.00000000000015344)]
- 177.** Abdollahia A, LeBouthillier DM, Najafic M, Asmundson GJ, Hosseini S, Shahid S, et al. Effect of exercise augmentation of cognitive behavioural therapy for treatment of suicidal ideation and depression. *Journal of Affective Disorders* 2017;**219**:58-63. [DOI: [10.1016/j.jad.2017.05.012](https://doi.org/10.1016/j.jad.2017.05.012)]
- 178.** Bademli K, Lök N, Lök S. The effect of a physical activity intervention on burden and depressive symptoms in depressed family caregivers of patients with schizophrenia: a randomized controlled trial. *Journal of Physical Activity & Health* 2023;**20**(12):1109-15. [DOI: [10.1123/jpah.2022-0335](https://doi.org/10.1123/jpah.2022-0335)]
- 179.** Belvederi Murri M, Amore M, Menchetti M, Toni G, Neviani F, Cerri M, et al. Physical exercise for late-life major depression. *British Journal of Psychiatry* 2015;**207**(3):235-42. [DOI: [10.1192/bjp.bp.114.150516](https://doi.org/10.1192/bjp.bp.114.150516)]
- 180.** Belvederi Murri M, Amore M, Menchetti M, Toni G, Neviani F, Cerri M, et al. Physical exercise for late-life major depression. *Focus (American Psychiatric Publishing)* 2021;**19**(3):365-73. [DOI: [1176/appi.focus.19306](https://doi.org/10.1176/appi.focus.19306)]
- 181.** Toni G, Belvederi Murri M, Piepoli M, Zanetidou S, Cabassi A, Squatrito S, et al; SEEDS study group. Physical exercise for late-life depression: effects on heart rate variability. *American Journal of Geriatric Psychiatry* 2016;**24**(11):989-97.
- 182.** Zanetidou S, Belvederi Murri M, Menchetti M, Toni G, Ascoli F, Bagnoli L, et al; Bertakis KD; Safety Efficacy of Exercise for Depression in Seniors Study Group. Physical exercise for late-life depression: customizing an intervention for primary care. *Journal of the American Geriatrics Society* 2017;**65**(2):348-55.
- 183.** Murri MB, Ekkekakis P, Menchetti M, Neviani F, Trevisani F, Tedeschi S, et al. Physical exercise for late-life depression: effects on symptom dimensions and time course. *Journal of Affective Disorders* 2018;**230**:65-70.
- 184.** Neviani F, Belvederi Murri M, Mussi C, Triolo F, Toni G, Simoncini E, et al. Physical exercise for late life depression: effects on cognition and disability. *International Psychogeriatrics/IPA* 2010;**29**(7):1105-12.
- 185.** Bose G. Changes in depression status in low socioeconomic perinatal subjects in rural India after supervised physical exercise: a randomized controlled study. *Indian Journal of Psychiatry* 2015;**57**(4):412-3.
- 186.** Brush CJ, Hajcak G, Bocchine AJ, Ude AA, Muniz KM, Foti D, et al. A randomized trial of aerobic exercise for major depression: examining neural indicators of reward and cognitive control as predictors and treatment targets. *Psychological Medicine* 2022;**52**(5):893-903. [DOI: [10.1017/S0033291720002573](https://doi.org/10.1017/S0033291720002573)]
- 187.** Buschert V, Prochazka D, Bartl H, Diemer J, Malchow B, Zwanzger P, et al. Effects of physical activity on cognitive performance: a controlled clinical study in depressive patients. *European Archives of Psychiatry and Clinical Neuroscience* 2019;**269**(5):1-9. [DOI: [10.1007/s00406-018-0916-0](https://doi.org/10.1007/s00406-018-0916-0)]
- 188.** Cheung LK, Lee S. A randomized controlled trial on an aerobic exercise programme for depression outpatients. *Sport Sciences for Health* 2018;**14**:173-81.
- 189.** De Groot M, Shubrook JH, Hornsby WG Jr, Pillay Y, Mather KJ, Fitzpatrick K, et al. Program ACTIVE II: outcomes from a randomized, multistate community-based depression treatment for rural and urban adults with type 2 diabetes. *Diabetes Care* 2019;**42**(7):1185-93.
- 190.** De Groot M, Shubrook J, Schwartz F, Guyton Hornsby Jr W, Pillay Y, Saha C. Program ACTIVE II: design and methods for a multi-center community-based depression treatment for rural and urban adults with type 2 diabetes. *Journal of Diabetes Research and Therapy* 2015;**1**(2):1-15.
- 191.** De Groot M, Hornsby G, Saha C, Yang Z, Pillay Y, Fitzpatrick K, et al. Program ACTIVE II: a comparative effectiveness trial to treat major depression in T2DM. Diabetes, Conference: 77th Scientific Sessions of the American Diabetes Association, ADA 2017. United States 2017;**66**:A99.
- 192.** Dose M, Ziegenbein M, Hoos O, Reim D, Stengert W, Hoffer N, et al. Self-selected intensity exercise in the treatment of major depression: a pragmatic RCT. *International Journal of Psychiatry in Clinical Practice* 2015;**19**(4):266-75. [PMID: [26265421](https://pubmed.ncbi.nlm.nih.gov/26265421/)]
- 193.** Eman AM, Mahmoud ME. Effect of aerobic exercises versus foot reflexology on post-menopausal depression. *Clinical and Experimental Obstetrics & Gynecology* 2017;**44**(5):726-9.
- 194.** Haller N, Lorenz S, Pfirrmann D, Koch C, Lieb K, Dettweiler U, et al. Individualized web-based exercise for the treatment of depression: randomized controlled trial. *JMIR Mental Health* 2018;**5**(4):e10698. [DOI: [10.2196/10698](https://doi.org/10.2196/10698)]

- 195.** Heinzel S, Schwefel M, Sanchez A, Heinen D, Fehm L, Henze R, et al. Physical exercise training as preceding treatment to cognitive behavioral therapy in mild to moderate major depressive disorder: a randomized controlled trial. *Journal of Affective Disorders* 2022;**319**:90-8. [DOI: [10.1016/j.jad.2022.09.024](https://doi.org/10.1016/j.jad.2022.09.024)]
- 196.** Heissel A, Vesterling A, White SA, Kallies G, Behr D, Arafat AM, et al. Feasibility of an exercise program for older depressive inpatients. *GeroPsych* 2015;**28**(4):163-71.
- 197.** Hidalgo JL, Sotos JR; DEP-EXERCISE Group. Effectiveness of physical exercise in older adults with mild to moderate depression. *Annals of Family Medicine* 2021;**19**(4):302-9. [DOI: [10.1370/afm.2670](https://doi.org/10.1370/afm.2670)]
- 198.** Ho CW, Chan SC, Wong JS, Cheung WT, Chung DW, Lau TF. Effect of aerobic exercise training on Chinese population with mild to moderate depression in Hong Kong. *Rehabilitation Research and Practice* 2014;**2014**:1-8. [DOI: [10.1155/2014/627376](https://doi.org/10.1155/2014/627376)] [PMID: 24800081]
- 199.** Chung WS, Ho CW, Chan PS, Wong MJ, Cheung AW, Lau TF. Augmentation with aerobic exercise to pharmacological treatment for Chinese inpatients with moderate to severe depression: a single-blinded randomised controlled study. *International Journal of Neuropsychopharmacology* 2014;**17**:60. [DOI: [10.1017/S1461145714000741](https://doi.org/10.1017/S1461145714000741)]
- 200.** Huang TT, Liu CB, Tsai YH, Chin YF, Wong CH. Physical fitness exercise versus cognitive behavior therapy on reducing the depressive symptoms among community-dwelling elderly adults: a randomized controlled trial. *International Journal of Nursing Studies* 2015;**52**(10):1542-52. [PMID: 26105535]
- 201.** Karg N, Dorscht L, Kornhuber J, Luttenberger K. Boulderling psychotherapy is more effective in the treatment of depression than physical exercise alone: results of a multicentre randomised controlled intervention study. *BMC Psychiatry* 2020;**20**(116):- . [DOI: [10.1186/s12888-020-02518-y](https://doi.org/10.1186/s12888-020-02518-y)]
- 202.** Luttenberger K, Karg-Hefner N, Berking M, Kind L, Weiss M, Kornhuber J, et al. Boulderling psychotherapy is not inferior to cognitive behavioural therapy in the group treatment of depression: a randomized controlled trial. *British Journal of Clinical Psychology* 2021;**61**(2):465-93. [DOI: [10.1111/bjc.12347](https://doi.org/10.1111/bjc.12347)]
- 203.** Dorscht L, Karg N, Book S, Graessel E, Kornhuber J, Luttenberger K. A German climbing study on depression: a boulderling psychotherapeutic group intervention in outpatients compared with state-of-the-art cognitive behavioural group therapy and physical activation - study protocol for a multicentre randomised controlled trial. *BMC Psychiatry* 2019;**19**(1):154. [DOI: [10.1186/s12888-019-2140-5](https://doi.org/10.1186/s12888-019-2140-5)]
- 204.** Schwarzkopf L, Dorscht L, Kraus L, Luttenberger K. Is boulderling-psychotherapy a cost-effective way to treat depression when compared to group cognitive behavioral therapy - results from a randomized controlled trial. *BMC Health Services Research* 2021;**21**(1):1162. [DOI: [10.1186/s12913-021-07153-1](https://doi.org/10.1186/s12913-021-07153-1)]
- 205.** Luttenberger K, Donath C, Graessel E, Kornhuber J, Schlüter A, Dorscht L, et al. Treating depression in an outpatient setting: predictors of patient response to boulderling psychotherapy, cognitive behavioural therapy or exercise alone. *Clinical Psychology & Psychotherapy* 2023;:-1-13. [DOI: [10.1002/cpp.2898](https://doi.org/10.1002/cpp.2898)]
- 206.** Khorvash M, Askari A, Rafiemanzelat F, Botshekan M, Khorvash F. An investigation on the effect of strength and endurance training on depression, anxiety, and C-reactive protein's inflammatory biomarker changes. *Journal of Research in Medical Sciences* 2012;**17**(11):1072-6. [PMID: 23833584]
- 207.** Khoshnab LP, Nikseresht A. Comparison of the effect of aerobic exercise and antidepressant medications on depression and sexual desire of depressed middle-aged women. *International Journal of Women's Health and Reproduction Sciences* 2017;**5**(2):119-22.
- 208.** La Rocque CL, Mazurka R, Stuckless TJ, Pyke K, Harkness KL. Randomized controlled trial of bikram yoga and aerobic exercise for depression in women: efficacy and stress-based mechanisms. *Journal of Affective Disorders* 2021;**280**(Pt A):457-66. [DOI: [10.1016/j.jad.2020.10.067](https://doi.org/10.1016/j.jad.2020.10.067)]
- 209.** Legrand FD. Effects of exercise on physical self-concept, global self-esteem, and depression in women of low socioeconomic status with elevated depressive symptoms. *Journal of Sport & Exercise Psychology* 2014;**36**(4):357-65. [PMID: 25226604]
- 210.** Legrand FD, Neff EM. Efficacy of exercise as an adjunct treatment for clinically depressed inpatients during the initial stages of antidepressant pharmacotherapy: an open randomised controlled trial. *Journal of Affective Disorders* 2016;**191**:139-44.
- 211.** Olson RL, Brush CJ, Ehmann PJ, Alderman BL. A randomized trial of aerobic exercise on cognitive in major depression. *Clinical Neurophysiology* 2017;**128**:903-13.
- 212.** Prakhinkit S, Suppakitiporn S, Tanaka H, Suksom D. Effects of Buddhism walking meditation on depression, functional fitness, and endothelium-dependent vasodilation in depressed elderly. *Journal of Alternative and Complementary Medicine (New York, NY)* 2014;**20**(5):411-6. [PMID: 24372522]
- 213.** Roy A, Govindan R, Muralidharan K. The impact of an add-on video assisted structured aerobic exercise module on mood and somatic symptoms among women with depressive disorders: study from a tertiary care centre in India. *Asian Journal of Psychiatry* 2018;**32**:118-22.
- 214.** Salehi I, Hosseine SM, Haghighi M, Jahangard L, Bajoghli H, Gerber M, et al. Electroconvulsive therapy (ECT) and aerobic exercise training (AET) increased plasma BDNF and ameliorated depressive symptoms in patients suffering from major depressive disorder. *Journal of Psychiatric Research* 2016;**76**:1-8.
- 215.** Schuch FB, Vasconcelos-Moreno MP, Borowsky C, Zimmermann AB, Rocha NS, Fleck MP. Exercise and severe major depression: effect on symptom severity and quality of life at discharge in an inpatient cohort. *Journal of Psychiatric Research* 2015;**61**:25-32. [PMID: 25439084]

- 216.** Schuch FB, Vasconcelos-Moreno MP, Borowsky C, Fleck MP. Exercise and severe depression: preliminary results of an add-on study. *Journal of Affective Disorders* 2011;**133**:615-8.
- 217.** Schuch FB, Vasconcelos-Moreno MP, Borowsky C, Zimmermann AB, Wollenhaupt-Aguiar B, Ferrari P, et al. The effects of exercise on oxidative stress (TBARS) and BDNF in severely depressed inpatients. *European Archives of Psychiatry and Clinical Neuroscience* 2014;**264**(7):605-13.
- 218.** Siqueira CC, Valiengo LL, Carvalho AF, Santos-Silva PR, Missio G, de Sousa RT, et al. Antidepressant efficacy of adjunctive aerobic activity and associated biomarkers in major depression: a 4-week, randomized, single-blind, controlled clinical trial. *PLoS-ONE* 2016;**11**(5: e0154195):1-11.
- 219.** Taheri M, Irandoust K, Sadeghi A, Yari S. The effect of omega-3 fatty acid supplement and aerobic exercise on lipid profile and depression in obese women. *Acta Medica Mediterranea* 2018;**34**:865-70.
- 220.** Wang J, Li Z. Effect of physical exercise on medical rehabilitation treatment of depression. *Revista Brasileira de Medicina do Esporte* 2022;**28**(3):- [DOI: [10.1590/1517-8692202228032021_0483](https://doi.org/10.1590/1517-8692202228032021_0483)]
- 221.** Yu, DJ, Yu, AP, Leung, CK, et al. Comparison of moderate and vigorous walking exercise on reducing depression in middle-aged and older adults: a pilot randomized controlled trial. *European Journal of Sport Science* 2023;**23**(6):1018-27. [DOI: [10.1080/17461391.2022.2079424](https://doi.org/10.1080/17461391.2022.2079424)]
- 222.** Blumenthal JA, Babyak MA, Doraiswamy PM, Watkins L, Hoffman BM, Barbour KA, et al. Exercise and pharmacotherapy in the treatment of major depressive disorder. *Psychosomatic Medicine* 2007;**69**(7):587-96.
- 223.** Sherwood A, Blumenthal JA, Smith PJ, Watkins LL, Hoffman BA, Hinderliter AL. Effects of exercise and sertraline on measures of coronary heart disease risk in patients with major depression: results from the SMILE-II randomized clinical trial. *Psychosomatic Medicine* 2016;**78**(5):602-9. [DOI: [10.1097/PSY.0000000000000301](https://doi.org/10.1097/PSY.0000000000000301)]
- 224.** Blumenthal JA, Babyak MA, Craighead WE, Davidson J, Hinderliter A, Hoffman B, et al. The role of comorbid anxiety in exercise and depression trials: secondary analysis of the SMILE-II randomized clinical trial. *Depression and Anxiety* 2021;**38**(2):124-33. [DOI: [10.1002/da.23088](https://doi.org/10.1002/da.23088)]
- 225.** Hoffman BM, Babyak MA, Craighead WE, Sherwood A, Doraiswamy PM, Coons MJ, et al. Exercise and pharmacotherapy in patients with major depression: one-year follow-up of the SMILE study. *Psychosomatic Medicine* 2011;**73**(2):127-33.
- 226.** Blumenthal JA, Babyak MA, Moore KA, Craighead WE, Herman S, Khatri P, et al. Effects of exercise training on older patients with major depression. *Archives of Internal Medicine* 1999;**159**(19):2349-56.
- 227.** Babyak M, Blumenthal JA, Herman S, Khatri P, Doraiswamy M, Moore K, et al. Exercise treatment for major depression: maintenance of therapeutic benefit at 10 months. *Psychosomatic Medicine* 2000;**62**(5):633-8.
- 228.** Herman S, Blumenthal JA, Babyak M, Khatri P, Craighead WE, Krishnan KR, et al. Exercise therapy for depression in middle-aged and older adults: predictors of early dropout and treatment failure. *Health Psychology* 2002;**21**(6):533-63.
- 229.** Khatri P, Blumenthal JA, Babyak MA, Craighead WE, Herman S, Baldewitz T, et al. Effects of exercise training on cognitive functioning among depressed older men and women. *Journal of Aging and Physical Activity* 2001;**9**:43-57.
- 230.** Blumenthal J, Sherwood A, Babyak M, Watkins L, Smith PJ, Hoffman B, et al. Exercise and pharmacological treatment of depressive symptoms in patients with coronary heart disease. *Journal of the American College of Cardiology* 2012;**60**(12):1053-63.
- 231.** Blumenthal JA, Sherwood A, Rogers SD, Babyak MA, Doraiswamy M, Watkins L, et al. Understanding prognostic benefits of exercise and antidepressant therapy for persons with depression and heart disease: the UPBEAT study-rationale, design, and methodological issues. *Clinical Trials (London, England)* 2007;**4**:548.
- 232.** Bonnet LH. Effects of Aerobic Exercise in Combination with Cognitive Therapy on Self Reported Depression [dissertation]. Hempstead, NY: Hofstra University, 2005.
- 233.** Brenes GA, Williamson JD, Messier SP, Rejeski WJ, Pahor M, Ip E, et al. Treatment of minor depression in older adults: a pilot study comparing sertraline and exercise. *Aging & Mental Health* 2007;**11**(1):61-8.
- 234.** Chu IH. Effect of Exercise Intensity during Aerobic Training on Depressive Symptoms in Initially Sedentary Depressed Women [dissertation]. Columbus, OH: The Ohio State University, 2008.
- 235.** Chu IH, Buckworth J, Kirby TE, Emery CF. Effect of exercise intensity on depressive symptoms in women. *Mental Health and Physical Activity* 2009;**2**(1):37-43.
- 236.** Doyne EJ, Ossip-Klein DJ, Bowman ED, Osborn KM, McDougall-Wilson IB, Neimeyer RA. Running versus weight lifting in the treatment of depression. *Journal of Consulting and Clinical Psychology* 1987;**55**(5):748-54.
- 237.** Ossip-Klein DJ, Doyne EJ, Bowman ED, Osborn KM, McDougall-Wilson IB, Neimeyer RA. Effects of running or weight lifting on self-concept in clinically depressed women. *Journal of Consulting and Clinical Psychology* 1989;**7**(1):158-61.
- 238.** Dunn AL, Trivedi MH, Kampert JB, Clark CG, Chambliss HO. The DOSE study: a clinical trial to examine efficacy and dose response of exercise as a treatment for depression. *Controlled Clinical Trials* 2002;**23**:584-603.
- 239.** Dunn AL, Trivedi MH, Kampert JB, Clark CG, Chambliss HO. Exercise treatment for depression: efficacy and dose response. *American Journal of Preventive Medicine* 2005;**28**(1):1-8.

- 240.** Dunn AL, Trivedi MH. Exercise and depression: meeting standards to establish treatment efficacy. In: Proceedings of the 63rd Annual Meeting of the American Psychosomatic Society, 2005 March 2-5, Vancouver, Canada. McLean, VA: American Psychosomatic Society, 2005:A19.
- 241.** Kitzman HE, Trivedi MH, Dunn AL, Galper DI, Kampert JB, Greer TL. The effect of exercise dose on quality of life in depressed adults. In: Proceedings of the 63rd Annual Meeting of the American Psychosomatic Society, 2005 March 2-5, Vancouver, Canada. McLean, VA: American Psychosomatic Society, 2005:A110.
- 242.** Epstein D. Aerobic Activity versus Group Cognitive Therapy: an Evaluative Study of Contrasting Interventions for the Alleviation of Clinical Depression [dissertation]. Reno: University of Nevada, 1986.
- 243.** Fetsch RJ. A Comparison of the Psychological Effects of Running and Transactional Analysis Stroking for the Relief of Reactive Depression in Adults [thesis]. *Dissertation Abstracts International* 1980;**40**(10-B):4999.
- 244.** Foley LS, Prapavessis H, Osuch EA, De Pace JA, Murphy BA, Podolinsky NJ. An examination of potential mechanisms for exercise as a treatment for depression: a pilot study. *Mental Health and Physical Activity* 2008;**1**(2):69-73.
- 245.** Fremont J, Wilcoxon Craighead L. Aerobic exercise and cognitive therapy in the treatment of dysphoric moods. *Cognitive Therapy and Research* 1987;**11**:241-51.
- 246.** Fremont J. Separate and Combined Effects of Cognitively Based Counseling and Aerobic Exercise for the Treatment of Mild and Moderate Depression [dissertation]. PA, USA: Pennsylvania State University, 1983.
- 247.** Gary RA, Dunbar SB, Higgins MK, Musselman DL, Smith AL. Combined exercise and cognitive behavioral therapy improves outcomes in patients with heart failure. *Journal of Psychosomatic Research* 2010;**69**(2):119-31.
- 248.** Greist JH, Klein MH, Eischens RR, Faris J, Gurman AS, Morgan WP. Running as treatment for depression. *Comprehensive Psychiatry* 1979;**20**(1):41-54.
- 249.** Greist JH, Klein MH, Eischens RR, Faris J, Gurman AS, Morgan WP. Running through your mind. *Journal of Psychosomatic Research* 1978;**22**(4):259-94.
- 250.** Greist JH, Klein MH, Eischens RR, Faris J, Gurman AS, Morgan WP. Running as a treatment for depression. In: Proceedings of the 131st Annual Meeting of the American Psychiatric Association; 1978 May 8-12; Atlanta, GA. 1978.
- 251.** Hemat-Far A, Shahsavari A, Mousavi SR. Effects of selected aerobic exercises on the depression and concentrations of plasma serotonin in the depressed female students aged 18 to 25. *Journal of Applied Research* 2012;**12**(1):47-52.
- 252.** Hess-Homeier MJ. A Comparison of Beck's Cognitive Therapy and Jogging as Treatments for Depression [dissertation]. Missoula: University of Montana, 1981.
- 253.** Hoffman JM, Bell KR, Powell JM, Behr J, Dunn EC, Dikmen S, et al. A randomized controlled trial of exercise to improve mood after traumatic brain injury. *American Academy of Physical Medicine and Rehabilitation* 2010;**2**:911-9.
- 254.** Wise ER, Hoffman JM, Powell JM, Bombadier CH, Bell KR. Benefits of exercise maintenance after traumatic brain injury. *Archives of Physical Medicine and Rehabilitation* 2012;**93**(8):1319-23.
- 255.** Klein MH, Greist JH, Gurman RA, Neimeyer RA, Lesser DP, Bushnell NJ. A comparative outcome study of group psychotherapy vs. exercise treatments for depression. *International Journal of Mental Health* 1985;**13**:148-77.
- 256.** Knubben K, Reischies FM, Adli M, Schlattmann P, Bauer M, Dimeo F. A randomised controlled study on the effects of a short-term endurance training programme in patients with major depression. *British Journal of Sports Medicine* 2007;**41**:29-33.
- 257.** Martinsen EW, Medhus A, Sandvik L. Effects of aerobic exercise on depression: a controlled study. *British Medical Journal* 1985;**291**(6488):109.
- 258.** Mather AS, Rodrigues C, Guthrie MF, McHarg AM, Reid IC, McMurdo ME. Effects of exercise on depressive symptoms in older adults with poorly responsive depressive disorder. *British Journal of Psychiatry* 2002;**180**:411-5.
- 259.** McCann IL, Holmes DS. Influence of aerobic exercise on depression. *Journal of Personality and Social Psychology* 1984;**46**(5):1142-7.
- 260.** McNeil JK, LeBlanc EM, Joyner M. The effect of exercise on depressive symptoms in the moderately depressed elderly. *Psychology and Aging* 1991;**6**(3):487-8.
- 261.** Mota-Pereira J, Silverio J, Carvalho S, Ribiero JC, Fonte D, Ramos J. Moderate exercise improves depression parameters in treatment-resistant patients with major depressive disorder. *Journal of Psychiatric Research* 2011;**45**:1005-11.
- 262.** Mutrie N. Exercise as a treatment for moderate depression in the UK National Health Service [abstract]. Proceedings of Sport, Health Psychology and Exercise Symposium, Sports Council and Health Education Authority, London 198896-105.
- 263.** Mutrie N. Exercise as a Treatment for Depression within a National Health Service [dissertation]. PA, USA: Pennsylvania State University, 1986.
- 264.** Nabkasorn C, Miyai N, Sootmongkol A, Junprasert S, Yamamoto H, Arita M, et al. Effects of physical exercise on depression, neuroendocrine stress hormones and physiological fitness in adolescent females with depressive symptoms. *European Journal of Public Health* 2005;**16**:179-84.
- 265.** Orth DK. Clinical Treatments for Depression [dissertation]. Morgantown, WV: West Virginia University, 1979.
- 266.** Pilu A, Sorba M, Hardoy MC, Floris AL, Mannu F, Seruis ML, et al. Efficacy of physical activity in the adjunctive treatment of major depressive disorders: preliminary results. *Clinical*

Practice and Epidemiology in Mental Health 2007;**3**(8):- [DOI: 10.1186/1745-0179-3-8]

267. Carta MG, Hardoy MC, Pilu A, Sorba M, Floris AL, Mannu FA, et al. Improving physical quality of life with group physical activity in the adjunctive treatment of major depressive disorder. *Clinical Practice and Epidemiology in Mental Health* 2008;**4**:1.

268. Pinchasov BB, Shurgaja AM, Grischin OV, Putilov AA. Mood and energy regulation in seasonal and non-seasonal depression before and after midday treatment with physical exercise or bright light. *Psychiatry Research* 2000;**94**:29-42.

269. Reuter M, Mutrie N, Harris DV. Running as an Adjunct to Counseling in the Treatment of Depression [unpublished manuscript]. Pennsylvania State University, 1984.

270. Setaro JL. Aerobic Exercise and Group Counseling in the Treatment of Anxiety and Depression [dissertation]. Maryland: University of Baltimore, 1985.

271. Shahidi M, Mojtahed A, Modabbernia A, Mojtahed M, Shafiabady A, Delavar A, et al. Laughter yoga versus group exercise program in elderly depressed women: a randomized controlled trial. *International Journal of Geriatric Psychiatry* 2011;**26**(3):322-7.

272. Shahidi M. Reply to the letter to the editor re Shahidi M, Mojtahed A, Modabbernia a, et al. 2011. Laughter yoga versus group exercise program in elderly depressed women: a randomized controlled trial. *Int J Geriatr Psychiatr* 26: 322-327. First things first! Caveats in research on "laughter yoga". *International Journal of Geriatric Psychiatry* 2012;**27**(8):875-6.

273. Sims J, Galea M, Taylor N, Dodd K, Jespersen S, Joubert L, et al. Regenerate: assessing the feasibility of a strength-training program to enhance the physical and mental health of chronic post stroke patients with depression. *International Journal of Geriatric Psychiatry* 2009;**24**(1):76-83.

274. Teoh V, Sims J, Milgrom J. Psychosocial predictors of quality of life following a stroke. *Topics in Stroke Rehabilitation* 2009;**16**(2):157-66.

275. Singh NA, Clements KM, Fiatarone MA. A randomized controlled trial of progressive resistance training in depressed elders. *Journals of Gerontology. Series A, Biological Sciences and Medical Sciences* 1997;**52**(1):M27-35.

276. Singh NA, Clements KM, Fiatarone MA. A randomized controlled trial of the effect of exercise on sleep. *Sleep* 1997;**20**(2):95-100.

277. Singh NA, Stavriou TM, Scarbek Y, Galambos G, Liber C, Fiatarone Singh MA. A randomised controlled trial of high versus low intensity weight training versus general practitioner care for clinical depression in older adults. *Journals of Gerontology. Series A, Biological Sciences and Medical Sciences* 2005;**60A**:768-76.

278. Veale D, Le Fevre K, Pantelis C, De Souza V, Mann A, Sargeant A. Aerobic exercise in the adjunctive treatment of

depression: a randomized controlled trial. *Journal of the Royal Society of Medicine* 1992;**85**(9):541-4.

279. Williams CL, Tappen RM. Exercise training for depressed older adults with Alzheimer's disease. *Aging & Mental Health* 2008;**12**(1):72-80.

280. Abascal L. The effect of depression and adherence in a dietary and physical activity intervention for overweight and obese adults [dissertation]. University of California, San Diego and San Diego State University 2008.

281. Abd El-Kader SM, Al-Jiffri OH. Exercise alleviates depression related systemic inflammation in chronic obstructive pulmonary disease patients. *African Historical Studies* 2016;**16**(4):1078-88.

282. Abt K. The effects of a group exercise intervention in the adjunctive treatment of clinical depression. Doctoral Dissertation (d-Scholarship.pitt.edu/8699/) 2005.

283. Aidar FJ, Jaco de Oliveira R, Gama de Matos D, Chilibeck PD, de Souza RF, Carneiro AL, et al. A randomized trial of the effects of an aquatic exercise program on depression, anxiety levels, and functional capacity of people who suffered an ischemic stroke. *Journal of Sports Medicine and Physical Fitness* 2018;**58**(7-8):1171-7.

284. Ankandere M, Demir B. The effect of dance over depression. *Collegium Antropologicum* 2011;**35**(3):651-6.

285. Alsaraireh FA, Aloush SM. Mindfulness meditation versus physical exercise in the management of depression among nursing students. *Journal of Nursing Education* 2017;**56**(10):599-604.

286. Annesi JJ, Gorjala S. Association of reduction in waist circumference with normalization of mood in obese women initiating exercise supported by the Coach Approach protocol. *Southern Medical Journal* 2010;**103**(6):517-21.

287. Arcos-Carmona IM, Castro-Sanchez AM, Mataran-Penarrocha GA, Gutierrez-Rubio AB, Ramos-Gonzalez E, Moreno-Lorenzo C. Effects of aerobic exercise program and relaxation techniques on anxiety, quality of sleep, depression, and quality of life in patients with fibromyalgia: A randomised controlled trial. *Medicina Clinica* 2011;**137**(9):398-401.

288. Armstrong K, Edwards H. The effects of exercise and social support on mothers reporting depressive symptoms: a pilot randomised controlled trial. *International Journal of Mental Health Nursing* 2003;**12**(2):130-8.

289. Armstrong K, Edwards H. The effectiveness of a pram-walking exercise programme in reducing depressive symptomatology for postnatal women. *International Journal of Nursing Practice* 2004;**10**:177-94.

290. Asbury EA, Kanji N, Ernst E, Barbir M, Collins P. Autogenic training to manage symptomatology in women with chest pain and normal coronary arteries. *Menopause (New York, N.Y.)* 2009;**16**(1):60-5.

- 291.** Aylin K, Arzu D, Sabri S, Handan TE, Ridvan A. The effect of combined resistance and home-based walking exercises in type 2 diabetes patients. *International Journal of Diabetes in Developing Countries* 2009;**29**(4):159-65.
- 292.** Badger T, Segrin C, Dorros SM, Meek P, Lopez AM. Depression with anxiety in women with breast cancer and their partners. *Nursing Research* 2007;**56**(1):44-53.
- 293.** Baker MK, Kennedy DJ, Bohle PL, Campbell DS, Knapman L, Grady J, et al. Efficacy and feasibility of a novel tri-modal robust exercise prescription in a retirement community: a randomised, controlled trial. *Journal of the American Geriatrics Society* 2007;**55**:1-10.
- 294.** Bashir N, Hassan Z, Zaryyab, Saleem M, Atif MM, Mehdi Z. Effects if aerobic exercises on postmenopausal depression: randomised controlled trial. *Pakistan Journal of Medical and Health Sciences* 2022;**16**(12):70-3.
- 295.** Berke EM, Gottlieb LM, Moudon AV, Larson EB. Protective association between neighborhood walkability and depression in older men. *Journal of the American Geriatrics Society* 2007;**55**:526-33.
- 296.** Blumenthal JA, Babyak MA, O'Connor C, Keteyian S, Landzberg J, Howlett J, et al. Effects of exercise training on depressive symptoms in patients with chronic heart failure. *JAMA* 2012;**308**(5):465-74.
- 297.** Bosch PR, Traustadóttir T, Howard P, Matt KS. Functional and physiological effects of yoga in women with rheumatoid arthritis: a pilot study. *Alternative Therapies in Health and Medicine* 2009;**15**(4):24-31.
- 298.** Boyll JF. Effects of Active and Passive Electronic Muscle Stimulation on Self-Concept, Anxiety and Depression [dissertation]. Flagstaff: Northern Arizona University, 1986.
- 299.** Brittle N, Patel S, Wright C, Baral S, Versfeld P, Sackley C. An exploratory cluster randomised controlled trial of group exercise on mobility and depression in care home residents. *Clinical Rehabilitation* 2009;**23**:146-54.
- 300.** Broberg L, Tabor A, Rosthøj S, Backhausen M, Frokjaer VG, Damm P, et al. Effect of supervised group exercise on psychological well-being among pregnant women with or at risk of depression (the EWE Study): A randomised controlled trial. *Acta Obstetrica et Gynecologica Scandinavica* 2021;**100**(1):129-38.
- 301.** Burton NW, Pakenham KI, Brown WJ. Evaluating the effectiveness of psychosocial resilience training for heart health, and the added value of promoting physical activity: a cluster randomized trial of the READY program. *BMC Public Health* 2009;**9**:427.
- 302.** Carneiro LS, Fonseca AM, Vieira-Coelho MA, Mota MP, Vasconcelos-Raposo J. Effects of structured exercise and pharmacotherapy vs. pharmacotherapy for adults with depressive symptoms: A randomized clinical trial. *Journal of Psychiatric Research* 2015;**71**:48-55. [PMID: 26522871]
- 303.** Carneiro LS, Fonseca AM, Serrao P, Mota MP, Vasconcelos-Raposo J, Vieira-Coelho MA. Impact of physical exercise on catechol-O-methyltransferase activity in depressive patients: A preliminary communication. *Journal of Affective Disorders* 2016;**193**:117-22.
- 304.** Carneiro LS, Mota MP, Vieira-Coelho MA, Alves RC, Fonseca AM, Vasconcelos-Raposo J. Monoamines and cortisol as potential mediators of the relationship between exercise and depressive symptoms. *European Archives of Psychiatry and Clinical Neuroscience* 2017;**267**(2):117-21.
- 305.** Carney RM, Templeton B, Hong BA, Harter HR, Hagberg JM, Schechtman KB, et al. Exercise training reduces depression and increases the performance of pleasant activities in haemodialysis patients. *Nephron* 1987;**47**:194-8.
- 306.** Chan JS M, Ho RT H, Wang CW, Yuen LP, Sham JS T, Chan CL W. Effects of qigong exercise on fatigue, anxiety, and depressive symptoms of patients with chronic fatigue syndrome-like illness: A randomized controlled trial. *Evidence-Based Complementary and Alternative Medicine* 2013;**doi: 10.1155/2013/485341**...
- 307.** Chang MC, Lee AY, Kwak S, Kwak SG. Effect of resistance exercise on depression in mild alzheimer disease patients with sarcopenia. *American Journal of Geriatric Psychiatry* 2020;**28**(5):587-9.
- 308.** Chang PS, Lu Y, Nguyen CM, Suh Y, Luciani M, Ofner S, et al. Effects of qigong exercise on physical and psychological health among african americans.. *Western Journal of Nursing Research* 2021;**43**(6):551-62.
- 309.** Chapman JJ, Coombes JS, Brown WJ, Khan A, Chamoli S, Pachana NA, et al. The feasibility and acceptability of high-intensity interval training for adults with mental illness: A pilot study. *Mental Health and Physical Activity* 2017;**13**:40-8.
- 310.** Chen KM, Chen MH, Chao HC, Hung HM, Lin HS, Li CH. Sleep quality, depression state, and health status of older adults after silver yoga exercises: cluster randomized trial. *International Journal of Nursing Studies* 2009;**46**(2):154-63.
- 311.** Christensen H. Beyond ageing project: a randomized controlled trial to assess the benefit of improving mental health literacy in depression as well as the effect of folate and B12, and physical activity, in preventing major depression and cognitive impairment among australians aged 65-74 years. <http://clinicaltrials.gov/ct2/show/NCT00214682> 2005;...
- 312.** Christensen SB, Dall CH, Prescott E, Pedersen SS, Gustafsson F. A high-intensity exercise program improves exercise capacity, self-perceived health, anxiety and depression in heart transplant recipients: A randomised, controlled trial. *Journal of Heart and Lung Transplantation* 2012;**31**(1):106-7.
- 313.** Clegg A, Barber S, Young J, Forster A, Iliffe S. The home-based older people's exercise (HOPE) trial: study protocol for a randomised controlled trial. *Trials (Electronic Resource)* 2011;**12**(143).
- 314.** Courneya KS, Segal RJ, Gelmon K, Reid RD, Mackey JR, Friedenreich CM, et al. Six-month follow-up of patient-related

outcomes in a randomized controlled trial of exercise training during breast cancer chemotherapy. *Cancer Epidemiology, Biomarkers and Prevention* 2007;**16**(12):2572-8.

315. Courneya KS, McKenzie DC, Gelmon K, MacKey JR, Reid RD, Yasui Y, et al. A multicenter randomized trial of the effects of exercise dose and type on psychosocial distress in breast cancer patients undergoing chemotherapy. *Cancer Epidemiology Biomarkers and Prevention* 2014;**23**(5):857-64.

316. Dalton RB. Effects of Exercise and Vitamin B12 Supplementation on the Depression Scale Scores of a Wheelchair Confined Population [dissertation]. Columbia, MO: University of Missouri, 1981.

317. Danielsson L, Waern M, Hensing G, Holmgren K. Work-directed rehabilitation or physical activity to support work ability and mental health in common mental disorders: a pilot randomized controlled trial. *Clinical Rehabilitation* 2020;**34**(2):170-81.

318. Demiralp M, Oflaz F. The effect of relaxation training on symptoms of anxiety and depression in patients with breast cancer. *TAF Preventive Medicine Bulletin* 2011;**10**(2):165-74.

319. DeVaney SB. Comparative Effects of Exercise Reduction and Relaxation Training on Type a Behaviour and Dysphoric Mood States in Habitual Aerobic Exercisers [dissertation]. Greensboro: University of North Carolina, 1991.

320. DiLorenzo TM, Bargman EP, Stucky-Ropp R, Brassington GS, Frensch PA, LaFontaine T. Long-term effects of aerobic exercise on psychological outcomes. *Preventive Medicine* 1999;**28**(1):75-85.

321. Dotson VM, Hsu FC, Langaee TY, McDonough CW, King AC, Cohen RA, et al. Genetic moderators of the impact of physical activity on depressive symptoms. *Journal of Frailty & Aging* 2016;**5**(1):6-14.

322. Eby JM. An Investigation into the Effects of Aerobic Exercise on Anxiety and Depression [dissertation]. Toronto, ON: University of Toronto, 1985.

323. Echeta Alice. Walking as a patient-centered exercise and its effects on depression in adults. *Dissertation Abstracts International: Section B: The Sciences and Engineering* 2018;**79**(2-B(E)):No Pagination Specified.

324. Edwards R, Linck P, Hounsborne N, Raisanen L, Williams N, Moore L, et al. Cost-effectiveness of a national exercise referral programme for primary care patients in Wales: results of a randomised controlled trial. *BMC Public Health* 2013;**13**(1):1-11.

325. Edwards MK, Loprinzi PD. Experimental effects of brief, single bouts of walking and meditation on mood profile in young adults. *Health Promotion Perspectives* 2018;**8**(3):171-8.

326. Elavsky S, McAuley E. Lack of perceived sleep improvement after 4-month structured exercise programs. *Menopause (New York, N.Y.)* 2007;**14**(3):535-40.

327. El-Rafie MM, Khafagy GM, Gamal MG. Effect of aerobic exercise during pregnancy on antenatal depression. *International Journal of Women's Health* 2016;**8**:53-7.

328. Emery CF, Blumenthal JA. Perceived change among participants in an exercise program for older adults. *Gerontologist* 1990;**30**(4):516-21.

329. Emery CF, Gatz M. Psychological and cognitive effects of an exercise program for community-residing older adults. *Gerontologist* 1990;**30**(2):184-8.

330. Ersek M, Turner JA, Cain KC, Kemp CA. Results of a randomized controlled trial to examine the efficacy of a chronic pain self-management group for older adults. *Pain* 2009;**138**(1):29-40.

331. Fisher E, Wood SJ, Upthegrove R, Aldred S. Designing a feasible exercise intervention in first-episode psychosis: Exercise quality, engagement and effect. *Psychiatry Research* 2020;**286**(doi: 10.1016/j.psychres.2020.112840):...

332. Fox KR, Stathi A, McKenna J, Davis MG. Physical activity and mental well-being in older people participating in the better ageing project. *European Journal of Applied Physiology* 2007;**100**:591-602.

333. Gao L, Zhang L, Qi H, Petridis L. Middle-aged female depression in perimenopausal period and square dance intervention. *Psychiatria Danubina* 2016;**28**(4):372-8.

334. Gary R, Lee SY. Physical function and quality of life in older women with diastolic heart failure: effects of a progressive walking program on sleep patterns. *Progress in Cardiovascular Nursing* 2007;**22**:72-80.

335. Ghroubi S, Elleuch H, Chikh T, Kaffel N, Abid M, Elleuch MH. Physical training combined with dietary measures in the treatment of adult obesity. A comparison of two protocols. *Annals of Physical and Rehabilitation Medicine* 2009;**52**:394-413.

336. Gottlieb SS, Kop WJ, Ellis SJ, Binkley P, Howlett J, O'Connor C, et al. Relation of depression to severity of illness in heart failure (from heart failure and a controlled trial investigating outcomes of exercise training [HF-ACTION]). *American Journal of Cardiology* 2009;**103**:1285-9.

337. Gusi N, Reyes MC, Gonzalez-Guerrero JL, Herrera E, Garcia JM. Cost-utility of a walking programme for moderately depressed, obese, or overweight elderly women in primary care: a randomised controlled trial. *BMC Public Health* 2008;**8**:231.

338. Villaverde Gutierrez C, Torres Luque G, Medina, Ábalos Medina GM, Argente del Castillo MJ, Guisado IM, et al. Influence of exercise on mood in postmenopausal women. *Journal of Clinical Nursing* 2012;**21**(7-8):923-8.

339. Haffmans PM, Kleinsman AC, Van Weelden C, Huijbrechts I, Hoencamp E. Comparing running therapy with physiotherapy in the treatment of mood disorders. *Acta Neuropsychiatrica* 2006;**18**:173-6.

- 340.** Hallgren M, Helgadottir B, Herring MP, Zeebari Z, Lindefors N, Kaldo V, et al. Exercise and internet-based cognitive-behavioural therapy for depression: multicentre randomised controlled trial with 12-month follow-up. *British Journal of Psychiatry* 2016;**209**(5):414-20. [PMID: 27609813]
- 341.** Hallgren M, Kraepelien M, Ojehagen A, Lindefors N, Zeebari Z, Kaldo V, et al. Physical exercise and internet-based cognitive-behavioural therapy in the treatment of depression: Randomised controlled trial. *British Journal of Psychiatry* 2015;**207**(3):227-34.
- 342.** Yacaman-Mendez D, Hallgren M, Forsell Y. Childhood adversities, negative life events and outcomes of non-pharmacological treatments for depression in primary care: A secondary analysis of a randomized controlled trial. *Journal of Psychiatric Research* 2019;**110**:152-8.
- 343.** Rahman MS, Millischer V, Zeebari Z, Forsell Y, Lavebratt C. BDNF Val66Met and childhood adversity on response to physical exercise and internet-based cognitive behavioural therapy in depressed Swedish adults. *Journal of Psychiatric Research* 2017;**93**:50-8.
- 344.** Bondesson E, Jöud A, Stigmar K, Ringqvist Å, Kraepelien M, Kaldo V, et al. Utilization of healthcare and prescription medicines after non-pharmacological interventions for depression - A 3-year register follow-up of an RCT in primary care. *Preventive Medicine Reports* 2021;**25**(101658 (doi: 10.1016/j.pmedr.2021.101658)):
- 345.** Helgadottir B, Hallgren M, Ekblom O, Forsell Y. Training fast or slow? Exercise for depression: A randomized controlled trial. *Preventive Medicine* 2016;**91**:123-31. [PMID: 27514246]
- 346.** Rahman MS, Zhao X, Liu JJ, Torres EQ, Tibert Bb, Kumar P, et al. Exercise reduces salivary morning cortisol levels in patients with depression. *Molecular Neuropsychiatry* 2018;**4**:196-203.
- 347.** Hallgren M, Helgadottir B, Herring MP, Zeebari Z, Lindefors N, Kaldo V, et al. Exercise and internet-based cognitive-behavioural therapy for depression: multicentre randomised controlled trial with 12-month follow-up. *British Journal of Psychiatry* 2016;**209**(5):414-20.
- 348.** Hannaford CP, Harrell EH, Cox K. Psychophysiological effects of a running program on depression and anxiety in a psychiatric population. *Psychological Record* 1988;**38**:37-48.
- 349.** Haugen TS, Stavem K. Rehabilitation in a warm versus a colder climate in chronic obstructive pulmonary disease. *Journal of Cardiopulmonary Rehabilitation and Prevention* 2007;**27**:50-6.
- 350.** Helgadottir B, Hallgren M, Ekblom O, Forsell Y. Training fast or slow? Exercise for depression: A randomized controlled trial. *Preventive Medicine* 2016;**91**:123-31.
- 351.** Helgadottir B, Owen N, Dunstan David W, Ekblom O, Hallgren Mats; Forsell Y. Changes in physical activity and sedentary behavior associated with an exercise intervention in depressed adults. *Psychology of Sport and Exercise* 2017;**30**:10-8.
- 352.** Helgadottir B, Forsell Y, Hallgren M, Moller J, Ekblom O. Long-term effects of exercise at different intensity levels on depression: A randomized controlled trial. *Preventive Medicine* 2017;**105**:37-46.
- 353.** Hembree LD. Exercise and Its Effect on Hopelessness and Depression in an Aging Female Population in Eastern Oklahoma [dissertation]. Fayetteville: University of Arkansas, 2001.
- 354.** Herrera F. Efficacy of a psychological intervention for depression in patients on haemodialysis [Eficacia de la intervención psicológica en la depresión del paciente en hemodiálisis]. *Psiquis* 1994;**15**(4):175-8.
- 355.** Huang TT, Liu CB, Tsai YH, Chin YF, Wong CH. Physical fitness exercise versus cognitive behavior therapy on reducing the depressive symptoms among community-dwelling elderly adults: A randomized controlled trial. *International Journal of Nursing Studies* 2015;**52**(10):1542-52.
- 356.** Hughes JR, Casal DC, Leon AS. Psychological effects of exercise: a randomised cross-over trial. *Journal of Psychosomatic Research* 1986;**30**:355-60.
- 357.** Imayama I, Alfano CM, Mason C, Wang C, Duggan C, Campbell KL, et al. Weight and metabolic effects of dietary weight loss and exercise interventions in postmenopausal antidepressant medication users and non-users: A randomized controlled trial. *Preventive Medicine* 2013;**57**(5):525-32.
- 358.** Imboden C, Gerber M, Beck J, Holsboer-Trachsler E, Pühse U, Hatzinger M. Aerobic exercise or stretching as add-on to inpatient treatment of depression: Similar antidepressant effects on depressive symptoms and larger effects on working memory for aerobic exercise alone. *Journal of Affective Disorders* 2020;**276**:866-876 (DOI: 10.1016/j.jad.2020.07.052).
- 359.** Jacobsen P, Phillips K, Jim H, Faul LA, Small B, Meade C, et al. Self-directed stress management and exercise training for cancer chemotherapy patients. In: *Psycho-Oncology (abstracts of the 9th Annual Conference of the American Psychosocial Oncology Society)*. Vol. 21:17. 2012.
- 360.** Johansson M, Hassmen P, Jouper J. Acute effects of Qigong exercise on mood and anxiety. *Sport, Exercise and Performance Psychology* 2011;**15**(2):199-207.
- 361.** Justine M, Hamid TA. A multicomponent exercise program for institutionalized older adults. *Journal of Gerontological Nursing* 2010;**36**(10):32-41.
- 362.** Karlsson MR, Edström-Plüss C, Held C, Henriksson P, Billing E, Wallén NH. Effects of expanded cardiac rehabilitation on psychosocial status in coronary artery disease with focus on type D characteristics. *Journal of Behavioral Medicine* 2007;**30**:253-61.
- 363.** Katayama A, Kanda K, Hase A, Miyatake N. Face-to-Face Exercises Performed by Instructors to Improve the Mental Health of Japanese in the Community-A Randomized Control Trial. *Medicina (Kaunas, Lithuania)* 2020;**56**(8):404.
- 364.** Kerr J, Patrick K, Norman G, Stein MB, Calfas K, Zabinski M, et al. Randomized control trial of a behavioral intervention

for overweight women: impact on depressive symptoms. *Depression and Anxiety* 2008;**25**:555-8.

365. Kim KB, Cohen SM, Oh HK, Sok SR. The effects of meridian exercise on anxiety, depression and self-esteem of female college students in Korea. *Holistic Nursing Practice* 2004;**18**:230-4.

366. Kim YS, O'Sullivan DM, Shin SK. Can 24weeks strength training reduce feelings of depression and increase neurotransmitter in elderly females? *Experimental Gerontology* 2019;**115**:62-8.

367. Kim, H, Lee, K, Lee, YH, et al. The effectiveness of a mobile phone-based physical activity program for treating depression, stress, psychological well-being, and quality of life among adults: quantitative study. *JMIR MHealth and UHealth* 2023;**11**:doi: 10.2196/46286.

368. Knapen J, Van De Vliet P, Van Coppenolle H, David A, Peuskens J, Knapen K, et al. The effectiveness of two psychomotor therapy programmes on physical fitness and physical concept in nonpsychotic psychiatric patients: a randomised controlled trial. *Clinical Rehabilitation* 2003;**17**:637-47.

369. Knapen J, Van Coppenolle H, Peuskens J, Pieters G, Knapen K. Comparison of changes in physical fitness, physical self-concept, global self-esteem, depression and anxiety following two different psychomotor therapy programs in non-psychotic psychiatric inpatients. In: *Concept of Self in Education, Family and Sports*. Nova, 2006:Chapter 4.

370. Krogh J, Gøtze JP, Jørgensen MB, Kristensen LO, Kistorp C, Nordentoft M. Copeptin during rest and exercise in major depression. *Journal of Affective Disorders* 2013;**151**(1):284-90.

371. Krogh J, Benros ME, Jørgensen MB, Vesterager L, Elfving B, Nordentoft M. The association between depressive symptoms, cognitive function, and inflammation in major depression. *Brain, Behavior, and Immunity* 2014;**35**:70-6.

372. Kubesch S, Bretschneider V, Freudenmann R, Weiderhammer N, Lehmann M, Spitzer M, et al. Aerobic endurance exercise improves executive function in depressed patients. *Journal of Clinical Psychiatry* 2003;**64**:1005-12.

373. Kulcu DG, Kurtas Y, Tur BS, Gülec S, Seckin B. The effect of cardiac rehabilitation on quality of life, anxiety and depression in patients with congestive heart failure. A randomized controlled trial, short-term results. *Europa Medicophysica* 2007;**43**:489-97.

374. Kupecz DB. Effects of a Structured Exercise Program in Older Veteran Patients [dissertation]. Greeley, CO: University of Northern Colorado, 2001.

375. Lai SM, Studenski S, Richards L, Perera S, Reker D, Rigler S, et al. Therapeutic exercise and depressive symptoms after stroke. *Journal of the American Geriatrics Society* 2006;**54**(2):240-7.

376. Latimer AE, Martin Ginis KA, Hicks AL, McCartney N. An examination of the mechanisms of exercise-induced change

in psychological well-being among people with spinal cord injury. *Journal of Rehabilitation Research and Development* 2004;**41**(5):643-51.

377. Lautenschlager NT, Cox KL, Flicker L, Foster JK, Van Bockxmeer FM, Xiao J, et al. Effect of physical activity on cognitive function in older adults at risk for Alzheimer disease. *JAMA* 2008;**300**(9):1027-37.

378. Leppämäki SJ, Partonen TT, Hurme J, Haukka JK, Lonnqvist JK. Randomised trial of the efficacy of bright-light exposure and aerobic exercise on depressive symptoms and serum lipids. *Journal of Clinical Psychiatry* 2002;**63**(4):316-21.

379. Leppämäki S, Haukka J, Lonnqvist J, Partonen T. Drop-out and mood improvement: a randomised controlled trial with light exposure and physical exercise. *BMC Psychiatry* 2004;**4**:22.

380. Levendoğlu F, Altintepe L, Okudan N, Uğurlu H, Gökbel H, Tonbul Z, et al. A twelve exercise program improves the psychological status, quality of life and work capacity in hemodialysis patients. *Journal of Nephrology* 2004;**17**:826-32.

381. Lever-van Milligen B. Mood treatment with antidepressants or running (MOTAR). <http://www.trialregister.nl/admin/rctview.asp?TC=3460> June 2012.

382. Levin G, Greenwood K, Newton R. Exercise decreases depression and improves quality of life in cancer survivors with elevated depressive symptomatology. *Supportive Care in Cancer* 2016;**24** (1 Supplement 1):S70.

383. Levin GT, Greenwood KM, Singh F, Newton RU. Modality of exercise influences rate of decrease in depression for cancer survivors with elevated depressive symptomatology. *Supportive Care in Cancer* 2018;**26**(5):1597-606.

384. Levinger I, Selig S, Goodman C, Jerums G, Stewart A, Hare DL. Resistance training improves depressive symptoms in individuals at high risk for type 2 diabetes. *Journal of Strength and Conditioning Research* 2011;**25**(8):2328-33.

385. Lin MR, Wolf SL, Hwang HF, Gong SY, Chen CY. A randomized, controlled trial of fall prevention programs and quality of life in older fallers. *Journal of the American Geriatrics Society* 2007;**55**:499-506.

386. Littbrand H, Carlsson M, Lundin-Olsson L, Lindelof N, Haglin L, Gustafson Y, et al. Effect of a high-intensity functional exercise program on functional balance: Preplanned subgroup analyses of a randomized controlled trial in residential care facilities. *Journal of the American Geriatrics Society* 2011;**59**(7):1274-82.

387. Littlecott HJ, Moore GF, Moore L, Murphy S. Psychosocial mediators of change in physical activity in the Welsh national exercise referral scheme: secondary analysis of a randomised controlled trial. *International Journal of Behavioral Nutrition and Physical Activity* 2014;**11**:109. [PMID: 25209188]

388. Lok N, Lok S, Canbaz M. The effect of physical activity on depressive symptoms and quality of life among elderly nursing home residents: Randomized controlled trial. *Archives of Gerontology and Geriatrics* 2017;**70**:92-8.

- 389.** Lolak S, Conors GL, Sherican MJ, Wise TN. Effects of progressive muscle relaxation training on anxiety and depression in patients enrolled in an outpatient pulmonary rehabilitation program. *Psychotherapy and Psychosomatics* 2008;**77**(2):119-25.
- 390.** Lovrenčić MV, Pibernik-Okanović M, Šekerija M, Prašek M, Ajduković D, Kos J, et al. Improvement in depressive symptoms is associated with reduced oxidative damage and inflammatory response in type 2 diabetic patients with subsyndromal depression: the results of a randomized controlled trial comparing psychoeducation, physical exercise, and enhanced treatment as usual. *International Journal of Endocrinology* 2015;doi: **10.1155/2015/210406**...
- 391.** Luo Y, Yang Z, Chen X, Huang Y. Effect of a video-based exercise intervention on depression and sleep conditions of peritoneal dialysis patients. *Clinical Nephrology* 2023;**99**(2):58-68.
- 392.** Machado LA, Azevedo DC, Capanema MB, Neto TN, Cerceau DM. Client-centered therapy vs exercise therapy for chronic low back pain: a pilot randomized controlled trial in Brazil. *Pain Medicine (Malden, Mass.)* 2007;**8**(3):251-8.
- 393.** Mackintosh S. Improving mood through physical activity for carers and care recipients (IMPACCT): protocol for a randomised trial. *Journal of Physiotherapy* 2013;**59**(2):125.
- 394.** Mailey EL, Wojcicki TR, Moti RW, Hu L, Strauser DR, Collins KD, et al. Internet-delivered physical activity intervention for college students with mental health disorders: a randomised pilot trial. *Psychology Health and Medicine* 2010;**15**(6):646-59.
- 395.** Majumder P, Sharma I, Vostanis P, Bone CT. The effect of aerobic exercise in the maintenance treatment of depression. *BJPsych International* 2015;**12**:S3-S6.
- 396.** Makizako H, Tsutsumimoto K, Doi T, Makino K, Nakakubo S, Liu-Ambrose T, et al. Exercise and horticultural programs for older adults with depressive symptoms and memory problems: a randomized controlled trial [2019 Dec 30;9(1): 99. doi: 10.3390/jcm9010099. PMID: 31906021; PMCID: PMC7019282]. *J Clin Med*. 2019;**9**(1):99.
- 397.** Martin JK, Church TS, Thompson AM, Earnest CP, Blair SN. Exercise dose and quality of life. *Archives of Internal Medicine* 2009;**169**(3):269-78.
- 398.** Matthews MM, Hsu FC, Walkup MP, Barry LC, Patel KV, Blair SN. Depressive symptoms and physical performance in the lifestyle interventions and independence for elders pilot study. *Journal of the American Geriatrics Society* 2011;**59**(3):495-500.
- 399.** Meyer JD, Murray TA, Brower CS, Cruz-Maldonado GA, Perez ML, Ellingson LD, et al. Magnitude, timing and duration of mood state and cognitive effects of acute moderate exercise in major depressive disorder. *Psychology of Sport and Exercise* 2022;**61**:https://doi.org/10.1016/j.psychsport.2022.102172.
- 400.** Midtgaard J, Stage M, Moller T, Andersen C, Quist M, Rorth M, et al. Exercise may reduce depression but not anxiety in self-referred cancer patients undergoing chemotherapy. Post-hoc analysis of data from the 'Body & Cancer' trial. *Acta Oncologia* 2011;**50**(5):660-9.
- 401.** Milani RV, Lavie CJ. Impact of cardiac rehabilitation on depression and its associated mortality. *American Journal of Medicine* 2007;**120**:799-806.
- 402.** Morey MC, Dubbert PM, Doyle ME, MacAller H, Crowley GM, Kuchibhatla M, et al. From supervised to unsupervised exercise: factors associated with exercise adherence. *Journal of Aging and Physical Activity* 2003;**11**(3):351-68.
- 403.** Motl RW, Konopack JF, McAuley E, Elavasky S, Jerome GJ, Marquez DX. Depressive symptoms among older adults: long-term reduction after a physical activity intervention. *Journal of Behavioral Medicine* 2005;**28**(4):384-94.
- 404.** Mudge A. The impact of a disease management programme including a supervised exercise programme versus disease management alone on death, readmissions, depression and functional status on patients with a recent hospitalisation for heart failure. Australian New Zealand Clinical Trials Registry 2008.
- 405.** Mutrie N, Campbell AM, Whyte F, McConnachie A, Emslie C, Lee L, et al. Benefits of supervised group exercise programme for women being treated for early stage breast cancer: pragmatic randomised controlled trial. *BMJ (Clinical Research Ed.)* 2007;**334**(7592):517.
- 406.** Nct02885285. The impact of three distinct exercise types on fatigue, anxiety, and depression in parkinson's disease. <https://clinicaltrials.gov/show/nct02885285> 2016.
- 407.** Nct03542396. Evaluation of a group therapeutic exercise program. <https://clinicaltrials.gov/show/nct03542396> 2018.
- 408.** NCT03833349. The impact of three distinct exercise types on fatigue, anxiety, and depression in parkinson's disease (pdexercise). [ClinicalTrials.gov](https://clinicaltrials.gov) 2016.
- 409.** Neidig JL. Aerobic Exercise Training: Effects on Depressive Symptoms in HIV Infected Adults (Immune Deficiency, Exercise Training) [dissertation]. Columbus, OH: The Ohio State University, 1998.
- 410.** Neidig JL, Smith BA, Brashers DE. Aerobic exercise training for depressive symptom management in adults living with HIV infection. *Journal of Association of Nurses in AIDS Care* 2003;**14**(2):30-40.
- 411.** Smith BA, Neidig JL, Nickel JT, Mitchell GL, Para MF, Fass RJ. Aerobic exercise: effects on parameters related to fatigue, dyspnea, weight and body composition in HIV-infected adults. *AIDS (London, England)* 2001;**15**(6):693-701.
- 412.** Netz Y, Yaretzki A, Salganik I, Jacob T, Finkeltov B, Argov E. The effect of supervised physical activity on cognitive and affective state of geriatric and psychogeriatric in-patients. *Clinical Gerontologist* 1994;**15**(1):47-56.
- 413.** Neuberger GB, Aaronson LS, Gajewski B, Embretson SE, Cagle PE, Loudon JK, et al. Predictors of exercise and effects of exercise on symptoms, function, aerobic fitness, and disease

outcomes of rheumatoid arthritis. *Arthritis and Rheumatism* 2007;**57**(6):943-52.

414. Neunhauserer D, Sturm J, Baumgartlinger MM, Niederseer D, Ledl-Kurkowski E, Steidle E, et al. Hiking in suicidal patients: neutral effects on markers of suicidality. *American Journal of Medicine* 2013;**126**(10):927-30. [PMID: 23953873]

415. Ng TP, Nyunt MS Z, Feng L, Feng L, Niti M, Tan BY, et al. Multi-domains lifestyle interventions reduces depressive symptoms among frail and pre-frail older persons: randomized controlled trial. *Journal of Nutrition, Health & Aging* 2017;**21**(8):918-26.

416. Nguyen HQ, Carrieri-Kohlman V, Demir-Deviren S, Stulbarg MS. Are the improvements in fatigue, vigor and depression sustained with a home walking programme? In: Proceedings of the American Thoracic Society 2001 International Conference; May 18-23; San Francisco, California. 2001.

417. Nitzsche N, Baumgartel L, Mueller K, Schulz H. Effects of exercise therapy in hospitalized patients with major depressive disorder on heart rate variability - A pilot study. *Journal of Experimental and Integrative Medicine* 2015;**5**(3):137-43.

418. Nyunt MS Z, Niti M, Feng L, Tan BY, Chan G, Chan SM, et al. Singapore frailty intervention trial: Effect of frailty reversal on reducing depressive symptoms. *Annals of the Academy of Medicine, Singapore* 2014; **Conference: NUHS Academic Psychiatry Conference 2014 Singapore Singapore. Conference Start: 20141031 Conference End: 20141101. Conference Publication:**S29-30.

419. Oeland AM, Laessoe U, Olesen AV, Munk-Jorgensen P. Impact of exercise on patients with depression and anxiety. *Nordic Journal of Psychiatry* 2010;**64**(3):210-7.

420. Oertel-Knochel V, Mehler P, Thiel C, Steinbrecher K, Malchow B, Tesky V, et al. Effects of aerobic exercise on cognitive performance and individual psychopathology in depressive and schizophrenia patients. *European Archives of Psychiatry and Clinical Neuroscience* 2014;**264**(7):589-604. [PMID: 24487666]

421. Ouzouni S, Kouidi E, Grekas D, Deligiannis A. Effect of intradialytic exercise training on health-related quality of life indices in haemodialysis patient. *Clinical Rehabilitation* 2009;**23**:53-63.

422. Pakkala I, Read S, Leinonen R, Hirvensalo M, Lintunen T, Rantanen T. The effects of physical activity counseling on mood among 75- to 81-year-old people: a randomized controlled trial. *Preventive Medicine* 2008;**46**:412-8.

423. Palmer JA, Michiels K, Thigpen B. Effects of type of exercise on depression in recovering substance abusers. *Perceptual and Motor Skills* 1995;**80**(2):523-30.

424. Pelham TW, Compagna PD, Ritvo PG, Birnie WA. The effects of exercise therapy on clients in a psychiatric rehabilitation program. *Psychosocial Rehabilitation Journal* 1993;**16**(4):75-83.

425. Penninx B, Rejeski WJ, Pandya J, Miller ME, Bari MD, Applegate WB, et al. Exercise and depressive symptoms: a comparison on aerobic and resistance exercise effects on emotional and physical function in older persons with high and low depressive symptomatology. *Journals of Gerontology. Series B, Psychological Sciences and Social Sciences* 2002;**57**(2):124-32.

426. Ettinger WH Jr, Burns R, Messier SP, Applegate W, Rejeski WJ, Morgan T, et al. A randomized trial comparing aerobic exercise and resistance exercise with a health education program in older adults with knee osteoarthritis. The Fitness Arthritis and Seniors Trial (FAST). *JAMA* 1997;**277**(1):25-31.

427. Penttinen HM, Saarto T, Kellokumpu-Lehtinen P, Blomqvist C, Huovinen R, Kautiainen H, et al. Quality of life and physical performance and activity of breast cancer patients after adjuvant treatments. *Psycho-oncology* 2011;**20**(11):1211-20.

428. Perales M, Refoyo I, Coteron J, Bacchi M, Barakat R. Exercise during pregnancy attenuates prenatal depression: A randomized controlled trial. *Evaluation & the Health Professions* 2015;**38**(1):59-72.

429. Pereira DS, De Queiroz BZ, Miranda AS, Rocha NP, Felicio DC, Mateo EC, et al. Effects of physical exercise on plasma levels of brain-derived neurotrophic factor and depressive symptoms in elderly women - A randomized clinical trial. *Archives of Physical Medicine and Rehabilitation* 2013;**94**(8):1443-50.

430. Perna FM, Craft L, Freund KM, Skrinar G, Stone M, Kachnic L, et al. The effect of a cognitive behavioral exercise intervention on clinical depression in a multiethnic sample of women with breast cancer: A randomized controlled trial. *International Journal of Sport and Exercise Psychology* 2010;**8**(1):36-47.

431. Peterson JC, Charlson ME, Wells MT, Altemus M. Depression, coronary artery disease, and physical activity: How much exercise is enough? *Clinical Therapeutics* 2014;**36**(11):1518-30.

432. Poelke Gina, Ventura Maria I, Byers Amy L, Yaffe Kristine, Sudore Rebecca, Barnes Deborah E. Leisure activities and depressive symptoms in older adults with cognitive complaints. *International psychogeriatrics* 2015;**28**(1):63-9.

433. Rashidi M, Rashidy-Pour A, Ghorbani R, Diyant H, Shahvaranian M. The comparison of aerobic and anaerobic exercise effects on depression and anxiety in students. [Persian]. *Koomesh* 2017;**19**(2):333-40.

434. Rawson RA, Chudzynski J, Gonzales R, Mooney L, Dickerson D, Ang A, et al. The impact of exercise on depression and anxiety symptoms among abstinent methamphetamine-dependent individuals in a residential treatment setting. *Journal of Substance Abuse Treatment* 2015;**57**:36-40. [PMID: 25934458]

435. Rebar AL, Boles C, Burton N, Duncan MJ, Short CE, Happell B, et al. Healthy mind, healthy body: A randomized trial testing the efficacy of a computer-tailored vs. interactive web-based intervention for increasing physical activity and

reducing depressive symptoms. *Mental Health and Physical Activity* 2016;**11**:29-37.

436. Rhodes DL. Mens Sana Corpore Sano: A Study of the Effect of Jogging on Depression Anxiety and Self Concept [dissertation]. Durham, NC: Duke University, 1980.

437. Robledo Colonia AF, Sandoval Restrepo N, Mosquera Valderrama YF, Escobar Hurtado C, Ramirez Velez R. Aerobic exercise training during pregnancy reduces depressive symptoms in nulliparous women: a randomised trial. *Journal of Physiotherapy* 2012;**58**(1):9-15.

438. Rogerson M, Wood C, Pretty J, Schoenmakers P, Bloomfield D, Barton J. Regular doses of nature: the efficacy of green exercise interventions for mental wellbeing.. *Int J Environ Res Public Health*. 2020;**17**(5):1526.

439. Ruunsunen A, Voutilainen S, Karhunen L, Lehto SM, Tolmunen T, Keinänen-Kiukaanniemi S, et al. How does lifestyle intervention affect depressive symptoms? Results from the Finnish Diabetes Prevention Study. *Diabetic Medicine* 2012;**29**(7):e126-e132.

440. Salminen M, Isoaho R, Vahlberg T, Ojanlatva A, Kivela SL. Effects of a health advocacy, counselling, and activation programme on depressive symptoms in older coronary heart disease patients. *International Journal of Geriatric Psychiatry* 2005;**20**(6):552-8.

441. Sarsan A, Ardiç F, Özgen M, Topuz O. The effects of aerobic and resistance exercises in obese women. *Clinical Rehabilitation* 2006;**20**:773-82.

442. Sims J, Hill K, Davidson S, Gunn J, Huang N. Exploring the feasibility of a community-based strength training program for older people with depressive symptoms and its impact on depressive symptoms. *BMC Geriatrics* 2006;**6**:18.

443. Skrinar GS, Huxley NA, Hutchinson DS, Menninger E, Glew P. The role of a fitness intervention on people with serious psychiatric disabilities. *Psychiatric Rehabilitation Journal* 2005;**29**(2):122-7.

444. Smith PS, Thompson M. Treadmill training post stroke: are there any secondary benefits? A pilot study. *Clinical Rehabilitation* 2008;**22**:997-1002.

445. Songoygard KM, Stafne SN, Evensen KA, Salvesen KA, Vik T, Morkved S. Does exercise during pregnancy prevent postnatal depression? : A randomized controlled trial. *Acta Obstetrica et Gynecologica Scandinavica* 2012;**91**(1):62-7.

446. Soucy I, Provencher M, Fortier M, McFadden T. Efficacy of guided self-help behavioural activation and physical activity for depression: a randomized controlled trial. *Cognitive Behaviour Therapy* 2017;**46**(6):493-506.

447. Soucy I, Provencher MD, Fortier M, McFadden T. Secondary outcomes of the guided self-help behavioral activation and physical activity for depression trial. *Journal of Mental Health (Abingdon, England)* 2019;**28**(4):410-8.

448. Stein PN, Motta RW. Effects of aerobic and nonaerobic exercise on depression and self-concept. *Perceptual and Motor Skills* 1992;**74**(1):79-89.

449. Stener-Victorin E, Holm G, Janson PO, Gustafson D, Waern M. Acupuncture and physical exercise for affective symptoms and health-related quality of life in polycystic ovary syndrome: secondary analysis from a randomized controlled trial. *BMC Complementary and Alternative Medicine* 2013;**13**(13):131.

450. Stern MJ, Gorman PA, Kaslow L. The group counseling v exercise therapy study. A controlled intervention with subjects following myocardial infarction. *Archives of Internal Medicine* 1983;**143**(9):1719-25.

451. Stern MJ, Plionis E, Kaslow L. Group process expectations and outcome with post-myocardial infarction patients. *General Hospital Psychiatry* 1984;**6**(2):101-8.

452. Strömbeck BE, Theander E, Jacobsson LT. Effects of exercise on aerobic capacity and fatigue in women with primary Sjögren's syndrome. *Rheumatology* 2007;**46**:868-71.

453. Sung K. The effects of 16-week group exercise program on physical function and mental health of elderly Korean women in long-term assisted living facility. *Journal of Cardiovascular Nursing* 2009;**24**(5):344-51.

454. Szuhany Kristin Lynn. Efficacy of adjunctive exercise for the behavioral treatment of major depression. *Dissertation Abstracts International: Section B: The Sciences and Engineering* 2018;**79**(8-B(E)):No Pagination Specified.

455. Tapps T. An Investigation into the Effects of Resistance Exercise Participation on the Perceived Depression Levels of Older Adults Residing in a Long-Term Care Facility over Time [dissertation]. Oklahoma State University, 2009.

456. Taylor CB, Houston Miller N, Ahn DK, Haskell W, DeBusk RF. The effects of exercise training programmes on psychosocial improvement in uncomplicated postmyocardial infarction patients. *Journal of Psychosomatic Research* 1986;**30**(5):581-7.

457. Tenorio LM. Effects of Aerobic Exercise on Symptoms of Depression in Women [dissertation]. Denton, TX: Texas Woman's University, 1986.

458. Thomas SA, Walker MF, Macniven JA, Haworth H, Lincoln NB. Communication and Low Mood (CALM): a randomized controlled trial of behavioural therapy for stroke patients with aphasia. *Clinical Rehabilitation* 2013;**27**(5):398-408.

459. Thomson RL, Buckley JD, Lim SS, Noakes M, Clifton PM, Norman RJ et al. Lifestyle management improves quality of life and depression in overweight and obese women with polycystic ovary syndrome. *Fertility and Sterility* 2010;**94**(5):1812-6.

460. Tomas-Carus P, Gusi N, Häkkinen A, Häkkinen K, Leal A, Ortega-Alonso A. Eight months of physical training in warm water improves physical and mental health in women with fibromyalgia: a randomized controlled trial. *Journal of Rehabilitation Medicine* 2008;**40**:248-52.

- 461.** Tsang HW, Mok CK, Au Yeung YT, Chan SY. The effect of Qigong on general and psychosocial health of elderly with chronic physical illnesses: a randomised clinical trial. *International Journal of Geriatric Psychiatry* 2003;**18**(5):441-9.
- 462.** Ellard DR, Taylor SJ, Parsons S, Thorogood M. The OPERA trial: a protocol for the process evaluation of a randomised trial of an exercise intervention for older people in residential and nursing accommodation. *Trials* 2011;**12**:28.
- 463.** Underwood M, Lamb SE, Eldridge S, Sheehan B, Slowther A, Spencer A, et al. Exercise for depression in care home residents: a randomised controlled trial with cost effectiveness analysis (OPERA). *Health Technology Assessment* 2013;**17**(18):1-281.
- 464.** Underwood M, Lamb SE, Eldridge S, Sheehan B, Slowther A-M, Spencer A, et al. Exercise for depression in elderly residents of care homes: a cluster-randomised controlled trial. *Lancet* 2013;**382**(9886):41-9.
- 465.** Underwood M, Eldridge S, Lamb S, Potter R, Sheehan B, Slowther AM, et al. The OPERA trial: protocol for a randomised trial of an exercise intervention for older people in residential and nursing accommodation. *Trials* 2011;**12**:27.
- 466.** van Beljouw Ilse MJ, van Exel Eric, van de Ven Peter M, Joling Karlijn J, Dhondt Ton DF, Stek Max L, et al. Does an outreaching stepped care program reduce depressive symptoms in community-dwelling older adults? A randomized implementation trial. *American Journal of Geriatric Psychiatry* 2015;**23**(8):807-17.
- 467.** Van der Waerden JE B, Hoefnagels C, Hosman CM H, Souren PM, Jansen MW J. A randomized controlled trial of combined exercise and psycho-education for low-SES women: Short- and long-term outcomes in the reduction of stress and depressive symptoms. *Social Science & Medicine* 2013;**91**:84-93.
- 468.** Vankova H, Holmerova I, Machacova K, Volicer L, Veleta P, Celko AM. The effect of dance on depressive symptoms in nursing home residents. *Journal of the American Medical Directors Association* 2014;**15**(8):582-7.
- 469.** Wadden TA. Impact of intensive lifestyle intervention on depression and health-related quality of life in type 2 diabetes: The lookahead trial. *Diabetes Care* 2014;**37**(6):1544-53.
- 470.** Weinstein AA, Deuster PA, Kop WJ. Heart rate variability as a predictor of negative mood symptoms induced by exercise withdrawal. *Medicine and Science in Sports and Exercise* 2007;**39**(4):735-41.
- 471.** White PD, Sharpe MC, Chalder T, DeCesare JC, Walwyn R; PACE trial group. Protocol for the PACE trial: a randomised controlled trial of adaptive pacing, cognitive behaviour therapy, and graded exercise as supplements to standardised specialist medical care versus standardised specialist medical care alone for patients with the chronic fatigue syndrome/myalgic encephalomyelitis or encephalopathy. *BMC Neurology* 2007;**7**:6.
- 472.** Whitham EA, Thommi SB, Holtzman NS, Ostacher MM, El-Mallakh RS, Baldassano CF, et al. Rapid cycling and antidepressants: Data from a STEP-BD randomized clinical trial. In: *Bipolar Disorders* [abstracts from the 9th International Conference on Bipolar Disorder]. Vol. 13. 2011:s1.
- 473.** Wilbur J, Zenk S, Wang E, Oh A, McDevitt J, Block D, et al. Neighborhood characteristics, adherence to walking, and depressive symptoms in midlife African American women. *Journal of Women's Health* 2009;**18**(8):1201-10.
- 474.** Wipfli BM. Serotonin and Psychological Variables in the Relationship between Exercise and Mental Health [dissertation]. Arizona State University, 2008.
- 475.** Wipfli B, Landers D, Nagoshi C, Ringenbach S. An examination of serotonin and psychological variables in the relationship between exercise and mental health. *Scandinavian Journal of Medicine & Science in Sports* 2011;**21**(3):474-81.
- 476.** Yacamán-Mendez D, Hallgren M, Forsell Y. Childhood adversities, negative life events and outcomes of non-pharmacological treatments for depression in primary care: A secondary analysis of a randomized controlled trial. *Journal of Psychiatric Research* 2019;**110**:152-8.
- 477.** Yeh SH, Lin LW, Chuang YK, Liu CL, Tsai LJ, Tsuei FS, et al. Effects of music aerobic exercise on depression and brain-derived neurotrophic factor levels in community dwelling women. *BioMed Research International* 2015;doi: **10.1155/2015/135893**...
- 478.** Zheng Y, Zhou Y, Lai Q. Effects of twenty-four move shadow boxing combined with psychosomatic relaxation on depression and anxiety in patients with type-2 diabetes. *Psychiatria Danubina* 2015;**27**(2):174-9.
- 479.** Beffert JW. Aerobic exercise as treatment of depressive symptoms in early adolescents [dissertation]. University of Northern Colorado 1993.
- 480.** Brown SW, Welsh MC, Labbé EE, Vitulli WE, Kulkarni P. Aerobic exercise in the psychological treatment of adolescents. *Perceptual and Motor Skills* 1992;**74**:555-60.
- 481.** Hughes CW, Trivedi MH, Cleaver J, Greer TL, Emslie GJ, Kennard B, et al. DATE: Depressed adolescents treated with exercise: study rationale and design for a pilot study. *Mental Health and Physical Activity* 2009;**2**:76-85.
- 482.** MacMahon JR, Gross RT. Physical and psychological effects of aerobic exercise in delinquent adolescent males. *American Journal of Diseases of Children* 1988;**142**:1361-6.
- 483.** NCT00964054. Adapting exercise treatment for depression to adolescents: a pilot study [NCT00964054]. <http://clinicaltrials.gov/show/NCT00964054> (accessed 1 March 2013).
- 484.** Rofey DL, Szigethy EM, Noll RB, Dahl RE, Lobst E, Arslanian SA. Cognitive-behavioral therapy for physical and emotional disturbances in adolescents with polycystic ovary syndrome: a pilot study. *Journal of Pediatric Psychology* 2009;**34**(2):156-63.
- 485.** Roshan VD, Pourasghar M, Mohammadian Z. The efficacy of intermittent walking in water on the rate of MHPG sulfate

and the severity of depression. *Iranian Journal of Psychiatry* 2011;**5**(2):26-31.

486. Silverman SC. The effects of exercise on mood and self-efficacy in emotionally disturbed adolescents. *Dissertation Abstracts International: Section B: The Sciences and Engineering*. 1998;**58**:Mar-B.

487. Attia E. In the clinic eating disorders. *Annals of Internal Medicine* 2012;**156**(7):1-16.

488. Biddle S. Exercise and the treatment of depression. *British Journal of Hospital Medicine* 1989;**42**(4):267.

489. Bromby S, Chandrasekara A. Study of the effect of physical activity on obese/overweight farm men and women on their psychological health and obesity related clinical co-morbidities. Available from <http://www.anzctr.org.au/ACTRN12610000827033.aspx> (accessed May 2013).

490. Broocks A, Meyer TF, George A, Pekrun G, Hillmer-Vogel U, Hajak G, et al. Value of sports in treatment of psychiatric illness. *Psychotherapie, Psychosomatik, Medizinische Psychologie* 1997;**47**(11):379-93.

491. Burbach FR. The efficacy of physical activity interventions within mental health services: anxiety and depressive disorders. *Journal of Mental Health (Abingdon, England)* 1997;**6**:543-66.

492. Chekroud AM. A mendelian randomization approach for assessing the relationship between physical activity and depression. *JAMA Psychiatry* 2019;**76**(4):361-2.

493. Deslandes AC, Morales H, Alves H, Pompeu F, Silveira, H, et al. Effect of aerobic training on EEG alpha asymmetry and depressive symptoms in the elderly: A 1-year follow-up study. *Brazilian Journal of Medical and Biological Research* 2010;**43**(6):585-92.

494. Hedayati SS, Yalamanchili, Finkelstein FO. A practical approach to the treatment of depression in patients with chronic kidney disease and end-stage renal disease. *Kidney International* 2012;**81**(3):247-55.

495. Jaffery A, Edwards MK, Loprinzi PD. Randomized control intervention evaluating the effects of acute exercise on depression and mood profile: solomon experimental design. *Mayo Clinic Proceedings* 2017;**92**(3):480-1.

496. Kinser PA, Thacker LR, Rider A, Moyer S, Amstadter AB, Mazzeo SE, et al. Feasibility, acceptability, and preliminary effects of "mindful moms": a mindful physical activity intervention for pregnant women with depression. *Nursing Research* 2021;**70**(2):95=105.

497. Knapen J, Vancampfort D, Morien Y, Marchal Y. Exercise therapy improves both mental and physical health in patients with major depression. *Disability and Rehabilitation* 2015;**37**(16):1490-5. [PMID: 25342564]

498. Korman N, Fox H, Skinner T, Dodd C, Suetani S, Chapman J, et al. Feasibility and acceptability of a student-led lifestyle (diet and exercise) intervention within a residential rehabilitation setting for people with severe mental illness, GO

HEART (group occupation, health, exercise and rehabilitation treatment). *Frontiers in Psychiatry / Frontiers Research Foundation* 2020;**11**:319.

499. Leibold ML. Activities and adaptive strategies in late life depression: A qualitative study [thesis]. *Dissertation Abstracts International: Section B: The Sciences and Engineering* 2011;**71**(9-B):5425.

500. Martinsen EW. Exercise intervention studies in patients with anxiety and depressive disorders. In: *Proceedings of Sport, Health Psychology and Exercise Symposium*. London: Sports Council and Health Education Authority, 1988:77-83.

501. Martinsen EW. Aerobic exercise in the treatment of nonpsychotic mental disorders: an exploratory study. *Nordic Journal of Psychiatry* 1989;**43**:521-9.

502. Martinsen EW. Therapeutic implications of exercise for clinically anxious and depressed patients. *International Journal of Sport Psychology* 1993;**24**:185-99.

503. Munro J, Brazier J, Davey R, Nicholl J. Physical activity for the over-65s: could it be a cost-effective exercise for the NHS? *Journal of Public Health Medicine* 1997;**19**(4):397-402.

504. NCT00416221. Development and a pilot study of the PACEPRO exercise and mood management program in depressed patients on escitalopram oxalate (Lexapro). <http://clinicaltrials.gov/show/NCT00416221> (accessed 2008).

505. Pourmousavi L, Abdoly M, Nikseresht A, Pourmousavi M. Comparison effects of aerobic exercise and sertraline on depression and sexual desire in women with depression. *Journal of Sexual Medicine* 2016;**2**:S97.

506. Raglin JS. Exercise and mental health. Beneficial and detrimental effects. *Sports Medicine (Auckland, N.Z.)* 1990;**9**(6):323-9.

507. Rao UT, Noronha JA, Adiga K. Effect of aerobic exercises on depressive symptoms, anxiety, self-esteem, and quality of life among adults with depression. *Clinical Epidemiology and Global Health* 2020;**8**(4):1147-51.

508. Salmon P. Effects of physical exercise on anxiety, depression, and sensitivity to stress: a unifying theory. *Clinical Psychology Review* 2001;**21**(1):33-61.

509. Silveira H, Deslandes AC, De Moraes H, Mouta R, Ribeiro P, Piedade R, et al. Effects of exercise on electroencephalographic mean frequency in depressed elderly subjects. *Neuropsychobiology* 2010;**61**(3):141-7.

510. Tsang Hector WH, Tsang William WN, Jones Alice YM, Fung Kelvin MT, Chan Alan HL, Chan Edward P, et al. Psycho-physical and neurophysiological effects of qigong on depressed elders with chronic illness. *Aging & Mental Health* 2013;**17**(3):336-48.

511. Van de Vliet P, Onghena P, Knapen J, Fox KR, Probst M, Van Coppenolle H, et al. Assessing the additional impact of fitness training in depressed psychiatric patients receiving multifaceted treatment: a replicated single-subject design. *Disability and Rehabilitation* 2003;**25**(24):1344-53.

- 512.** Weiss CR, Jamieson NB. Women, subjective depression, and water exercise. *Health Care for Women International* 1989;**10**(1):75-88.
- 513.** Alexopoulos George S, Kiosses Dimitris N, Sirey Jo Anne, Kanellopoulos Dora, Seirup Joanna K, Novitch Richard S, et al. Untangling therapeutic ingredients of a personalized intervention for patients with depression and severe COPD. *American Journal of Geriatric Psychiatry* 2014;**22**(11):1316-24.
- 514.** Bowden D, Gaudry C, An SC, Gruzelier J. A comparative randomised controlled trial of the effects of brain wave vibration training, Iyengar yoga, and mindfulness on mood, well-being, and salivary cortisol. Evidence-Based Complementary and Alternative Medicine 2012 Dec 15 [Epub ahead of print]. [DOI: [10.1155/2012/234713](https://doi.org/10.1155/2012/234713)]
- 515.** Busch Andrew M, Whited Matthew C, Appelhans Bradley M, Schneider Kristin L, Waring Molly E, DeBiaise Michele A, et al. Reliable change in depression during behavioral weight loss treatment among women with major depression. *Obesity Obesity Research* 2013;**21**(3):E211-8.
- 516.** Chan AS, Cheung M-C, Tsui WJ, Sze SL, Shi D. Dejian mind-body intervention on depressive mood of community-dwelling adults: A randomised controlled trial. Evidence-Based Complementary and Alternative Medicine 2011;1473961.
- 517.** Chu I Hua. Effect of exercise intensity during aerobic training on depressive symptoms in initially sedentary depressed women. *Dissertation Abstracts International: Section B: The Sciences and Engineering* 2018;**78**(11-B(E)):No Pagination Specified.
- 518.** Ciocon JO, Galindo-Ciocon D. Loneliness and depression in nursing home setting: the effect of a restorative program. In: International Psychogeriatrics [abstracts from the International Psychogeriatric Association 11th International Congress, Chicago, United States. 17-22 August 2003]. Vol. 15. 2003:S027-04.
- 519.** Duarte A, Walker J, Walker S, Richardson G, Holm Hansen C, Martin P, et al. Cost-effectiveness of integrated collaborative care for comorbid major depression in patients with cancer. *Journal of Psychosomatic Research* 2015;**79**(6):465-70.
- 520.** Fitzsimmons S. Easy rider wheelchair biking: a nursing-recreation therapy clinical trial for the treatment of depression. *Journal of Gerontological Nursing* 2001;**27**:14-23.
- 521.** Grosse, J, Huppertz, C, Roh, A, et al. Step away from depression-results from a multicenter randomized clinical trial with a pedometer intervention during and after inpatient treatment of depression. *European Archives of Psychiatry and Clinical Neuroscience* 2023;doi: [10.1007/s00406-023-01646-2](https://doi.org/10.1007/s00406-023-01646-2)...
- 522.** He X. Physical activity in the treatment of depression in college students. *Revista Brasileira de Medicina do Esporte* 2022;**28**(1):68-71.
- 523.** Keller-Varady K, Haufe S, Schieffer E, Kerling A, Tegtbu U, Kahl KG. Personalized training as a promoter for physical activity in people with depressive disorder-a randomized controlled trial in Germany. *Frontiers in Psychiatry / Frontiers Research Foundation* 2023;**14**:doi: [10.3389/fpsy.2023.1158705](https://doi.org/10.3389/fpsy.2023.1158705).
- 524.** Kleinstaub M, Reuter M, Doll N, Fallgatter Andreas J. Rock climbing and acute emotion regulation in patients with major depressive disorder in the context of a psychological inpatient treatment: A controlled pilot trial. *Psychology Research and Behavior Management Vol 10* 2017, ArtID 277-281 2017;**10**:277-81.
- 525.** Nystrom MB T, Stenling A, Sjoström E, Neely G, Lindner P, Hassmen P, et al. Behavioral activation versus physical activity via the internet: A randomized controlled trial. *Journal of Affective Disorders* 2017;**215**:85-93.
- 526.** Van der Merwe I, Naude S. Exercise and depression: a treatment manual. *Health SA Gesondheid* 2004;**9**(4):28-41.
- 527.** Walker J, Hansen CH, Martin P, Symeonides S, Gourley C, Wall L, et al. Integrated collaborative care for major depression comorbid with a poor prognosis cancer (SMaRT Oncology-3): A multicentre randomised controlled trial in patients with lung cancer. *Lancet Oncology* 2014;**15**(10):1168-76.
- 528.** Chou K-L, Lee PW, Yu EC, MacFarlane D, Cheng Y-H, Chan SS, et al. Effect of Tai Chi on depressive symptoms amongst Chinese older patients with depressive disorders: a randomised clinical trial. *International Journal of Geriatric Psychiatry* 2004;**19**:1105-7.
- 529.** Srinivasan S, Reagan LP, Hardin JW, Matthews M, Leaphart E, Grillo CA, et al. Adjunctive tai chi in geriatric depression with comorbid arthritis: A randomized, controlled trial abstract. *American Journal of Geriatric Psychiatry [abstracts of the American Association for Geriatric Psychiatry, AAGP Annual Meeting; 2014 Mar 14-17; Orlando, FL United States]* 2014;**22**(3):S135-6.
- 530.** Chan ES, Koh D, Teo YC, Hj Tamin R, Lim A, Fredericks S. Biochemical and psychometric evaluation of Self-Healing Qigong as a stress reduction tool among first year nursing and midwifery students. *Complementary Therapies in Clinical Practice* 2013;**19**(4):179-83.
- 531.** Chow YW, Dorcas A, Siu AM. The effects of qigong on reducing stress and anxiety and enhancing body-mind well-being. *Mindfulness* 2012;**3**(1):51-9.
- 532.** Tsang HW, Fung KM, Chan AS, Lee G, Chan F. Effect of qigong exercise programme on elderly with depression. *International Journal of Geriatric Psychiatry* 2006;**21**:890-7.
- 533.** Immink MA, Hillier SL. Yoga for chronic post-stroke hemiparesis: A pilot randomised controlled trial. In: International Journal of Stroke. Vol. Abstract of the 22nd Stroke Society of Australasia Annual Scientific Meeting. 2011.
- 534.** Kahl KG, Kerling A, Tegtbu U, Gutzlaff E, Herrmann J, Borchert L, et al. Effects of additional exercise training on epicardial, intra-abdominal and subcutaneous adipose tissue in major depressive disorder: A randomized pilot study. *Journal of Affective Disorders* 2016;**192**(1):91-7.

- 535.** Oretzky S. The effects of yoga on elevated depressive and somatic symptoms in young adults [dissertation]. California School of Professional Psychology, Alliant International University 2006.
- 536.** Rashidi Z, Rashidi A, Rouzbahani R, Rezaei F. Effects of group exercise on women's depressive symptoms. *Journal of Isfahan Medical School* 2013;**30**(212):1856-61.
- 537.** Bartholomew JB, Morrison D, Ciccolo JT. Effects of acute exercise on mood and well-being in patients with major depressive disorder. *Medicine and Science in Sports and Exercise* 2005;**37**:2032-7.
- 538.** Bodin T, Martinensen EW. Mood and self-efficacy during acute exercise in clinical depression. A randomised controlled study. *Journal of Sport & Exercise Psychology* 2004;**26**:623-33.
- 539.** Gustafsson G, Lira CM, Johansson J, Wisén A, Wohlfart, Ekman R, et al. The acute response of plasma brain-derived neurotrophic factor as a result of exercise in major depressive disorder. *Psychiatry Research* 2009;**169**:244-8.
- 540.** Legrand Fabien D, Race Maximillian, Herring Matthew P. Acute effects of outdoor and indoor exercise on feelings of energy and fatigue in people with depressive symptoms. *Journal of Environmental Psychology* 2018;**56**:91-6.
- 541.** Berlin B, Moul DE, LePage JP, Mogge NL, Sellers DG. The effect of aquatic therapy interventions on patients with depression: a comparison study. *Annual in Therapeutic Recreation* 2003;**12**:7-13.
- 542.** Chalder M, Wiles NJ, Campbell J, Hollinghurst SP, Searle A, Haase AM, et al. A pragmatic randomised controlled trial to evaluate the cost-effectiveness of a physical activity intervention as a treatment for depression: The treating depression with physical activity (TREAD) trial. *Health Technology Assessment* 2012;**16**(10):1-164.
- 543.** Baxter H, Winder R, Chalder M, Wright C, Sherlock S, Haase A, et al. Physical activity as a treatment for depression: the TREAD randomised trial protocol. *Trials* 2010;**11**:105.
- 544.** Chalder M, Wiles NJ, Campbell J, Hollinghurst SP, Haase AM, Taylor AH, et al. Facilitated physical activity as a treatment for depressed adults: a randomised controlled clinical trial. *BMJ (Clinical Research Ed.)* 2012;**344**:e2758.
- 545.** Haase AM, Taylor AH, Fox KR, Thorp H, Lewis G. Rationale and development of the physical activity counselling intervention for a pragmatic TRial of Exercise and Depression in the UK (TREAD-UK). *Mental Health and Physical Activity* 2010;**3**(2):85-91.
- 546.** Searle A, Calnan M, Lewis G, Campbell J, Taylor A, Turner K. Patients' views of physical activity as treatment for depression: a qualitative study. *British Journal of General Practice* 2011;**61**(585):149-56.
- 547.** Searle A, Calnan M, Turner KM, Lawlor DA, Campbell J, Chalder M, et al. General Practitioners' beliefs about physical activity for managing depression in primary care [ISRCTN16900744;TREAD UK]. *Mental Health and Physical Activity* 2012;**5**(1):13-9.
- 548.** Greer TL, Grannemann BD, Chansard M, Karim AI, Trivedi MH. Dose-dependent changes in cognitive function with exercise augmentation for major depression: results from the TREAD study. *European Neuropsychopharmacology* 2015;**25**(2):248-56. [PMID: 25453481]
- 549.** Kratz Anna L, Ehde Dawn M, Bombardier Charles H. Affective mediators of a physical activity intervention for depression in multiple sclerosis. *Rehabilitation Psychology Psychological Aspects of Disability* 2014;**59**(1):57-67.
- 550.** Piette JD, Valenstein M, Himle J, Duffy S, Torres T, Vogel M, et al. Clinical complexity and the effectiveness of an intervention for depressed diabetes patients. *Chronic Illness* 2011;**7**(4):267-78.
- 551.** Vickers KS, Patten CA, Lewis BA, Clark MM, Ussher M, Ebbert JO, et al. Feasibility of an exercise counseling intervention for depressed women smokers. *Nicotine & Tobacco Research* 2009;**11**(8):985-95.
- 552.** Chau RM, Tsui AY, Wong EY, Cheung EY, Chan DY, Lau PM, et al. Effectiveness of a structured physical rehabilitation program on the physical fitness, mental health and pain for Chinese patients with major depressive disorders in Hong Kong - a randomized controlled trial with 9-month follow-up outcomes. *Disability and Rehabilitation* 2020;<https://doi.org/10.1080/09638288.2020.1800833>:1-11.
- 553.** Chau, RM W, Tsui, AY Y, Wong, EY W, et al. Effectiveness of a structured physical rehabilitation program on the physical fitness, mental health and pain for Chinese patients with major depressive disorders in Hong Kong - a randomized controlled trial with 9-month follow-up outcomes. *Disability and Rehabilitation* 2022;**44**(8):1294-304.
- 554.** Danielsson L, Papoulias I, Petersson EL, Carlsson J, Waern M. Exercise or basic body awareness therapy as add-on treatment for major depression: A controlled study. *Journal of Affective Disorders* 2014;**168**:98-106.
- 555.** Farrand P, Pentecost C, Greaves C, Taylor RS, Warren F, Green C, et al. A written self-help intervention for depressed adults comparing behavioural activation combined with physical activity promotion with a self-help intervention based upon behavioural activation alone: Study protocol for a parallel group pilot randomised controlled trial (BACpAc). *Trials* 2014;**196**(15):doi: 10.1186/1745-6215-15-196.
- 556.** Garcia-Toro M, Ibarra O, Gili M, Salva J, Monzon S, Vives M, et al. Effectiveness of hygienic-dietary recommendations as enhancers of antidepressant treatment in patients with depression: study protocol of a randomized controlled trial. *BMC Public Health* 2010;**10**:404. [PMID: 20618920]
- 557.** Garcia-Toro M, Gili M, Ibarra O, Monzon S, Vives M, Garcia-Campayo J, et al. Metabolic syndrome improvement in depression six months after prescribing simple hygienic-dietary recommendations. *BMC Research Notes* 2014;**7**(5):339.

- 558.** Gyurak A, Gross J, Chan L, Etkin A. Cognitive-affective remediation training intervention in anxiety and depression. *Neuropsychopharmacology [abstracts of the 52nd Annual Meeting of the American College of Neuropsychopharmacology, ACNP; 2013 Dec 8-12; Hollywood, FL United States]* 2013;**38**:S231.
- 559.** McClure JB, Catz SL, Ludman EJ, Richards J, Riggs K, Grothaus L. Feasibility and acceptability of a multiple risk factor intervention: the Step Up randomized pilot trial. *BMC Public Health* 2011;**11**(167):doi:10.1186/1471-2458-11-167.
- 560.** NCT00644995. Step up wellness program for depression, physical inactivity, and smoking. clinicaltrials.gov (<http://www.clinicaltrials.gov>) (accessed 2008).
- 561.** Meyer JD, Koltyn KF, Stegner AJ, Kim JS, Cook DB. Influence of exercise intensity for improving depressed mood in depression: a dose-response study. *Behavior Therapy* 2016;**47**(4):527-37.
- 562.** O'Neil A, Hawkes AL, Chan B, Sanderson K, Forbes A, Hollingsworth B, et al. A randomised, feasibility trial of a tele-health intervention for Acute Coronary Syndrome patients with depression ('Moodcare') : Study protocol. *BMC Cardiovascular Disorders* 2011 25th Feb [Epub ahead of print];**11**(8). [DOI: [10.1186/1471-2261-11-8](https://doi.org/10.1186/1471-2261-11-8)]
- 563.** Patten CA, Bronars CA, Vickers Douglas KS, Ussher M, Levine JA, Tye SJ, et al. Supervised, vigorous intensity exercise intervention for depressed femalesmokers: a pilot study. *Nicotine & Tobacco Research* 2017;**19**(1):77-86.
- 564.** Shachar-Malach T, Cooper Kazaz R, Constantini N, Lifschytz T, Lerer B. Effectiveness of aerobic exercise as an augmentation therapy for inpatients with major depressive disorder: a preliminary randomized controlled trial. *Israel Journal of Psychiatry and Related Sciences* 2015;**52**(3):65-70.
- 565.** Szuhany KL, Otto MW. Assessing BDNF as a mediator of the effects of exercise on depression. *Journal of Psychiatric Research* 2020;**123**:114-8.
- 566.** Szuhany KL, Otto MW. Efficacy evaluation of exercise as an augmentation strategy to brief behavioral activation treatment for depression: a randomized pilot trial. *Cognitive Behaviour Therapy* 2020;**49**(3):228-41.
- 567.** Patten CA, Vickers KS, Lewis BA, Finnie SB, Wheeldon TJ, Decker PA, et al. Feasibility trial of an unsupervised, facility-based exercise programme for depressed outpatients. *Psychology, Health & Medicine* 2019;**24**(3):320-32.
- 568.** Sneider KL, Bodenlos JS, Ma Y, Olendzki B, Oleski J, Merriam P, et al. Design and methods for a randomised clinical trial treating comorbid obesity and major depressive disorder. *BMC Psychiatry* 2008;**8**:77.
- 569.** Verrall AM, Bombardier H, Ehde M, Gibbons E, Kraft GH. The effects of telephone-based physical activity counselling on fatigue, pain, symptoms and quality of life in people with multiple sclerosis and major depression abstract. *Multiple Sclerosis (Houndmills, Basingstoke, England)* 2012;**28th Congress of the European Committee for Treatment and Research in Multiple Sclerosis Lyon France. Conference Start: 20121010 Conference End: 20121013**:245.
- 570.** Balchin R, Linde J, Blackhurst D, Rauch HL, Schonbachler G. Sweating away depression? The impact of intensive exercise on depression. *Journal of Affective Disorders* 2016;**200**:218-21.
- 571.** Bosscher RJ. Running and mixed physical exercises with depressed psychiatric patients. *International Journal of Sport Psychology* 1993;**24**:170-84.
- 572.** Gerber M, Imboden C, Beck J, Brand S, Colledge F, Eckert A, et al. Effects of aerobic exercise on cortisol stress reactivity in response to the trier social stress test in inpatients with major depressive disorders: a randomized controlled trial. *Journal of Clinical Medicine* 2020;**9**(5):1-23 (doi: [10.3390/jcm9051419](https://doi.org/10.3390/jcm9051419)).
- 573.** Hanssen H, Minghetti A, Faude O, Schmidt-Trucksass A, Zahner L, Beck J, et al. Effects of endurance exercise modalities on arterial stiffness in patients suffering from unipolar depression: a randomized controlled trial. *Frontiers in Psychiatry Frontiers Research Foundation* 2017;**8**:311.
- 574.** Hanssen H, Minghetti A, Faude O, Schmidt-Trucksass A, Zahner L, Beck J, et al. Effects of different endurance exercise modalities on retinal vessel diameters in unipolar depression. *Microvascular Research* 2018;**120**:111-6.
- 575.** Hanssen H, Minghetti A, Faude O, Schmidt-Trucksass A, Zahner L, Beck J, et al. Effects of different endurance exercise modalities on retinal vessel diameters in unipolar depression. *Microvascular Research* 2018;**120**:111-6.
- 576.** Haussleiter IS, Bolsinger B, Assion HJ, Juckel G. Adjuvant guided exercise therapy versus self-organized activity in patients with major depression. *Journal of Nervous and Mental Disease* 2020;**208**(12):982-8.
- 577.** Imboden C, Beck J, Gerber M, Puhse U, Holsboer-Trachsler E, Eckert A, et al. Aerobic exercise decreases cortisol awakening reaction in depressed inpatients. *European Neuropsychopharmacology* 2017;**27** (Supplement 4):S810.
- 578.** Imboden C, Beck J, Gerber M, Puhse U, Holsboer-Trachsler E, Hatzinger M. Aerobic exercise as add-on treatment in depressed inpatients improves cognitive domains but has no additional effect on symptom severity. *European Neuropsychopharmacology* 2016;**26** (Supplement 2):S395-6.
- 579.** Imboden C, Beck J, Gerber M, Puhse U. Aerobic exercise decreases cortisol awakening reaction in depressed inpatients. *European Neuropsychopharmacology* 2017;**27**:S810.
- 580.** Judith Haffmans PM, Kleinsman AC, van Weelden C, Huijbrechts IP, Hoencamp E. Comparing running therapy with physiotherapy in the treatment of mood disorders. *Acta Neuropsychiatrica* 2006;**18**(3):173-6.
- 581.** Kerling A, Tegtbu U, Gutzlaff E, Kuck M, Borchert L, Ates Z, et al. Effects of adjunctive exercise on physiological and psychological parameters in depression: A randomized pilot trial. *Journal of Affective Disorders* 2015;**177**(15):1-6.

- 582.** Kerling A, Hartung D, Stubbs B, Kück M, Tegtbur U, Grams L, et al. Impact of aerobic exercise on muscle mass in patients with major depressive disorder: a randomized controlled trial. *Neuropsychiatric Disease and Treatment* 2018;**14**:1969–74.
- 583.** Kerling A, Kück M, Tegtbur U, Grams L, Weber-Spickschen S, Hanke A, et al. Exercise increases serum brain-derived neurotrophic factor in patients with major depressive disorder. *Journal of Affective Disorders* 2017;**215**:152–5. [DOI: [10.1016/j.jad.2017.03.034](https://doi.org/10.1016/j.jad.2017.03.034)]
- 584.** Kerling A, Kück M, Tegtbur U, Grams L, Weber-Spickschen S, Hanke A, et al. Exercise increases serum brain-derived neurotrophic factor in patients with major depressive disorder. *Journal of Affective Disorders* 2017;**215**:152–5.
- 585.** Kerling A, Dagmar Hartung D, Stubbs B, Kück M, Tegtbur U, Grams L, et al. Impact of aerobic exercise on muscle mass in patients with major depressive disorder: a randomized controlled trial. *Neuropsychiatric Disease and Treatment* 2018;**14**:1969–74.
- 586.** Labbe EE, Welsh MC, Delaney D. Effects of consistent aerobic exercise on the psychological functioning of women. *Perceptual and Motor Skills* 1988;**67**(3):919–25.
- 587.** Lacombe JB. Role of Aerobic Conditioning and Psychosocial Factors in Mediating the Effect of Exercise on Depression [dissertation]. Hempstead, NY: Hofstra University, 1987.
- 588.** Lavebratt C, Herring MP, Liu JJ, Wei YB, Bossoli D, Hallgren M, et al. Interleukin-6 and depressive symptom severity in response to physical exercise. *Psychiatry Research* 2017;**252**:270–6.
- 589.** Lavretsky H, Alstein LL, Olmstead RE, Ercoli LM, Riparetti-Brown M, Cyr NS, et al. Complementary use of tai chi chih augments escitalopram treatment of geriatric depression: A randomized controlled trial. *American Journal of Geriatric Psychiatry* 2011;**19**(10):839–50.
- 590.** Legrand FD, Mille CR. The effects of 60 minutes of supervised weekly walking (in a single vs. 3-5 session format) on depressive symptoms among older women: findings from a pilot randomized trial. *Mental Health and Physical Activity* 2009;**2**:71–5.
- 591.** Martinsen EW. Comparing aerobic and non-aerobic forms of exercise in the treatment of clinical depression: a randomised trial. In: Proceedings of Sport, Health Psychology and Exercise Symposium. London: Sports Council and Health Education Authority, 1988:84–95.
- 592.** Moraes HS, Silveira HS, Oliveira NA, Portugal EM, Araújo NB, Vasques PE, et al. Is strength training as effective as aerobic training for depression in older adults? A randomized controlled trial. *Neuropsychobiology* 2020;**79**(2):141–149 (doi: [10.1159/000503750](https://doi.org/10.1159/000503750)).
- 593.** NCT00546221. Pragmatic randomised controlled trial of a preferred intensity exercise programme to improve physiological and associated psychological, social, and wellbeing outcomes of women living with depression. *clinicaltrials.gov* (<http://www.clinicaltrials.gov>) 2007.
- 594.** NCT01152086. The effects of regular mountain hiking on hopelessness in chronically suicidal patients. *clinicaltrials.gov* (<http://www.clinicaltrials.gov>) (accessed 1st March 2013).
- 595.** Niederer D, Vogt L, Staschke V, Maulbecker-Armstrong C, Beck V, Banzer W. [Activity trails in the therapy of clinical depression: A randomized controlled equivalence trial]. *Zeitschrift fuer Psychosomatische Medizin und Psychotherapie* 2017;**63**(2):163–75.
- 596.** Passmore T, Lane S. Exercise as a treatment for depression: a therapeutic recreation intervention. *American Journal of Recreation Therapy* 2006;**5**(3):31–41.
- 597.** Sexton H, Maere A, Dahl NH. Exercise intensity and reduction in neurotic symptoms. A controlled follow-up study. *Acta Psychiatrica Scandinavica* 1989;**80**(3):231–5.
- 598.** Trivedi MH, Greer TL, Church TS, Carmody TJ, Grannemann BD, Galper DI, et al. Exercise as an augmentation treatment for nonremitted major depressive disorder: a randomized, parallel dose comparison. *Journal of Clinical Psychiatry* 2011;**72**(5):677–84.
- 599.** Rethorst CD, Tu J, Carmody TJ, Greer TL, Trivedi MH. Atypical depressive symptoms as a predictor of treatment response to exercise in Major Depressive Disorder. *Journal of Affective Disorders* 2016;**200**:156–8.
- 600.** Rethorst CD, Sunderajan P, Greer TL, Grannemann BD, Nakonezny PA, Carmody TJ, et al. Does exercise improve self-reported sleep quality in non-remitted major depressive disorder? *Psychological Medicine* 2013;**43**(4):699–709.
- 601.** Rethorst CD, Toups MS, Greer TL, Nakonezny PA, Carmody TJ, Grannemann BD, et al. Pro-inflammatory cytokines as predictors of antidepressant effects of exercise in major depressive disorder. *Molecular Psychiatry* 2013;**18**(10):1119–24.
- 602.** Greer TL, Grannemann BD, Chansard M, Karim AI, Trivedi MH. Dose-dependent changes in cognitive function with exercise augmentation for major depression: results from the TREAD study. *European Neuropsychopharmacology* 2015;**25**(2):248–56. [PMID: 25453481]
- 603.** Rethorst CD, Greer TL, Toups MS P, Bernstein I, Carmody TJ, Trivedi MH. IL-1 β and BDNF are associated with improvement in hypersomnia but not insomnia following exercise in major depressive disorder. *Translational Psychiatry* 2015 2015;**5**(8):1–6 (doi: [10.1038/tp.2015.104](https://doi.org/10.1038/tp.2015.104)).
- 604.** Greer TL, Trombello JM, Rethorst CD, Carmody TJ, Jha MK, Liao A, et al. Improvements in psychosocial functioning and health-related quality of life following exercise augmentation in patients with treatment response but nonremitted major depressive disorder: results from the tread study. *Depression and Anxiety* 2016;**33**(9):870–81.
- 605.** Trivedi MH, Greer TL, Grannemann BD, Church TS, Galper DI, Sunderajan P, et al, Wisniewski SR, Chambliss HO, Jordan AN, Finley C, Carmody TI. TREAD: Treatment with

Exercise Augmentation for Depression: study rationale and design. *Clinical Trials (London, England)* 2006;**3**(3):291-305.

606. National Institute of Mental Health (NIMH). Treatment with exercise augmentation for depression (TREAD). [clinicaltrials.gov](http://www.clinicaltrials.gov) (<http://www.clinicaltrials.gov>) 2003.

607. Trivedi MH, Greer TL, Church TS, Carmody TJ, Grannemann BD, Galper DI, et al. Exercise as an augmentation treatment for nonremitted major depressive disorder: a randomized, parallel dose comparison. *Journal of Clinical Psychiatry* 2011;**72**(5):677-84.

608. Uebelacker LA, Sillice MA, Epstein-Lubow G, Battle CL, Anderson B, Caviness C, et al. Combined intervention approaches for initiating and maintaining physical activity in depressed individuals: design and rationale of the Project MOVE randomized clinical trial. *Contemporary Clinical Trials* 2020;**91**:doi: 10.1016/j.cct.2020.105974.

609. Wieman JB. Running As a Treatment for Depression: A Theoretical Basis [dissertation]. Fresno, CA: California School of Professional Psychology, 1980.

610. Williams VL. Acute Mood and EEG Effects of Aerobic Exercise in Depressed and Non-Depressed Adults [dissertation]. Birmingham, AL: The University of Alabama, 1992.

611. Kratzer A, Luttenberger K, Karg-Hefner N, Weiss M, Dorscht L. Boulderling psychotherapy is effective in enhancing perceived self-efficacy in people with depression: results from a multicenter randomized controlled trial. *BMC Psychology* 2021;**9**(1):126.

612. Patten CA, Bronars CA, Vickers Douglas KS, Ussher MH, Levine JA, Tye SJ, et al. Supervised, vigorous intensity exercise intervention for depressed female smokers: a pilot study. *Nicotine & Tobacco Research* 2017;**19**(1):77-86. [PMID: 27613946]

613. Perri S 2nd, Temper DL. The effects of an aerobic exercise programme on psychological variables in older adults. *International Journal of Aging & Human Development* 1984;**20**(3):1984-5.

614. Roth DL, Holmes DS. Influence of aerobic exercise training and relaxation training on physical and psychologic health following stressful life events. *Psychosomatic Medicine* 1987;**49**(4):355-65.

615. Chau RM W, Ng RM K, Tsang AW K, Yu ZH S, Ng KM, Lam MM Y, et al. Effectiveness of a structured physical rehabilitation program for chinese population with depressive disorders-a randomised controlled trial. *Physiotherapy (united Kingdom)* 2015;**101**(1):World Confederation for Physical Therapy Congress 2015 Singapore Singapore. Conference Start 20150501 Conference End 20150504. Conference Publication (var.pagings). 101 (pp eS218).

616. Effect of exercises of mental health. <http://www.who.int/trialsearch/Trial2.aspx?TrialID=CTRI/2020/04/024632> 2020.

617. López Moral A. Exercise in severe mental illness. The psychiactive project 2.0. *ClinicalTrials.gov* 2020.

618. Pila E. Exercise + self-compassion intervention for depression. *ClinicalTrials.gov* 2020.

619. Peacock J. A feasibility study to analyse the psychosocial benefits of green exercise (GE) in comparison with cognitive behavioural therapy (CBT) with patients with mild to moderate depression. www.controlled-trials.com 2007.

620. Schwarz MJ, Hennings A, Riemer S, Stapf T, Selberdinger V, Gil FP, et al. Effect of physical exercise on psychoneuroimmunological parameters in patients with depression and patients with somatoform disorder. In: Neurology Psychiatry and Brain Research [abstracts of the 11th Psychoimmunology Expert Meeting]. 2012.

621. Skuhareuskaya, M, Yahlouskaya, A, Skugarevsky, O, et al. In-patient physical exercise program as an adjuvant therapy for depression. *E-Poster Viewing. European Psychiatry* 2020;**63**(S1 (<https://doi.org/10.1192/j.eurpsy.2020.6>)):S368.

622. Veloz AP, Padilla CP, Villa MF. Effectiveness of physical exercise in the elderly with depression [Efectividad del ejercicio físico en ancianos con depresión]. *NeuroQuantology* 2022;**20**(13):183-190 (DOI: 10.14704/NQ.2022.20.13.NQ88026).

623. Heinzel S, Rapp MA, Fydrich T, Strohle A, Teran C, Kallies G, et al. Neurobiological mechanisms of exercise and psychotherapy in depression: The SPeED study-Rationale, design, and methodological issues. *Clinical Trials (London, England)* 2018;**15**(1):53-64.

624. Heissel A, Pietrek A, Schwefel M, Abula K, Wilbertz G, Heinzel S, et al. STEP.De study-a multicentre cluster-randomised effectiveness trial of exercise therapy for patients with depressive symptoms in healthcare services: study protocol. *BMJ Open* 2020;**10**(4):<http://dx.doi.org/10.1136/bmjopen-2019-036287>.

625. Lopez-Torres Hidalgo J; Group Dep-Exercise. Effectiveness of physical exercise in the treatment of depression in older adults as an alternative to antidepressant drugs in primary care. *BMC Psychiatry* 2019;**19**(1):21.

626. Martland R, Stubbs B, Gaughran F, Onwumere J. A pilot randomised controlled trial to assess the feasibility and acceptability of high intensity interval training (HIIT) among inpatients with severe mental illness (SMI). *Schizophr Bull.* 2020;**46**(1):S207-S208 (doi: 10.1093/schbul/sbaa030.501).

627. NCT02407704. A multi-level life-span characterization of adult-depression and effects of medication and exercise. <https://clinicaltrials.gov/show/NCT02407704> (accessed 25 January 2017).

628. NCT02679053. Aerobic exercise as add-on treatment for inpatients with depression. <https://clinicaltrials.gov/show/NCT02679053> (accessed 25 January 2017).

629. Gerencia de Atención Primaria A. Effectiveness of exercise in the treatment of depression. <https://ClinicalTrials.gov/show/NCT03358433>.

- 630.** Nct03056287. Exercise and brain stimulation for post-stroke depression. <https://clinicaltrials.gov/show/nct03056287> 2017.
- 631.** Nct03107039. Exploring the effects of strength training on depressive symptoms and cardiovascular risk in black men. <https://clinicaltrials.gov/show/nct03107039> 2017.
- 632.** Safakar MY, Hrabok M, Urchuk L, Juhas M, Shalaby R, Parmar D, et al. Effectiveness of group cognitive behavioral therapy and exercise in the management of major depressive disorder: protocol for a pilot randomized controlled trial. *JMIR Research Protocols* 2020;**9**(5):e14309 (doi: 10.2196/14309).
- 633.** Siu PM. Effectiveness of walking exercise in improving depression in older adults with major depressive disorder, a pilot study. <https://clinicaltrials.gov/ct2/show/NCT04403373> 2020.
- 634.** NCT05703776. Physical exercise and depressive symptoms in moderately or severely depressed adults. <https://clinicaltrials.gov/show/NCT05703776> (accessed 08/09/2025).
- 635.** NCT05136027. Exercise program in patients with treatment-resistant depressive disorder compared to usual practice. <https://clinicaltrials.gov/study/NCT05136027> (accessed 08/09/2025).
- 636.** NCT05313542. Effect of physical activity in the format of self-defence training for depressive symptoms - a pilot randomised controlled trial. <https://clinicaltrials.gov/study/NCT05313542> (accessed 08/09/2025).
- 637.** NCT05812001. The efficacy of exergames-acceptance and commitment therapy program for treatment of major depressive disorder: comparison with acceptance and commitment therapy alone and treatment-as-usual in a randomized controlled trial. <https://clinicaltrials.gov/study/NCT05812001> (accessed 08/09/2025).
- 638.** NCT06387732. The mechanisms underlying the antidepressant effects of physical activity. <https://clinicaltrials.gov/study/NCT06387732> (accessed 08/09/2025).
- 639.** NCT06398496. The effects of exercise on gut bacteria, mood and cognition in depression. <https://clinicaltrials.gov/ct2/show/NCT06398496> (accessed 08/09/2025).
- 640.** NCT06450704. Cerebral and anti-inflammatory response through exercise - mechanisms in depressive disorders. <https://clinicaltrials.gov/ct2/show/NCT06450704> (accessed 08/09/2025).
- 641.** NCT06594588. Individualised exploration of aerobic exercise-assisted treatment of depression. <https://clinicaltrials.gov/ct2/show/NCT06594588> (accessed 08/09/2025).
- 642.** Neves LM, Silva-Batista C, Marquesini R, Da Cunha TF, Dimateo E, Nascimento L, et al. Aerobic exercise program with or without motor complexity as an add-on to the pharmacological treatment of depression - study protocol for a randomized controlled trial. *Trials [electronic Resource]* 2018;**19**(1):545.
- 643.** Safakar MY, Hrabok M, Urchuk L, Juhas M, Shalaby R, Parmar D, et al. Effectiveness of group cognitive behavioral therapy and exercise in the management of major depressive disorder: protocol for a pilot randomized controlled trial. *JMIR Research Protocols* 2020;**9**(5):e14309.
- 644.** Schmitter M, Spijker J, Smit F, Tendolkar I, Derksen A-M, Oostelbos P, et al. Exercise enhances: study protocol of a randomized controlled trial on aerobic exercise as depression treatment augmentation. *BMC Psychiatry* 2020;**20**(585):<https://doi.org/10.1186/s12888-020-02989-z>.
- 645.** Nayereh Kiani Shahvandi. Effect of combined training and detraining on type 2 diabetic women with depression. NIPH Clinical Trials Registry 2020.
- 646.** A randomised feasibility study assessing the effect of an active virtual reality gaming intervention on physical activity and mood in young men with mild to moderate depression. <http://www.who.int/trialssearch/Trial2.aspx?TrialID=ACTRN12620001372976>.
- 647.** ACTRN12624001150538. Creatine and exercise as a novel treatment strategy for depression in adults aged 18-39 years. <https://anzctr.org.au/ACTRN12624001150538.aspx> (accessed 08/09/2025).
- 648.** Aerobic and mind body exercise interventions for adults with anxiety and depressive disorders under primary care settings in Hong Kong - a randomized control trial. <http://www.who.int/trialssearch/Trial2.aspx?TrialID=ChiCTR-IOR-14005463> 2014.
- 649.** ChiCTR2200058744. Effects of table tennis and brisk walking on outpatients with mild to moderate depression. <https://www.chictr.org.cn/showproj.html?proj=160053> (accessed 08/09/2025).
- 650.** ChiCTR2300074393. Effects of high-intensity interval training on depressive symptoms in hong kong older adults with mild and moderate depressive symptoms: a cluster-randomized controlled trial. <https://www.chictr.org.cn/showproj.html?proj=201899> (accessed 08/09/2025).
- 651.** ChiCTR2300075559. Study on the impact of exercise intervention on depression in university students. <https://www.chictr.org.cn/showproj.html?proj=205746> (accessed 08/09/2025).
- 652.** ChiCTR2300077020. Mindfulness and Exercise Combined Intervention for Depressive and Insomnia Symptoms in Chinese Students: A Four-arm Randomized Controlled Study. <https://www.chictr.org.cn/showproj.html?proj=206531> (accessed 08/09/2025).
- 653.** ChiCTR2400087257. The effect of 8-week taekwondo exercise on the response inhibition of college students with depression symptoms - evidence based on ERPs. <https://www.chictr.org.cn/showproj.html?proj=228375> (accessed 08/09/2025).
- 654.** ChiCTR2400087261. The effect of 8 weeks of taekwondo training on conflict inhibition in depressive symptoms in university students: evidence from behavioral and ERP studies.

<https://www.chictr.org.cn/showproj.html?proj=237573> (accessed 08/09/2025).

655. ChiCTR2500097477. Effect of exercise on cognitive function in patients with mild to moderate depression. <https://www.chictr.org.cn/showproj.html?proj=257008> (accessed 08/09/2025).

656. ChiCTR2400083854. Individualized exploration of aerobic exercise-assisted treatment of depression monitored by smart band: a randomized controlled trial. <https://www.chictr.org.cn/showproj.html?proj=228691> (accessed 08/09/2025).

657. ChiCTR2400084484. Digital exercise intervention therapy for residual symptoms of depression: A randomized double-blind controlled trial. <https://www.chictr.org.cn/showproj.html?proj=229578> (accessed 08/09/2025).

658. ChiCTR2400091623. Fitness for body and mind: a study of online aerobic fitness combined with a digital psychological intervention to improve depression and enhance mental health in university students. <https://www.chictr.org.cn/showproj.html?proj=248745> (accessed 08/09/2025).

659. ChiCTR2500107313. Analysis of the effect of exercise prescription on depressive symptoms and severity in patients with mild-to-moderate depression. <https://www.chictr.org.cn/showproj.html?proj=274657> (accessed 08/09/2025).

660. To find out the effect of 2 different types of exercise on young men with depression. <http://www.who.int/trialsearch/Trial2.aspx?TrialID=CTRI/2016/06/007055> 2016.

661. Internet-based walking intervention for obese pregnant women with prenatal depression: a randomized controlled trial. <http://www.who.int/trialsearch/Trial2.aspx?TrialID=DRKS00005823> 2014.

662. Neurobiological correlates and mechanisms of the augmentation of psychotherapy with endurance exercise in mild to moderate depression. <http://www.who.int/trialsearch/Trial2.aspx?TrialID=DRKS00008869> 2015.

663. Promoting physical activity: individual training for people with depressive illness. <http://www.who.int/trialsearch/Trial2.aspx?TrialID=DRKS00023257> 2020.

664. Physical exercise and internet-based cognitive behavioral therapy in the treatment of depression (Regassa): a multi-centre, single blind, randomized controlled trial. <http://www.who.int/trialsearch/Trial2.aspx?TrialID=DRKS00008745> 2015.

665. DRKS00026117. Short-term aerobic exercise for depression in acute geriatric psychiatry: a randomized controlled trial. <http://drks.de/search/en/trial/DRKS00026117> (accessed 08/09/2025).

666. DRKS00032065. Short-term aerobic exercise for depression in an outpatient geriatric psychiatry: a multicenter RCT. <http://drks.de/search/en/trial/DRKS00032065> (accessed 08/09/2025).

667. Investigation the effectiveness of sport program on depression and fatigue in women with breast cancer under chemotherapy. <http://www.who.int/trialsearch/Trial2.aspx?TrialID=IRCT20130812014333N86> 2018.

668. Walking with pedometer, Depression, General health in postmenopausal women. <http://www.who.int/trialsearch/Trial2.aspx?TrialID=IRCT2014020616501N1> 2014.

669. The effectiveness of group aerobic exercise on depressed students' cognitive change process. <http://www.who.int/trialsearch/Trial2.aspx?TrialID=IRCT2015093024282N1> 2015.

670. The effect of aerobic exercise on depression. <http://www.who.int/trialsearch/Trial2.aspx?TrialID=IRCT2016082929598N1> 2016.

671. Effects of exercise in the treatment of depression. <http://www.who.int/trialsearch/Trial2.aspx?TrialID=IRCT2016111630924N1> 2017.

672. IRCT20190805044438N4. The effect of planned walking on the depression level of the depressed elderly. <http://en.irct.ir/trial/73916> (accessed 08/09/2025).

673. IRCT20210617051606N1. Effect of exercise training on depression. <http://en.irct.ir/trial/57026> (accessed 08/09/2025).

674. IRCT20211011052727N1. Effectiveness of aerobic exercises as an adjunctive therapy in depression: a randomized controlled trial. <http://en.irct.ir/trial/63467> (accessed 09/09/2025).

675. IRCT20241206063964N1. Comparison of the effect of moderate walking exercise and reminiscence therapy on the symptoms of anxiety and depression in the older adult. <https://irct.behdasht.gov.ir/trial/80590> (accessed 08/09/2025).

676. IRCT20250402065189N1. Effects of eight weeks of aerobic exercise and vitamin E supplementation in women with depressive symptoms. <https://irct.behdasht.gov.ir/trial/82801> (accessed 08/09/2025).

677. Lifestyle physical activity counselling for in-patients with major depressive disorder. <http://www.who.int/trialsearch/Trial2.aspx?TrialID=ISRCTN10469580> 2018.

678. Brief exercise for depressive and anxiety symptoms among inpatients with severe mental illness. <http://www.who.int/trialsearch/Trial2.aspx?TrialID=ISRCTN14914508> 2020.

679. The effect of cognitive rehabilitation vs. Low intensity exercise for post - stroke depression in long - term care health facilities: a randomized controlled trial. <http://www.who.int/trialsearch/Trial2.aspx?TrialID=JPRN-UMIN000029117> 2018.

680. The effects of aerobic exercise and computer-based cognitive training on depressive symptoms and cognitive function of major depressive disorder and bipolar disorder: a randomized, open-label, parallel group trial. <http://www.who.int/trialsearch/Trial2.aspx?TrialID=JPRN-UMIN000036468> 2019.

- 681.** Shahvandi NK. Effect of combined training and detraining on type 2 diabetic women with depression. *NIPH Clinical Trials* 2020.
- 682.** Movement for depression - an RCT assessing the effects of physical activity promotion for people with MDD. <https://ClinicalTrials.gov/show/NCT03403881> 2018.
- 683.** Physical exercise for treatment of depression and anxiety - RCT. <https://ClinicalTrials.gov/show/NCT04714528> 2020.
- 684.** Exercise Enhances: a randomized controlled trial on aerobic exercise as depression treatment augmentation strategy. <http://www.who.int/trialssearch/Trial2.aspx?TrialID=NL8432> 2020.
- 685.** Physical Activity and Sleep Hygiene as a treatment of Depression in the elderly. <http://www.who.int/trialssearch/Trial2.aspx?TrialID=RBR-248x3v> 2018.
- 686.** Effect of strength training in older depressed. <http://www.who.int/trialssearch/Trial2.aspx?TrialID=RBR-38nkj6> 2014.
- 687.** Effect of different types of physical exercise in SUS patients with depression. <http://www.who.int/trialssearch/Trial2.aspx?TrialID=RBR-9zgxzd> accessed 14/07/21.
- 688.** RBR-4pgd3ct. Complementary therapies for depression. <https://ensaiosclinicos.gov.br/rg/RBR-4pgd3ct> (Accessed 08/09/2025).
- 689.** Egger M, Davey Smith G, Minder C. Bias in meta-analysis detected by a simple, graphical test. *BMJ (Clinical Research Ed.)* 1997;**315**(7109):629-34.
- 690.** Duval S, Tweedie R. Trim and fill: a simple funnel-plot-based method of testing and adjusting for publication bias in meta-analysis. *Biometrics* 2000;**56**(2):455-63.
- 691.** Department of Health. Exercise referral systems: a national quality assurance framework. Available from http://www.dh.gov.uk/en/Publicationsandstatistics/Publications/PublicationsPolicyAndGuidance/DH_4009671 2001.
- 692.** Pavey TG, Taylor AH, Fox KR, Hillsdon M, Anoyke N, Campbell L, et al. Effect of exercise referral schemes in primary care on physical activity and improving health outcomes: systematic review and meta-analysis. *BMJ (Clinical Research Ed.)* 2011;**343**:d6462.
- 693.** Sorensen JB, Skovgaard T, Puggaard L. Exercise on prescription in general practice: a systematic review. *Scandinavian Journal of Primary Health Care* 2006;**24**(2):69-74.
- 694.** Liu Y, Zhao G, Guo J, Qu H, Kong L, Yue W. The efficacy of exercise interventions on depressive symptoms and cognitive function in adults with depression: an umbrella review. *Journal of Affective Disorders* 2024;**368**:779-88.
- 695.** Seshadri A, Adaji A, Orth SS, Singh B, Clark MM, Frye MA, et al. Exercise, yoga, and Tai Chi for treatment of major depressive disorder in outpatient settings: a systematic review and meta-analysis. *Primary Care Companion for CNS Disorders* 2020;**23**(1):20r02722.
- 696.** Hooper N, Johnson T, Sachs M, Silverio A, Zhu L, Bhimla A, et al. Comparative efficacy of exercise training and conventional psychotherapies for adult depression: a network meta-analysis. *Commonhealth (Philadelphia, Pa.)* 2022;**3**(2):47-64.
- 697.** Md Zemberi NF, Ismail MM, Abdullah M. Exercise interventions as the primary treatment for depression: evidence from a narrative review. *Malaysian Journal of Medical Sciences* 2020;**27**(5):5-23.
- 698.** Pérez Bedoya ÉA, Puerta-López LF, López Galvis DA, Rojas Jaimes DA, Moreira OC. Physical exercise and major depressive disorder in adults: systematic review and meta-analysis. *Scientific Reports* 2023;**13**(1):13223.
- 699.** Tian S, Liang Z, Qui F, Yu Y, Wang C, Zhang M, et al. Optimal exercise modality and dose to improve depressive symptoms in adults with major depressive disorder: a systematic review and Bayesian model-based network meta-analysis of RCTs. *Journal of Psychiatric Research* 2024;**176**:384-92.
- 700.** Rossi FE, Dos Santos GG, Rossi PA, Stubbs B, Barreto Schuch F, Neves LM. Strength training has antidepressant effects in people with depression or depressive symptoms but no other severe diseases: a systematic review with meta-analysis. *Psychiatry Research* 2024;**334**:115805.
- 701.** Carneiro L, Afonso J, Ramirez-Campillo R, Murawska-Ciałowicz E, Marques A, Clemente FM. The effects of exclusively resistance training-based supervised programs in people with depression: a systematic review and meta-analysis of randomized controlled trials. *International Journal of Environmental Research and Public Health* 2020;**17**(18):6715.

ADDITIONAL TABLES

Table 1. Participant numbers in included studies

Trial ID	Screened	Ran- domised	Allocated exercise	Completed trial	Completed com- parator group, e.g. control, oth- er treatment (as a proportion of those allocated)	Completed exercise (as a proportion of those allocated)
Abdelbasset 2019	57	46	23	46	23/23	23/23

Table 1. Participant numbers in included studies *(Continued)*

Abdollahi 2017	77	70	35	54	29/35	25/35
Bademli 2023	124	76	31	62	31/31	31/31
Barclay 2014	Not reported	11	5	11	6/6	5/5
Belvederi Murri 2015	177	121	42	102	27/37	34/42
Blumenthal 1999	604 underwent tele- phone screening	156	55	133	41/48 (medica- tion)	44/55 (exercise plus medication) 39/53 (exercise alone)
Blumenthal 2007	457	202	51 (super- vised), 53 home- based	183	42/49 (placebo) 45/49 (sertraline)	45/51 (supervised), 51/53 home-based
Blumenthal 2012b	1680 enquired about the study	101	37	95	23/24 complet- ed placebo and 36/40 completed the medication	36/37 completed the exercise
Bonnet 2005	Not reported	11	5	7	4/6	3/5
Bose 2015	146	60	30	54	28/30	26/30
Brenes 2007	Not reported	37	14	Not report- ed	Not reported	Not reported
Brush 2022	Not reported	66	35	51	26/35	25/31
Buschert 2019	43	38	18	30	15/20	15/18
Cheung 2018	44	34	17	32	17/17	15/17
Chu 2008	104 responded to ad- verts	54	36	38	12/18	26/36 (both ex- ercise arms com- bined) 15/18 in the high-in- tensity arm
De Groot 2019	180	140	34	123	28/36	30/34
Doose 2015	106	46	30	35	12/16	23/30
Doyne 1987	285 responded to ad- verts	57	Not report- ed	40 complet- ed treat- ment or control	27 (denominator not known)	13 (denominator not known)
Dunn 2005	1664 assessed for eli- gibility	80	17	45	9/13	11/17 (public health dose 3 times per week)

Table 1. Participant numbers in included studies (Continued)

Eman 2017	Not reported	40	20	34	14/20	10/20
Epstein 1986	250 telephone enquiries received	33	7	Not reported	Not reported	7
Fetsch 1979	Not reported	21	10	16	8/11	8/10
Foley 2008	215 responded to adverts	23	10	13	5/13	8/10
Fremont 1987	72 initially expressed an interest	61	21	49	31/40	18/21
Gary 2010	982 referred, 242 had heart failure, 137 had a BDI > 10 and 74 eligible and consented	74	20	68/74 completed post-intervention assessments and 62 completed follow-up assessments	Usual care 15/17	Exercise only: 20/20
Greist 1979	Not reported	28	10	22	15/18	8/10
Haller 2018	22	20	14	17	6/6	11/14
Heinzel 2022	467	120	45	90	23/30	22/45
Heissel 2015	26	14	7	12	6/7	6/7
Hemat-Far 2012	350 screened	20	10		Not stated	Not stated
Hess-Homeier 1981	Not reported	17	5	Not reported	Not reported	Not reported
Hidalgo 2021	432	347	176	159	93/171	66/176
Ho 2014	96	52	26	40	21/26	19/26
Hoffman 2010	253 screened, 58 ineligible	84	42	76	39/42 (2 were excluded by the trialists and 1 did not attend follow-up)	37/42 of exercise group provided data for analysis
Huang 2015	479	57	19	57	20/20	19/19
Karg 2020	332	233	77	198	65/77	69/77
Khorvash 2012	300	120	60	Not reported	Not reported	Not reported
Khoshnab 2017	Not reported	60	15	Not reported	Not reported	Not reported

Table 1. Participant numbers in included studies (Continued)

Klein 1985	209 responded to an advert	74	27	42	11/23 (meditation) 16/24 (group therapy)	15/27
Knubben 2007	Not reported	39 (note data on only 38 reported)	20	35	16/18	19/20
Krogh 2009	390 referred	165	110	137	42/55	95/110 (both exercise arms combined) 47/55 (strength) 48/55 (aerobic)
Kruisdijk 2018	183	48	25	46	4/23 at 12 months follow-up	5/25 at 12 months follow-up
La Rocque 2021	98	53	20	42	12/15	15/20
Legrand 2014	49	44	22	27	12/22	15/22
Legrand 2016	124	35	14	31	9/10	13/14
Martinsen 1985	Not reported	43	24	37	17/19	20/24
Mather 2002	1185 referred or screened	86	43	86	42/43	43/43
McCann 1984	250 completed BDI, 60 contacted	47	16	43	14/15 completed placebo 14/16 completed 'no treatment'	15/16
McNeil 1991	82	30	10	30	10/10 (waiting list) 10/10 (social contact)	10/10
Mota-Pereira 2011	150	33	22	29/33	10/11	19/22
Mutrie 1988	36	24	9	24	7/7	9/9
Nabkasorn 2005	266 volunteers screened	59	28	49	28/31	21/28
Olson 2017	56	50	25	30	15/25	15/25
Orth 1979	17	11	3	7	2/2	3/3
Pfaff 2014	426	200	108	172	81/92	91/108
Pilu 2007	Not reported	30	10	30	20/20	10/10

Table 1. Participant numbers in included studies (Continued)

Pinchasov 2000	Not reported	18	9	Not reported	Not reported	Not reported
Prakhinkit 2014	Not reported	45	15	40	13/15	14/15
Reuter 1984	Not reported	Not reported	9	Not reported	Not reported	9
Roy 2018	54	46	23	40	20/23	20/23
Salehi 2016	70	60	20	60	20/20	20/20
Schuch 2015	106	50	25	47	24/25	23/25
Setaro 1985	211 responses to advertisement	180	30	150	Not reported	25/30
Shahidi 2011	70 older depressed women chosen from 500 members of a district using the Geriatric Depression Scale	70	23	60/70	20/24	20/23
Sims 2009	1550 invitations, 233 responded	45	23	43	22/22	21/23
Singh 1997	Letters sent to 2953 people, 884 responded	32	17	32	15/15	17/17
Singh 2005	451	60	20	54	19/20 (GP standard care)	18/20 (high-intensity training)
Siqueira 2016	112	57	29	40	20/28	20/29
Taheri 2018	Not reported	32	8	32	8/8	8/8
Veale 1992	Not reported	83	48	57	29/35	36/48
Wang 2022	Not reported	Not reported	Not reported	Not reported	30 (denominator not reported)	30 (denominator not reported)
Williams 2008	96 in parent study	43	33	34	8/10	26/33 (both exercise groups combined) 15/16 exercise 11/17 walking
Yu 2023	67	35	11	30	10/11	10/11

BDI: Beck Depression Inventory

Table 2. Characteristics of included studies and synthesis

Study ID (country)	Participants	Intervention	Comparators	Outcomes (depression)	Meta-analyses
Abdelbasset 2019 (Saudi Arabia)	Source: clinical (people with New York Heart Association Class II to III congestive heart failure); Age: range 40 to 60 years; Sex: 24% women	Aerobic exercise (treadmill walk)	Health education (patient recommendations by psychosocial support)	Patient Health Questionnaire-9 (PHQ-9)	Analysis 4.1; Analysis 4.3
Abdollahi 2017 (Iran)	Source: clinical (psychology clinics); Age: mean 49.67 years; Sex: 47.1% women	Aerobic exercise plus Cognitive Behavioural Therapy (group - clapping and walking)	Cognitive Behavioural Therapy	1. Beck Depression Inventory-II 2. Beck Scale for Suicidal Ideation	Analysis 1.1; Analysis 1.3; Analysis 5.1; Analysis 5.2; Analysis 5.3; Analysis 5.4; Analysis 5.5; Analysis 6.1; Analysis 6.7
Bademli 2023 (Turkey)	Source: clinical (caregivers for people with schizophrenia); Age: > 18 years; Sex: 59.7% women	Aerobic exercise (group - rhythmic exercise and walking)	No treatment, waiting list, usual care or self-management (no change in normal routine)	Beck Depression Inventory	Analysis 1.1; Analysis 1.3; Analysis 5.1; Analysis 5.2; Analysis 5.3; Analysis 5.4; Analysis 5.5; Analysis 6.1; Analysis 6.4; Analysis 6.7
Barclay 2014 (USA)	Source: clinical and non-clinical (occupational health practice); Age: range 18 to 65 years; Sex: 100% women	Combined exercise (group - anaerobic and resistance)	No treatment, waiting list, usual care or self-management (treatment as normal)	Beck Depression Inventory-II	Analysis 1.1; Analysis 1.3; Analysis 5.1; Analysis 5.2; Analysis 5.3; Analysis 5.4; Analysis 5.5; Analysis 6.1; Analysis 6.4; Analysis 6.7
Belvederi Murri 2015 (Italy)	Source: clinical (sedentary adults); Age: range 65 to 85 years; Sex: 71% women	Progressive exercise (exercise bike) plus sertraline	1. Sertraline plus non-progressive exercise; 2. Sertraline	1. Hamilton Rating Scale for Depression (HRSD) score 2. Global improvement of depression (CGI)	Analysis 1.1; Analysis 1.3; Analysis 5.1; Analysis 5.2; Analysis 5.3; Analysis 5.4; Analysis 5.5; Analysis 6.1; Analysis 6.4; Analysis 6.5; Analysis 6.7
Blumenthal 1999 (USA)	Source: non-clinical (community volunteers); Age: mean 70 years; Sex: 63% women	Aerobic exercise (group - walking or jogging)	1. Aerobic exercise plus sertraline; 2. Sertraline	1. Clinical diagnosis of depression using DSM-IV 2. Hamilton Rating Scale for Depression 3. Beck Depression Inventory	Analysis 1.1; Analysis 1.2; Analysis 1.3; Analysis 3.1; Analysis 3.2; Analysis 3.3; Analysis 5.1; Analysis 5.2; Analysis 5.3; Analysis 5.4; Analysis 5.5; Analysis 6.1; Analysis 6.3; Analysis 6.4; Analysis 6.5; Analysis 6.6; Analysis 6.7
Blumenthal 2007 (USA)	Source: non-clinical (recruited via media); Age: mean 52 years; Sex: 76% women	Aerobic exercise (supervised - individual - walk or jogging)	1. Placebo; 2. Sertraline; 3. Home-based aerobic exercise	Primary endpoint was remission (no Major Depressive Disorder)	Analysis 1.1; Analysis 1.3; Analysis 3.1; Analysis 3.3; Analysis 5.1; Analysis 5.2; Analysis 5.3; Analysis 5.4; Analysis 5.5; Analysis 6.1;

Table 2. Characteristics of included studies and synthesis (Continued)

				and a Hamilton Rating Scale score of < 8, and also a continuous severity score on the Hamilton Rating Scale for Depression	Analysis 6.3; Analysis 6.4; Analysis 6.5; Analysis 6.6; Analysis 6.7
Blumenthal 2012a (USA)	Source: clinical and non-clinical (with coronary heart disease); Age: 35 years or older; Sex: NR	Aerobic exercise (group - walking, running or jogging on treadmill)	1. Placebo; 2. Sertraline	Depression as diagnosed by DSM-IV and severity of depression as rated on Hamilton Rating Scale for Depression	Analysis 1.1; Analysis 1.3; Analysis 3.1; Analysis 3.3; Analysis 5.1; Analysis 5.2; Analysis 5.3; Analysis 5.4; Analysis 5.5; Analysis 6.1; Analysis 6.3; Analysis 6.4; Analysis 6.5; Analysis 6.6; Analysis 6.7
Bonnet 2005 (USA)	Source: non-clinical (university counselling); Age: mean 23.3 years; Sex: 82% women	Aerobic exercise (treadmill walk) plus Cognitive Behavioural Therapy	Cognitive Behavioural Therapy	1. Diagnostic and Statistical Manual of Mental Disorder (IV) diagnosis of major depressive disorder, dysthymia or depressive disorder 2. Above cut-off depression on Beck Depression Inventory and Center for Epidemiological Studies-Depression Scale	Analysis 1.1; Analysis 1.3; Analysis 5.1; Analysis 5.2; Analysis 5.3; Analysis 5.4; Analysis 5.5; Analysis 6.1; Analysis 6.7
Bose 2015 (India)	Source: unclear (pregnant women); Age: 17 to 40 years; Sex: 100% women	Aerobic exercise (walking, stretching, cycling)	Health education (patient handout)	Physical Health Questionnaire-9	Analysis 4.1; Analysis 4.3
Brenes 2007 (USA)	Source: non-clinical (community dwelling and nursing home people); Age: over 65 years; Sex: NR	Combined exercise (group - aerobic and resistance)	1. No treatment, waiting list, usual care or self-management (discussion of general health); 2. Sertraline	Hamilton Rating Scale for Depression	Analysis 1.1; Analysis 1.4; Analysis 3.1; Analysis 3.4; Analysis 5.1; Analysis 5.2; Analysis 5.3; Analysis 5.4; Analysis 5.5; Analysis 6.1; Analysis 6.5; Analysis 6.7
Brush 2022 (USA)	Source: non-clinical; Age: mean 20 years; Sex: 74.24% women	Aerobic exercise (treadmill or cycle ergometer)	Stretching	Beck Depression Inventory-II	Analysis 1.1; Analysis 1.3; Analysis 5.1; Analysis 5.2; Analysis 5.3; Analysis 5.4; Analysis 5.5; Analysis 6.1; Analysis 6.4; Analysis 6.7

Table 2. Characteristics of included studies and synthesis (Continued)

Buschert 2019 (Germany)	Source: clinical (Hospital); Age: mean 47.37 years; Sex: 63% women	Aerobic exercise (group - walk &/or running)	Occupational or art therapy	1. Beck Depression Inventory I-II 2. Toronto Hamilton Rating Scale for Depression	Analysis 4.1; Analysis 4.3
Cheung 2018 (China)	Source: clinical (Hospital); Age: mean 47.8 years; Sex: 79.4% women	Aerobic exercise (individual - jog/walk/cross trainer/stationary bike)	Control given no instructions to change physical activity	1. Hamilton Rating Scale for Depression 2. Hospital Anxiety and Depression Scale	Analysis 1.1; Analysis 1.3; Analysis 5.1; Analysis 5.2; Analysis 5.3; Analysis 5.4; Analysis 5.5; Analysis 6.1; Analysis 6.3; Analysis 6.4; Analysis 6.5; Analysis 6.6; Analysis 6.7
Chu 2008 (USA)	Source: clinical and non-clinical (volunteers - university and physician referral); Age: mean 26.4 years; Sex: 100% women	Aerobic exercise (participant preference - individual - high intensity)	1. Aerobic (participant preference - low intensity); 2. Stretching	Beck Depression Inventory-II	Analysis 1.1; Analysis 1.3; Analysis 5.1; Analysis 5.2; Analysis 5.3; Analysis 5.4; Analysis 5.5; Analysis 6.1; Analysis 6.7
De Groot 2019 (USA)	Source: clinical and non-clinical (diabetic patients); Age: mean 56 years; Sex: 77% women	Aerobic exercise (group - unspecified)	1. No treatment, waiting list, usual care or self-management (usual care); 2. Cognitive Behavioural Therapy; 3. Aerobic (unspecified) plus Cognitive Behavioural Therapy	Depression remission	Analysis 1.1; Analysis 1.3; Analysis 2.1; Analysis 2.3; Analysis 5.1; Analysis 5.2; Analysis 5.3; Analysis 5.4; Analysis 5.5; Analysis 6.1; Analysis 6.3; Analysis 6.5; Analysis 6.7
Doose 2015 (Germany)	Source: non-clinical (general public); Age: mean 46.07 years; Sex: 63% women	Aerobic exercise (group - walk and/or running)	No treatment, waiting list, usual care or self-management (usual care)	1. Hamilton Rating Scale for Depression (HRSD-17) 2. Becks Depression Inventory II	Analysis 1.1; Analysis 1.3; Analysis 5.1; Analysis 5.2; Analysis 5.3; Analysis 5.4; Analysis 5.5; Analysis 6.1; Analysis 6.4; Analysis 6.7
Doyne 1987 (USA)	Source: non-clinical (community volunteers); Age: mean 28.5 years; Sex: 100% women	Aerobic exercise (individual - running or walking)	1. No treatment, waiting list, usual care or self-management (waiting list); 2. Weightlifting	1. Beck Depression Inventory 2. Lubin's Depression Adjective List 3. Hamilton Rating Scale for Depression	Analysis 1.1; Analysis 5.1; Analysis 5.2; Analysis 5.3; Analysis 5.4; Analysis 5.5; Analysis 6.1; Analysis 6.7
Dunn 2005 (USA)	Source: non-clinical (community volunteers); Age: mean 35.9 years; Sex: 75% women	Aerobic exercise (treadmill walk - public health dose - individual - 3 session/week)	1. Aerobic (treadmill walk - public health dose - 5 session/week); 2. Aerobic (treadmill walk - low dose - 5 session/week); 3.	Hamilton Rating Scale for Depression	Analysis 1.1; Analysis 1.3; Analysis 5.1; Analysis 5.3; Analysis 5.4; Analysis 5.5; Analysis 6.1; Analysis 6.3; Analysis 6.4; Analysis 6.5; Analysis 6.6; Analysis 6.7

Table 2. Characteristics of included studies and synthesis (Continued)

			Aerobic (treadmill walk - low dose - 5 session/week); 4. Flexibility (3 sessions/week)		
Eman 2017 (Egypt)	Source: clinical (post-menopausal women); Age: mean 56.77 years; Sex: 100% women	Aerobic exercise (walking)	Reflexology	Beck Depression Inventory (BDI-II)	Analysis 4.1
Epstein 1986 (USA)	Source: non-clinical (community volunteers); Age: mean 39.4 years; Sex: 92% women	Aerobic exercise (walking or jogging)	1. No treatment, waiting list, usual care or self-management (waiting list); 2. Cognitive Behavioural Therapy	1. Beck Depression Inventory 2. Zung Self Rating Depression Scale	Analysis 1.1; Analysis 2.1; Analysis 5.1; Analysis 5.2; Analysis 5.3; Analysis 5.4; Analysis 5.5; Analysis 6.1; Analysis 6.7
Fetsch 1979 (USA)	Source: non-clinical (iniversity counselling - advertisements); Age: NR; Sex: NR	Aerobic exercise (group - running)	Cognitive Behavioural Therapy (stroking therapy - type of 'talking' therapy)	Beck Depression inventory	Analysis 2.1; Analysis 2.3
Foley 2008 (New Zealand)	Source: clinical and non-clinical (media adverts and psychiatric referrals); Age: 18 to 55 years; Sex: NR	Aerobic exercise (unspecified)	Stretching	1. Beck Depression Inventory; 2. Montgomery-Asberg Depression Rating scale	Analysis 1.1; Analysis 1.3; Analysis 5.1; Analysis 5.2; Analysis 5.3; Analysis 5.4; Analysis 5.5; Analysis 6.1; Analysis 6.7
Fremont 1987 (USA)	Source: non-clinical (community volunteers); Age: NR; Sex: NR	Aerobic exercise (group - running)	1. Aerobic (running) plus Cognitive Behavioural Therapy; 2. Cognitive Behavioural Therapy	Beck Depression Inventory	Analysis 1.1; Analysis 1.2; Analysis 1.3; Analysis 2.1; Analysis 2.2; Analysis 2.3; Analysis 5.1; Analysis 5.2; Analysis 5.3; Analysis 5.4; Analysis 5.5; Analysis 6.1; Analysis 6.7
Gary 2010 (USA)	Source: clinical (New York Heart Association II and III); Age: 30 to 70 years; Sex: 58% women	Aerobic exercise (walking)	1. No treatment, waiting list, usual care or self-management (usual care); 2. Cognitive Behavioural Therapy; 3. Aerobic plus Cognitive Behavioural Therapy	Hamilton Rating Scale for Depression	Analysis 1.1; Analysis 1.2; Analysis 1.3; Analysis 1.4; Analysis 2.1; Analysis 2.2; Analysis 2.3; Analysis 2.4; Analysis 5.1; Analysis 5.2; Analysis 5.3; Analysis 5.4; Analysis 5.5; Analysis 6.1; Analysis 6.5; Analysis 6.7
Greist 1979 (USA)	Source: non-clinical (community volunteers); Age: 18 to 30 years; Sex: 53.4% women	Aerobic exercise (individual - running)	1. Low-intensity Cognitive Behavioural Therapy; 2. High-intensity Cognitive Behavioural Therapy	Symptom checklist score	

Table 2. Characteristics of included studies and synthesis (Continued)

Haller 2018 (Germany)	Source: clinical (out-patient psychiatrists); Age (mean): 45 years; Sex: 65% women	Aerobic exercise (individual - endurance and strength)	No treatment, waiting list, usual care or self-management (treatment as usual)	Quick Inventory of Depressive Symptomatology (QIDS)	Analysis 1.3
Heinzel 2022 (Germany)	Source: clinical and non-clinical (recruited through media); Age: mean 39.46 years; Sex: 52.2% women	High-intensity exercise (cycle ergometer, running or Nordic walking, aerobic workout)	1. Low intensity (cycle ergometer, running, stretching, relaxation); 2. No treatment, waiting list, usual care or self-management (waiting list)	Hamilton Rating Scale for Depression	Analysis 1.1; Analysis 1.2; Analysis 1.3; Analysis 5.1; Analysis 5.2; Analysis 5.3; Analysis 5.4; Analysis 5.5; Analysis 6.1; Analysis 6.4; Analysis 6.7
Heissel 2015 (Germany)	Source: clinical (inpatients); Age: (mean) 66.75 years; Sex: 58.3% women	Combined exercise (group - aerobic and resistance)	Stretch and relaxation	Beck Depression Inventory (BDI II)	Analysis 1.1; Analysis 1.3; Analysis 5.1; Analysis 5.2; Analysis 5.3; Analysis 5.4; Analysis 5.5; Analysis 6.1; Analysis 6.7
Hemat-Far 2012 (Iran)	Source: non-clinical (university students); Age: 18 to 25 years; Sex: 100% women	Aerobic exercise (running)	No treatment, waiting list, usual care or self-management (no active intervention)	Beck Depression Inventory	Analysis 1.1; Analysis 1.3; Analysis 5.1; Analysis 5.2; Analysis 5.3; Analysis 5.4; Analysis 5.5; Analysis 6.1; Analysis 6.7
Hess-Homeier 1981 (USA)	Source: non-clinical (community volunteers via media); Age: NR; Sex: NR	Aerobic exercise (walking or running)	1. No treatment, waiting list, usual care or self-management (waiting list); 2. Cognitive Behavioural Therapy	Beck Depression Inventory	Analysis 1.1; Analysis 2.1; Analysis 5.1; Analysis 5.2; Analysis 5.3; Analysis 5.4; Analysis 5.5; Analysis 6.1; Analysis 6.7
Hidalgo 2021 (Spain)	Source: clinical (out-patient family medicine clinic); Age: mean 72 years; Sex: 78.59% women	Combined exercise (group - aerobic, strength, flexibility, balance)	Antidepressant therapy	Montgomery-Åsberg Depression Rating Scale (MADRS)	Analysis 3.3
Ho 2014 (Hong Kong)	Source: clinical (in-patients psychiatric unit); Age: mean 46.15 years; Sex: 67% women	Aerobic exercise (interval training)	Stretching	1. Montgomery-Åsberg Depression Rating Scale (MADRS) 2. Beck Depression Inventory (Chinese)	Analysis 1.1; Analysis 1.3; Analysis 5.1; Analysis 5.2; Analysis 5.3; Analysis 5.4; Analysis 5.5; Analysis 6.1; Analysis 6.3; Analysis 6.5; Analysis 6.7
Hoffman 2010 (USA)	Source: clinical and non-clinical (traumatic brain injury); Age: NR; Sex: NR	Aerobic exercise (participants preference)	No treatment, waiting list, usual care or self-management (no intervention)	Beck Depression Inventory	Analysis 1.1; Analysis 1.3; Analysis 5.1; Analysis 5.2; Analysis 5.3; Analysis 5.4; Analysis 5.5; Analysis 6.1; Analysis 6.3; Analysis 6.7

Table 2. Characteristics of included studies and synthesis (Continued)

Huang 2015 (Taiwan)	Source: non-clinical; Age: mean 76.54 years; Sex: 52.62% women	Combined exercise (group - aerobic and resistance)	1. No treatment, waiting list, usual care or self-management; 2. Cognitive Behavioural Therapy	Geriatric Depression Scale (GDS) - 15	Analysis 1.1; Analysis 1.2; Analysis 1.3; Analysis 1.4; Analysis 2.1; Analysis 2.2; Analysis 2.3; Analysis 2.4; Analysis 5.1; Analysis 5.2; Analysis 5.3; Analysis 5.4; Analysis 5.5; Analysis 6.1; Analysis 6.4; Analysis 6.5; Analysis 6.7
Karg 2020 (Germany)	Source: clinical and non-clinical; Age: mean 42 years; Sex: 69.2% women	Combined exercise (aerobic, resistance, psychoeducation)	1. CBT; 2. Boulder-ing psychotherapy	Montgomery-Asberg Depression Rating Scale	
Khorvash 2012 (Iran)	Source: non-clinical (university physical education); Age: mean 25.1 years; Sex: 0% women	Strength exercise	1. No treatment, waiting list, usual care or self-management; 2. Endurance training	1. Beck questionnaire - depression 2. Kettle questionnaire - anxiety	
Khoshnab 2017 (Iran)	Source: unclear; Age: mean 51.2 years; Sex: 100% women	Aerobic exercise (treadmill and sertraline)	1. No treatment, waiting list, usual care or self-management (control); 2. Sertraline; 3. Aerobic exercise plus sertraline	Beck Depression Inventory	Analysis 1.1; Analysis 1.3; Analysis 3.1; Analysis 3.3; Analysis 5.1; Analysis 5.2; Analysis 5.3; Analysis 5.4; Analysis 5.5; Analysis 6.1; Analysis 6.7
Klein 1985 (USA)	Source: non-clinical (community volunteers via media); Age: mean 30.1 years; Sex: 72% women	Aerobic exercise (individual - running)	1. Placebo; 2. Cognitive Behavioural Therapy	1. Symptom checklist 2. Target symptoms 3. Structural Analysis of Social Behaviour 4. Social Adjustment Self-reported Questionnaire 5. Cornell Medical Index 6. Role Rating Questionnaire 7. Hamilton Rating Scale 8. Global Assessment Scale	Analysis 1.1; Analysis 1.2; Analysis 1.3; Analysis 2.1; Analysis 2.2; Analysis 2.3; Analysis 5.1; Analysis 5.2; Analysis 5.3; Analysis 5.4; Analysis 5.5; Analysis 6.1; Analysis 6.7
Knubben 2007 (Germany)	Source: clinical (in-patients); Age: mean 49 years; Sex: 55% women	Aerobic exercise (walking on treadmill)	Placebo	1. Bech-Rafaelsen Mania Scale (BRMS) 2. Center for Epidemiologic Studies Depression Scale (CES-D)	Analysis 1.1; Analysis 1.3; Analysis 5.1; Analysis 5.2; Analysis 5.3; Analysis 5.4; Analysis 5.5; Analysis 6.1; Analysis 6.3; Analysis 6.5; Analysis 6.7

Table 2. Characteristics of included studies and synthesis (Continued)

Krogh 2009 (Denmark)	Source: clinical and non-clinical (general practitioners, private psychiatrists, psychologists and psychiatric ward institutions); Age: mean 38.9 years; Sex: 73.9% women	Resistance exercise (group)	1. Combination aerobic (cycling, running, stepping, rowing); 2. Relaxation	Hamilton Rating Scale for Depression	Analysis 1.1; Analysis 1.2; Analysis 1.3; Analysis 5.1; Analysis 5.2; Analysis 5.3; Analysis 5.4; Analysis 5.5; Analysis 6.1; Analysis 6.3; Analysis 6.4; Analysis 6.5; Analysis 6.6; Analysis 6.7
Kruisdijk 2018 (Netherlands)	Source: clinical (inpatient, day case, outpatient); Age: mean 41.6 years; Sex: 62.5% women	Aerobic exercise plus treatment as usual (Group - Nordic Walk or Running)	Combined therapy (pharmacotherapy, supportive psychotherapy and daily activation for hospitalised or those in day treatment, and psychotherapy and Cognitive Behavioural Therapy for out-patients)	Hamilton Rating Scale of Depression	Analysis 1.1; Analysis 1.3; Analysis 5.1; Analysis 5.2; Analysis 5.3; Analysis 5.4; Analysis 5.5; Analysis 6.1; Analysis 6.5; Analysis 6.7
La Rocque 2021 (Canada)	Source: non-clinical; Age: mean range 29.4 to 34.85 years; Sex: 100% women	Combined exercise (group - aerobic, muscular conditioning, stretching)	1. No treatment, waiting list, usual care or self-management (waitlist control); 2. Yoga	Hamilton Rating Scale for Depression (HAMD)	Analysis 1.1; Analysis 1.3; Analysis 5.1; Analysis 5.2; Analysis 5.3; Analysis 5.4; Analysis 5.5; Analysis 6.1; Analysis 6.3; Analysis 6.4; Analysis 6.5; Analysis 6.6; Analysis 6.7
Legrand 2014 (France)	Source: non-clinical (adverts and word of mouth); Age: median 23 years; Sex: 100% women	Fast walking (group)	1. No treatment, waiting list, usual care or self-management; 2. Zumba; 3. Jogging	Beck Depression Inventory (BDI-II)	Analysis 1.1; Analysis 1.3; Analysis 5.1; Analysis 5.2; Analysis 5.3; Analysis 5.4; Analysis 5.5; Analysis 6.1; Analysis 6.7
Legrand 2016 (France)	Source: clinical (inpatients); Age: median 45.4 years; 68.6% women	Aerobic exercise (individual - walk and/or running)	1. No treatment, waiting list, usual care or self-management (no intervention); 2. Stretching	Beck Depression Inventory (BDI-II)	Analysis 1.1; Analysis 1.3; Analysis 5.1; Analysis 5.2; Analysis 5.3; Analysis 5.4; Analysis 5.5; Analysis 6.1; Analysis 6.3; Analysis 6.4; Analysis 6.7
Martinsen 1985 (Norway)	Source: clinical (psychiatric hospital inpatients); Age: mean 40 years; Sex: NR	Aerobic exercise (instructor)	Placebo (occupational therapy)	Beck Depression Inventory	Analysis 4.1; Analysis 4.3
Mather 2002 (UK)	Source: clinical and non-clinical (primary care, psychiatric services); Age: mean 63.7 to 66.2 years; Sex: 68.6% women	Combined exercise (group - endurance, muscle strengthening, stretching)	Placebo (health education)	1. Hamilton Rating Scale for Depression 2. Geriatric Depression Scale	Analysis 4.1; Analysis 4.2; Analysis 4.3
McCann 1984 (USA)	Source: non-clinical (undergraduate stu-	Aerobic exercise (group -	1. Placebo; 2. No treatment, waiting	Beck Depression Inventory	

Table 2. Characteristics of included studies and synthesis (Continued)

	dents); Age: NR; Sex: 100% women	running, jogging, dancing)	list, usual care or self-management (waiting list)		
McNeil 1991 (Canada)	Source: non-clinical (community volunteers); Age: mean 72.5 years; Sex: NR	Aerobic exercise (individual - walking)	1. Placebo (social contact group); 2. No treatment, waiting list, usual care or self-management (waiting list)	Beck Depression Inventory	Analysis 1.1; Analysis 1.3; Analysis 5.1; Analysis 5.2; Analysis 5.3; Analysis 5.4; Analysis 5.5; Analysis 6.1; Analysis 6.4; Analysis 6.7
Mota-Pereira 2011 (Portugal)	Source: clinical (outpatients); Age: 18 to 60 years; Sex: 65% women	Aerobic exercise (individual - walking)	No treatment, waiting list, usual care or self-management (no exercise)	Hamilton Depression Scale	Analysis 1.1; Analysis 1.3; Analysis 5.1; Analysis 5.2; Analysis 5.3; Analysis 5.4; Analysis 5.5; Analysis 6.1; Analysis 6.5; Analysis 6.7
Mutrie 1988 (UK)	Source: clinical (primary care); Age: mean 42.1 years; Sex: 83% women	Combined exercise (Individual - Endurance, muscle strengthening, stretching)	1. Placebo; 2. No treatment, waiting list, usual care or self-management (waiting list)	Beck Depression Inventory	Analysis 1.1; Analysis 1.3; Analysis 5.1; Analysis 5.2; Analysis 5.3; Analysis 5.4; Analysis 5.5; Analysis 6.1; Analysis 6.4; Analysis 6.7
Nabkasorn 2005 (Thailand)	Source: non-clinical (student nurses); Age: 18 to 20 years; Sex: 100% women	Aerobic exercise (group - running)	No treatment, waiting list, usual care or self-management (usual care)	Center for Epidemiological Studies Depression (CESD)	Analysis 1.1; Analysis 1.3; Analysis 5.1; Analysis 5.2; Analysis 5.3; Analysis 5.4; Analysis 5.5; Analysis 6.1; Analysis 6.7
Olson 2017 (USA)	Source: clinical and non-clinical; Age: mean 21.1 years; Sex: 78.57% women	Aerobic exercise (treadmill or cycle ergometer)	Stretching	Beck Depression Inventory-II	Analysis 1.1; Analysis 1.3; Analysis 5.1; Analysis 5.2; Analysis 5.3; Analysis 5.4; Analysis 5.5; Analysis 6.1; Analysis 6.7
Orth 1979 (USA)	Source: non-clinical (college students); Age: mean 22 years; Sex: 27% women	Aerobic exercise (individual - jogging)	1. No treatment, waiting list, usual care or self-management (self monitoring); 2. Meditation; 3. Self-chosen activity	1. Depression Adjective Checklist 2. Minnesota Multiphasic Personality Inventory	Analysis 1.1; Analysis 1.3; Analysis 5.1; Analysis 5.2; Analysis 5.3; Analysis 5.4; Analysis 5.5; Analysis 6.1; Analysis 6.4; Analysis 6.7
Pfaff 2014 (Australia)	Source: non-clinical; Age: mean 60.97 years; Sex: 62.9% women	Medical care plus resistance training	No treatment, waiting list, usual care or self-management (medical care)	1. Montgomery-Asberg Depression Rating Scale (SIGMA) percent reduction in depressive symptomatology 2. Structured Clinical Interview for DSM-IV Axis I Disorders (SCID-I)	

Table 2. Characteristics of included studies and synthesis (Continued)

				- remission of symptoms (no longer meeting the criteria for major and minor depression)	
Pilu 2007 (Italy)	Source: clinical (university psychiatric unit); Age: 40 to 60 years; Sex: 100% women	Resistance training (Group) plus pharmacological treatment	Pharmacological treatment	Hamilton Rating Scale for Depression	Analysis 1.1; Analysis 1.3; Analysis 1.4; Analysis 5.1; Analysis 5.2; Analysis 5.3; Analysis 5.4; Analysis 5.5; Analysis 6.1; Analysis 6.4; Analysis 6.7
Pinchasov 2000 (Russia)	Source: non-clinical; Age: mean 35.2 years; Sex: 100% women	Aerobic exercise (stationary cycling)	Bright light therapy	Hamilton Depression Rating Scale	Analysis 4.1
Prakhinkit 2014 (Thailand)	Source: non-clinical (welfare centres); Age: mean 76.5 years; Sex: 100% women	Aerobic exercise (Walking - Group)	1. No treatment, waiting list, usual care or self-management; 2. Buddhist walking meditation	Geriatric Depression Scale	Analysis 1.1; Analysis 1.3; Analysis 5.1; Analysis 5.2; Analysis 5.3; Analysis 5.4; Analysis 5.5; Analysis 6.1; Analysis 6.7
Reuter 1984 (USA)	Source: clinical (university students); Age: NR; Sex: NR	Aerobic exercise (running) plus Cognitive Behavioural Therapy (counselling)	Cognitive Behavioural Therapy (counselling)	Beck Depression Inventory	Analysis 1.1; Analysis 5.1; Analysis 5.2; Analysis 5.3; Analysis 5.4; Analysis 5.5; Analysis 6.2; Analysis 6.7
Roy 2018 (India)	Source: clinical (inpatient); Age: mean 32 years; Sex: 100% women	Aerobic exercise (group - unspecified)	No treatment, waiting list, usual care or self-management	Hamilton Rating Scale for Depression	Analysis 1.1; Analysis 1.3; Analysis 5.1; Analysis 5.2; Analysis 5.3; Analysis 5.4; Analysis 5.5; Analysis 6.1; Analysis 6.7
Salehi 2016 (Iran)	Source: clinical (university council service); Age: mean 29.67 years; Sex: 30% women	Aerobic exercise (individual - cycling on treadmill)	1. Aerobic exercise plus electroconvulsive therapy; 2. Electroconvulsive therapy	Beck Depression Inventory (BDI-II)	Analysis 1.1; Analysis 1.3; Analysis 5.1; Analysis 5.3; Analysis 5.4; Analysis 5.5; Analysis 6.1; Analysis 6.3; Analysis 6.4; Analysis 6.5; Analysis 6.6; Analysis 6.7
Schuch 2015 (Brazil)	Source: clinical (inpatient); Age: mean 40.3 years; Sex: 74% women	Aerobic exercise (individual - participant preference)	No treatment, waiting list, usual care or self-management (treatment as usual)	Hamilton Scale for Depression (HAM-D)	Analysis 1.1; Analysis 1.3; Analysis 1.4; Analysis 5.1; Analysis 5.2; Analysis 5.3; Analysis 5.4; Analysis 5.5; Analysis 6.1; Analysis 6.4; Analysis 6.7 Authors contacted and provided additional unpublished data.
Setaro 1985 (USA)	Source: non-clinical (community volunteers); Age: 18 to 35 years; Sex: 26% women	Aerobic exercise (group - dance)	1. No treatment, waiting list, usual care or self-management; 2. Cognitive Behavioural Therapy; 3. Non-aerobic exercise; 4. Cognitive Be-	Minnesota Multiphasic Personality Inventory	Analysis 1.1; Analysis 2.1; Analysis 5.1; Analysis 5.2; Analysis 5.3; Analysis 5.4; Analysis 5.5; Analysis 6.1; Analysis 6.7

Table 2. Characteristics of included studies and synthesis *(Continued)*

			havioural Therapy plus non-aerobic exercise; 5. Cognitive Behavioural Therapy plus aerobic exercise		
Shahidi 2011 (Iran)	Source: non-clinical (local cultural community); Age: 60 to 80 years; Sex: 100% women	Aerobic exercise (group - running)	1. No treatment, waiting list, usual care or self-management; 2. Yoga	Geriatric Depression Scale	Analysis 1.1; Analysis 1.3; Analysis 5.1; Analysis 5.2; Analysis 5.3; Analysis 5.4; Analysis 5.5; Analysis 6.1; Analysis 6.7
Sims 2009 (Australia)	Source: clinical and non-clinical (people 6 months post-stroke recruited from hospital databases, general practitioners, newspaper articles); Age: mean 67.13 years; Sex: 40% women	Resistance training (group)	Waiting list receiving usual care	Center for Epidemiologic Studies for Depression Scale	Analysis 1.1; Analysis 1.2; Analysis 1.3; Analysis 5.1; Analysis 5.2; Analysis 5.3; Analysis 5.4; Analysis 5.5; Analysis 6.1; Analysis 6.3; Analysis 6.7
Singh 1997 (USA)	Source: non-clinical (community volunteers); Age: mean 70 years; Sex: 63% women	Resistance training (group)	Placebo	1. Beck Depression Inventory 2. Hamilton Rating Scale of Depression	Analysis 4.1; Analysis 4.2; Analysis 4.3
Singh 2005 (Australia)	Source: non-clinical (postal questionnaire responders); Age: mean 69 years; Sex: 55% women	Resistance training (group - high intensity)	1. No treatment, waiting list, usual care or self-management; 2. Resistance training (low intensity)	1. Hamilton Rating Scale for depression 2. Geriatric Depression Scale	Analysis 1.1; Analysis 1.3; Analysis 5.1; Analysis 5.2; Analysis 5.3; Analysis 5.4; Analysis 5.5; Analysis 6.1; Analysis 6.3; Analysis 6.5; Analysis 6.7
Siqueira 2016 (Brazil)	Source: clinical (out-patient clinics); Age: mean 38.8 years; Sex: 71.9% women	Aerobic exercise (individual - treadmill) plus sertraline	No treatment, waiting list, usual care or self-management (sertraline)	1. Hamilton Depression Rating Scale; 2. Beck Depression Inventory	Analysis 1.1; Analysis 1.3; Analysis 5.1; Analysis 5.2; Analysis 5.3; Analysis 5.4; Analysis 5.5; Analysis 6.1; Analysis 6.3; Analysis 6.4; Analysis 6.5; Analysis 6.6; Analysis 6.7
Taheri 2018 (Iran)	Source: unclear; Age: mean 45 years; Sex: 100% women	Aerobic exercise (jogging)	1. Placebo; 2. Exercise plus omega; 3. Omega	Beck Depression Inventory	Analysis 1.1; Analysis 1.3; Analysis 5.1; Analysis 5.2; Analysis 5.3; Analysis 5.4; Analysis 5.5; Analysis 6.1; Analysis 6.7
Veale 1992 (UK)	Source: clinical (psychiatric hospital out-patients and day patients); Age: mean 35.5 years; Sex: 64% women	Aerobic exercise (group - running)	No treatment, waiting list, usual care or self-management (routine care)	1. Beck Depression Inventory 2. State-Trait Anxiety Inventory	Analysis 1.1; Analysis 1.3; Analysis 5.1; Analysis 5.2; Analysis 5.3; Analysis 5.4; Analysis 5.5; Analysis 6.1; Analysis 6.3; Analysis 6.7
Wang 2022 (China)	Source: clinical; Age: NR; Sex: NR	Aerobic exercise (jogging)	Pharmacotherapy (Biyoujie)	Hamilton Depression Rating Scale	Analysis 1.1; Analysis 1.2; Analysis 5.1; Analysis 5.2; Analysis 5.3;

Table 2. Characteristics of included studies and synthesis *(Continued)*

				(HAMD, 17 items)	Analysis 5.4; Analysis 6.1; Analysis 6.7
Williams 2008 (USA)	Source: non-clinical (long term care - Alzheimer's dementia); Age: mean 87.9 years; Sex: 89% women	Walking and strength training (individual)	1. Social conversation; 2. Walking alone	Cornell Scale for Depression in Dementia	Analysis 1.1; Analysis 1.3; Analysis 5.1; Analysis 5.2; Analysis 5.3; Analysis 5.4; Analysis 5.5; Analysis 6.1; Analysis 6.3; Analysis 6.5; Analysis 6.7
Yu 2023 (Hong Kong)	Source: non-clinical; Age: mean 60 years; Sex: 73.33 % women	Aerobic exercise (individual - treadmill walk - vigorous)	1. No treatment, waiting list, usual care or self-management; 2. Aerobic exercise (treadmill walk - moderate)	Beck Depression Inventory (BDI)	Analysis 1.1; Analysis 1.3; Analysis 5.1; Analysis 5.2; Analysis 5.3; Analysis 5.4; Analysis 5.5; Analysis 6.1; Analysis 6.5; Analysis 6.7

All studies are RCTs.

Scale names and acronyms are reported as designated by the trial authors.

Abbreviations: ID: identification; NR: not reported